Isomorphism of Finite Groups Given by Multiplication Tables

Abstract

We give a deterministic algorithm for isomorphism of finite groups specified by multiplication tables. For groups of order N, the algorithm runs in $N^{O(1)}$ time and returns either a negative decision or an actual isomorphism together with generators for the full automorphism group. The reduction preserves the complete isomorphism coset along characteristic elementary abelian subgroups. Its principal algebraic step is simultaneous equivalence of a module action and an extension class under the same quotient automorphisms. We construct the required representations and cohomology maps explicitly, retain the kernels of all induced actions, and bound the entire recursion in terms of the original group order. The necessary tensor-isomorphism and affine-action algorithms, including their representation-theoretic ingredients, are proved in this full version.

1 Introduction

Let G and H be finite groups specified by their multiplication tables. The task is to decide whether $G \cong H$ and, in the affirmative case, construct an isomorphism. A complete solution also constructs generators for $\mathrm{Aut}(G)$; together with one isomorphism $f : G \to H$, these describe every isomorphism as the coset $f\,\mathrm{Aut}(G)$. This complete output is essential for recursive algorithms. An isomorphism of two quotient groups need not extend to an isomorphism of the original groups, and selecting a single quotient isomorphism can discard every valid extension.

The multiplication-table model permits algorithms polynomial in the group order, rather than in its logarithm. This distinction makes several exponentially bounded linear-algebraic procedures affordable, provided that their exponential parameter is at most a constant multiple of the logarithm of the original group order. It does not permit enumerating arbitrary changes of basis on a vector space of logarithmic dimension: over $\mathbb{F}_p$, such an enumeration can cost $p^{\Theta((\log_p |G|)^2)}$.

Main result. Theorem 2.1 proves the stated polynomial bound for arbitrary finite groups, with no restriction on their composition factors, nilpotency class, or Sylow subgroups. The output describes the entire isomorphism coset. All unpublished ingredients required by the argument are proved within this paper; the bibliography contains the published foundations and related publicly available work.

The algebraic difficulty. Suppose $A \lhd G$ is an elementary abelian characteristic subgroup and $Q = G/A$. After choosing a basis of A and a section of the quotient map, the group is described by a representation of Q on A and a two-cocycle for that representation. An isomorphism must transform both data using the same automorphism of Q and the same linear map on A. In particular, solving a module-isomorphism problem for one fixed quotient automorphism is insufficient. Likewise, replacing the allowed quotient automorphisms by independently chosen transformations on separate coefficient spaces can introduce false isomorphisms.

The proof below addresses this simultaneous action directly. It separates explicit coordinate length from the dimensions on which the supplied groups act, represents cohomology by kernels and quotients of concrete linear maps, and passes complete isomorphism cosets between consecutive steps. Every kernel discarded by a linear representation is retained in the original group until its action on the remaining data has been handled.

Uniformity in the coefficient space. The intermediate algorithm concerns an affine group action on an invariant subquotient of a direct sum of tensor products of at most three given representation spaces. If their total dimension over $\mathbb{F}_p$ is d, its running time is $p^{O(d)}\,\mathrm{poly}(L)$, where L includes the explicit subspace, quotient, cocycle, and original-group data; see Theorem 15.4 for the precise input conditions. In particular, the dimension of a repeated coefficient space does not enter the exponent when the group fixes that space pointwise. This restriction is essential: an independently moving coefficient space must be included among the representation spaces counted in d.

For a group extension of order $N = p^a|Q|$, the constructions below use representation spaces of total dimension at most $C(2a + \log_p |Q|)$, for an absolute constant C. Hence the exponential factor is polynomial in N. The coordinate spaces, including the cochain spaces, are explicitly represented and have polynomial size in N. A bound on one call is not enough: the proof also bounds the total number of cohomology and module calls and gives only one recursive group isomorphism call, on a quotient of order at most $N/2$.

Preserving exact solution sets. Two features of the intermediate algorithm are important for the group application. First, when a representation has a nontrivial kernel, that kernel can still act by translations in the affine problem. The algorithm computes its entire translation image and retains the corresponding kernel in each isomorphism coset. Second, a filtration is used through successive exact quotient problems and their fibers. It is never identified with a direct sum of its factors. In particular, a group that acts naturally on a quotient representation is not assumed to have a compatible action on a previously chosen nonsplit representation.

The recursion on evaluation arguments always restricts actual invariant summands. Subsequent changes of presentation may change the evaluations used in the remaining problem, but cannot restore a summand that an ancestor case has excluded. This gives a single weighted recursion-tree bound, including all unsuccessful target cases. Changes in the supplied subgroup and invariant subquotients are accounted for in their original representations, so auxiliary additive groups do not create new progress steps or a fresh exponential allowance.

Related work. The multiplication-table isomorphism problem has a classical $N^{O(\log N)}$-time algorithm. A direct explanation of this bound is that an N-element group has a generating set of size at most $\log_2 N$, so one can enumerate the images of a generating tuple and check the resulting maps. This elementary bound is not optimal: subsequent improvements use bidirectional collision detection and polynomial-time isomorphism testing with fixed composition series [Ros13, Luk15]. The extension-and-cohomology approach of Grochow and Qiao [GQ17] separates compatible actions from compatible extension classes. Babai, Codenotti, and Qiao [BCQ12] give a polynomial-time algorithm for groups with no nontrivial abelian normal subgroup. The complete quotient-isomorphism coset formulation needed here is stated explicitly in [GQ15, Section 1.1].

The tensor viewpoint connects several algebraic isomorphism problems. Grochow and Qiao [GQ23] establish tensor-isomorphism completeness results, while their linear-length reductions [GQ25] preserve parameters that are crucial when translating exponential-in-dimension

bounds into multiplication-table bounds. These reductions do not by themselves give the general polynomial-time algorithm sought here. For odd primes, Sun [Sun23] gives an $N^{O((\log N)^{5/6})}$-time algorithm for groups of class two and exponent p. Ivanyos, Mendoza, Qiao, Sun, and Zhang [IMQ+24] improve this to $N^{\widetilde{O}(\sqrt{\log N})}$ for groups of Frattini class two at odd primes. A recent preprint of Grochow, Ivanyos, Qiao, and Sun [GIQS26] gives an $N^{O(\sqrt{\log N})}$ bound for the corresponding class of 2-groups. These are related results rather than suppliers of the new general affine transporter theorem used below.

Outline. The first part gives the group-theoretic reduction and the cohomological constructions. The subsequent sections prove the representation-theoretic and tensor-isomorphism algorithms used by that reduction. Their order is important: algorithms for tensor products of at most two factors are proved before general group intersections; ordinary three-tensor isomorphism is then proved using only those earlier algorithms; finally, special representation-theoretic constructions yield the affine three-factor problems needed for cohomology. There is no invocation of an unproved general group-isomorphism procedure in these intermediate algorithms.

The evaluation procedures used in the proof work with actual invariant subspaces and quotients. Passing to associated graded spaces alone does not preserve every equation observed earlier in a recursion. Accordingly, the algorithm first computes the exact remaining affine equation under the full returned stabilizer, and only then changes its representation. Its recursion bound counts all target choices, unsuccessful branches, and stabilizer computations. A separate invariant bounds the total number of changes of representation; the exponential allowance is not restarted at each recursive depth.

Notation and input conventions. Throughout, N denotes the order of an input group. Letters p and $q = p^e$ denote a prime and a prime power, respectively. Vector spaces and linear maps are specified by bases and matrices over an explicit finite field. Groups acting linearly are supplied by generators and by the additional original permutation or additive representations specified at each call. The symbol L denotes the length of this complete explicit input, including any retained representation. All algorithms are deterministic unless explicitly stated otherwise. Products of maps act from right to left. For a left action of a group Γ, the transporter from x to y is

$$\{g \in \Gamma : gx = y\}.$$

If it is nonempty and $tx = y$, it equals $t\Gamma_x$.

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2 From tensor transport to finite group isomorphism

This section proves the group-theoretic reduction, including its output and uniform running time. Throughout, N is the order of an input group; at one step of the recursion we write $m = p^a n$, where the selected normal subgroup has order p^a and the quotient has order n. All vector spaces are over $\mathbb{F}_p$. A section of a module means an explicitly given subquotient U/P, with $P \leq U$ invariant. Its matrices and the maps to and from its chosen quotient coordinates are part of its description.

Theorem 2.1 (Complete finite-group isomorphism)**.** *There is a deterministic algorithm which, given multiplication tables of finite groups G and H of order at most N, decides whether they are isomorphic in time $N^{O(1)}$. If they are isomorphic, it returns an isomorphism $G \to H$ and generators, as permutations of G, of the entire* $\mathrm{Aut}(G)$*. Consequently it computes the full isomorphism coset and, by applying it to (G, G), the automorphism group of any finite group. The exponent is absolute and independent of every prime dividing N.*

Proof. The internally proved Theorem 15.4, with arity three, gives the complete affine transporter in (2.1), including all original comparison kernels. Proposition 2.2, proved in this section, reduces finite-group isomorphism to this transporter and the published complete radical-free algorithm [BCQ12, GQ15]. Its dimension, input-length, and total-call bounds are uniform in the prime. The resulting single-quotient recurrence is polynomial, and its complete extension lifts give the asserted witness and entire automorphism group. □

We use the following precise consequence of Theorem 15.4. Let $X_1, \dots, X_s$ be specified vector spaces of total dimension d. Consider an invariant section of a finite direct sum of tensor products, each with at most three occurrences of the X_i or their duals. Repetitions of a space use the same matrix; the direct-sum multiplicity labels have trivial action. Given the actual matrices of a comparison group, its faithful permutation representation, an affine action on the section with that induced linear action, and two points, one can compute their complete transporter in

$$p^{cd}(L \log p)^k \tag{2.1}$$

for absolute constants c, k. Here L includes all explicit subspace, quotient, affine-action, and original permutation data. The output is empty, or consists of an element taking the first point to the second and generators of its full stabilizer in the original comparison group. In particular, a kernel of the action on the X_i is not silently removed. We only use prime-field natural and dual representations in this section. The following reduction does not assume that an abstract representation of a large coefficient space has such a tensor description. Every use of this description is constructed below.

Proposition 2.2 (Complete group reduction)**.** *The complete tensor transporter with the bound* (2.1)*, together with the published complete polynomial-time isomorphism algorithm for groups with trivial solvable radical, yields a deterministic $N^{O(1)}$ algorithm for isomorphism of arbitrary finite groups given by multiplication tables. On isomorphic inputs it returns an isomorphism and generators, as permutations of the source group, of its entire automorphism group. The exponent is independent of every prime dividing N.*

The algorithm makes one recursive call on a quotient by a characteristic nonzero elementary abelian subgroup. It then solves conjugation-action compatibility, transports an extension class, and lifts all stabilizer generators. The radical-free case is handled by the published base algorithm. At an elementary extension step $m = p^a n$, the reduction uses polynomially many complete tensor transports with

$$d \le O(2a + \log_p n), \qquad p^d \le m^{O(1)}, \qquad L \le m^{O(1)}.$$

These bounds hold at every prime with absolute constants. They give one extension-step bound m^C and the single-call recurrence $T(m) \le T(m/2) + m^C$, rather than a recurrence that repeats an exponential factor at each cohomological depth. The proofs below establish the small tensor realizations, the polynomial total number of their uses, and all original kernels required for this conclusion.

The radical-free algorithm is the algorithm of Babai, Codenotti, and Qiao [BCQ12]. Its complete-coset output is explicitly recorded before Problem 1 and in the proof of Corollary 2

of Grochow and Qiao [GQ15]; thus we do not infer a generating-set algorithm from a decision-only statement. We use the standard deterministic permutation-group algorithms for membership, point stabilizers, homomorphism kernels with an explicit action, normal closure, and generator compression [Ser03]. Their cost is polynomial in the faithful permutation degree and the supplied generator length. An explicit action on an exponentially large vector set is used only when that set has size $p^{O(d)}$ within (2.1).

2.1 Characteristic elementary subgroups and extension lifting

The solvable radical $\mathrm{rad}(G)$ is computable in polynomial table time. For each $g \in G$, compute its normal closure by repeatedly adjoining conjugates and products, and compute the derived series of that closure. Generate the product of the closures found to be solvable. Products of solvable normal subgroups are solvable: the product of two has one as a normal subgroup and a quotient of the other as its quotient. Conversely, every element of $\mathrm{rad}(G)$ has solvable normal closure. The computed product is therefore exactly $\mathrm{rad}(G)$.

If $\mathrm{rad}(G) \neq 1$, let D be its last nonidentity derived subgroup. It is characteristic in G and abelian. Let p be the least prime dividing $|D|$, let D_p be its unique Sylow p-subgroup, and set

$$A(G) = \{x \in D_p : x^p = 1\}.$$

This is a nonzero characteristic elementary abelian subgroup. Factoring integers at most N by trial division, enumerating D_p, and finding a basis of $A(G)$ all cost polynomial time. The construction commutes with every group isomorphism; choosing merely some elementary characteristic subgroup of the appropriate order would not suffice.

Let $A = A(G) \cong \mathbb{F}_p^a$, $Q = G/A$, and choose a set-theoretic section $s : Q \to G$ with $s(1) = 1$. In coordinates $A \times Q$, multiplication is

$$(u, q)(v, r) = (u + \theta(q)v + f(q, r), qr), \tag{2.2}$$

where $\theta : Q \to \mathrm{GL}(A)$ is conjugation and f is a normalized 2-cocycle. We use the cochain convention

$$\begin{aligned}(\delta b)(q, r) &= \theta(q)b(r) - b(qr) + b(q),\\ (\delta f)(q, r, t) &= \theta(q)f(r, t) - f(qr, t) + f(q, rt) - f(q, r).\end{aligned}$$

Changing s changes f by a coboundary. Cochains of degree at most three have at most $a|Q|^3$ coordinates, so bases for cycles, boundaries, and their quotient are computable by Gaussian elimination. Their coordinate dimension will be charged to L, not to d.

After a quotient isomorphism identifies the two quotient tables, every isomorphism preserving the selected subgroups has the form

$$F(u, q) = (\alpha u + b(q), \beta q), \qquad \alpha \in \mathrm{GL}(A), \quad \beta \in \mathrm{Aut}(Q).$$

Substitution in (2.2) proves that it is an isomorphism if and only if

$$\alpha\theta_G(q) = \theta_H(\beta q)\alpha, \tag{2.3}$$

$$b(q) + \theta_H(\beta q)b(r) - b(qr) = \alpha f_G(q, r) - f_H(\beta q, \beta r). \tag{2.4}$$

The compatible-pair action on cochains is $((\alpha, \beta)f)(q, r) = \alpha f(\beta^{-1}q, \beta^{-1}r)$. Thus the two tasks, in this order, are complete transport of module actions and complete transport of their extension classes. A first compatible pair identifies the two coefficient modules; subsequent class transport is under its full source stabilizer.

Lemma 2.3 (All extension lifts)**.** *A complete compatible-pair transporter of the two extension classes, followed by linear algebra, gives the complete group-isomorphism coset.*

Proof. Solve (2.4) for its witness, and for every generator of the source class stabilizer with $G = H$. These systems are consistent by the definition of class transport. The kernel of projection to the compatible-pair group consists exactly of

$$(u, q) \longmapsto (u + z(q), q), \qquad z \in Z^1(Q, A).$$

Compute a vector-space basis of this kernel and include the corresponding permutations. The lifted generators project onto the whole class stabilizer and include its entire projection kernel. They therefore generate the whole automorphism group preserving A. Since A is characteristic, this is $\mathrm{Aut}(G)$. Converting all maps to the original table labels and checking their multiplication identities is polynomial time. Notice that the kernel is $Z^1(Q, A)$, not $H^1(Q, A)$; in particular it is not generally $\mathrm{Hom}(Q, A)$. □

2.2 Filtrations, affine fibers, and invisible kernels

For a left action, write $\mathrm{Trans}_\Gamma(x, y) = t\Gamma_x$ when this set is nonempty. Complete transport on an invariant filtration is done from its quotient downward. Once a quotient condition has been solved, normalize its witness so that the two points lie in the same fiber. The complete quotient stabilizer acts affinely on that fiber. Relative to a point x in the fiber its translation is $g \mapsto gx - x$, an explicitly computable 1-cocycle with values in the filtration subspace. Solving this affine condition and multiplying the witnesses gives the complete transporter at the next level. This proves by induction the validity of descending filtration transport, including arbitrary supplied affine translations.

For an invariant section U/P of a filtered space W, the induced filtration is $((U \cap F_i) + P)/P$. Each of its factors is a section of F_i/F_{i-1}: explicitly, use the numerator $U \cap F_i$ and denominator $(U \cap F_{i-1}) + (P \cap F_i)$ before passing to the appropriate image in F_i/F_{i-1}. Images, inverse images, intersections, and denominator sums are Gaussian-elimination operations. This retains correlations between all coordinates; it does not transport separate generators of U and P.

Here is an explicit procedure for a kernel that is invisible in a particular small linear representation. It also explains why complete outputs at every intermediate step are necessary.

Lemma 2.4 (Affine kernel correction)**.** *Let Γ be supplied faithfully as a permutation group and let $\rho : \Gamma \to \mathrm{GL}(X)$ have $\dim X = d$. Suppose Γ acts affinely on W, with linear part induced from a specified tensor section on X, and write its action at 0 as $g \mapsto b(g)$. Then the full condition $b(g) = w$ reduces to complete affine transport in a section with enlarged denominator, together with $p^{O(d)}\,\mathrm{poly}(L)$ work. All original kernels and constructive preimages are retained.*

Proof. Compute $K = \ker \rho$ by successively stabilizing the images of a basis of X. Each vector orbit has at most p^d elements. Since K acts linearly trivially on W, the restriction $b|_K$ is a homomorphism to the additive group. Its image B is the $\mathbb{F}_p$-span of the displacements of its generators. Row reduction records a lift in K of every vector in a basis of B. The cocycle identity gives

$$b(gkg^{-1}) = g\,b(k) \qquad (k \in K),$$

so B is invariant. The affine action on W/B factors through $\rho(\Gamma)$. If $W = U/P$, use the preimage of B in U as the new denominator, without changing the tensor realization.

Lift a witness from the quotient action to $g \in \Gamma$. Its error $w - b(g)$ lies in B, and a left multiplier from K with that displacement corrects it. Correct each lifted source-stabilizer generator

in the same way, now making its displacement zero. It remains to compute all of $\ker(b|_K)$ without enumerating the vector set B.

Let $t_1, \dots, t_e \in K$ lift a basis of B. For each original generator k of K, express $b(k) = \sum_i a_i b(t_i)$ and include the relation $k(t_1^{a_1} \cdots t_e^{a_e})^{-1}$. Also include t_i^p and all commutators $[t_i, t_j]$. Let J be the normal closure of these elements in K. They lie in $\ker(b|_K)$. The quotient K/J is generated by e commuting elements of order dividing p, and it maps onto B, which has order p^e. Thus $K/J \cong B$ and $J = \ker(b|_K)$. Normal closure in the retained permutation representation computes this kernel in polynomial time. It and the corrected lifts generate the complete answer. □

The lemma is applied to the *whole current condition*. At a return that has matched only a quotient of the current space, elements whose translations lie in the remaining fiber are retained. Removing them before matching that fiber would lose valid solutions.

2.3 Module action equivalence with a common quotient automorphism

We need a slightly more general formulation than (2.3). Let an actual permutation group Γ have a given homomorphism $\beta : \Gamma \to \mathrm{Aut}(Q)$. For two modules M, M' of dimension a, compute all pairs (α, g) satisfying

$$\alpha\theta(q) = \theta'(\beta_g q)\alpha. \tag{$*$}$$

The same procedure applies to a supplied coset after normalizing one representative. Extra components of g are never forgotten. We first give elementary module operations whose cost is polynomial in $|Q| + p^a$; they do not enumerate the vector set of $\mathrm{End}(M)$.

For each nonzero $v \in M$, span its Q-orbit. A nonzero cyclic submodule of minimum dimension is simple, since a proper nonzero submodule would contain a smaller cyclic submodule. Repeating in successive quotients produces a composition series using at most a enumerations, each of size at most p^a. For simple modules S, T, solve the linear equations for $\mathrm{Hom}_Q(S, T)$. They are isomorphic exactly when this space is nonzero, and every nonzero map is then an isomorphism.

Compute $\mathrm{Hom}_Q(M, S)$ for each simple type occurring in a composition series of M, and intersect the kernels of bases of these spaces. The result is $\mathrm{rad}(M)$: every maximal submodule is the kernel of a surjection onto a simple composition factor, and intersecting a basis of homomorphisms has the same common kernel as intersecting all of them. In particular, $M/\mathrm{rad}(M)$ embeds into a finite direct sum of simple modules and is semisimple. This construction is invariant under module isomorphisms and twisting by automorphisms of Q.

A semisimple module can be decomposed constructively. Find a simple submodule S as above and solve linearly for a Q-linear map $M \to S$ whose restriction to S is the identity. Such a projection exists; its kernel is a complement. Iterate. This also constructs isomorphisms between semisimple modules by matching their simple summands.

For completeness, generators of their full automorphism groups need no large-algebra enumeration. On an isotypic summand $S^{\oplus e}$ put $D = \mathrm{End}_Q(S)$. Evaluation at a fixed nonzero vector of S injects D into S, by simplicity, so $|D| \le p^{\dim S} \le p^a$. Enumerate D from a computed basis; all its nonzero elements are invertible. The group $\mathrm{GL}_e(D)$ is generated by elementary transvections with entries from an additive basis of D, coordinate interchanges, and diagonal matrices with a nonzero entry from D. Gaussian elimination over the division ring D proves this generating claim. Transvections with arbitrary entries are products of those with basis entries. Taking these generators on every isotypic summand gives $\mathrm{Aut}_Q(M)$ in polynomial time.

Lemma 2.5 (Small-set transport)**.** *For a permutation group $P \le \mathrm{Sym}(\Omega)$ and subsets $S, T \subseteq \Omega$ of size b, their complete set transporter is computable in $2^{O(b)}\,\mathrm{poly}(|\Omega|)$ time.*

Proof. The cases $b = 0, 1$ use the whole group or a point orbit and stabilizer. Partition $S = S_1 \sqcup S_2$ as evenly as possible. For every $U \subseteq T$ of size $|S_1|$, recursively compute $\mathrm{Trans}_P(S_1, U) = g_U P_{S_1}$. Within P_{S_1} compute the complete transporter of S_2 to $g_U^{-1}(T \setminus U)$. Every solution occurs in exactly one of these branches. Each successful branch is a coset of the same subgroup $P_{S_1} \cap P_{S_2}$, supplied by its recursive answer; no general subgroup-intersection algorithm is invoked. Given one branch witness t, all other witnesses t_i, and that common subgroup, the elements $t^{-1}t_i$ together with its generators generate P_S. Indeed they stabilize S, and every element of P_S is the ratio of t to an element of one of the branch cosets. Compression keeps the returned generators polynomial in $|\Omega|$. The recurrence

$$T(b, u) \leq 2\binom{b}{\lfloor b/2 \rfloor} T(\lceil b/2 \rceil, u) + 2^{O(b)} \operatorname{poly}(u)$$

has solution $2^{O(b)} \operatorname{poly}(u)$, by summing its geometric sequence of exponents. Original permutations can be carried along throughout. □

For semisimple M, M', close their occurring simple isomorphism types under the generators of $\beta(\Gamma)$, using the preceding simple isomorphism test. There are at most $|Q|$ simple $\mathbb{F}_p Q$-module types: every simple module is a quotient of the regular module, by sending 1 to a nonzero vector, and hence is a composition factor of that $|Q|$-dimensional module. Thus this closure is found by polynomially many steps. Its representations have dimension at most a, since twisting preserves dimension. The induced label permutations are computed while retaining the original elements of Γ.

For each multiplicity, apply Lemma 2.5 to its two subsets of simple labels. There are at most a labels of positive multiplicity. Make the calls sequentially under the complete subgroup returned by the previous call. For one surviving witness and each stabilizer generator, match the simple summands to construct an intertwiner. Add the whole $\mathrm{Aut}_Q(M)$ as the kernel over Γ. This computes exactly all pairs in $(*)$ for semisimple modules, including every element of $\ker \beta$.

For general M, set $U = \mathrm{rad}(M)$ and $V = M/U$, and similarly for M'. Unequal dimensions cause rejection. Recursively match U, U' once, retaining pairs (α_U, g) rather than just α_U or β_g. Now apply the semisimple construction to V, V' with this returned pair group as its comparison group. A lift of each surviving generator and the entire $\mathrm{Aut}_Q(V)$ give exactly the complete triples (α_U, α_V, g) with the *same* β_g on both modules. This is a constructive preimage calculation, not an intersection of two independently computed quotient-automorphism images.

Choose vector-space splittings and write

$$\theta(q) = \begin{pmatrix} \theta_U(q) & c(q) \\ 0 & \theta_V(q) \end{pmatrix}, \qquad z(q) = c(q)\theta_V(q)^{-1}.$$

Multiplication of these matrices gives $z(qr) = z(q) + qz(r)$, where $qT = \theta_U(q) T \theta_V(q)^{-1}$ on $\mathrm{Hom}(V, U)$. A change of splitting changes z by a coboundary. A block upper triangular intertwiner has arbitrary off-diagonal block t; substitution shows that such a block exists exactly when the two classes in

$$H^1(Q, \mathrm{Hom}(V, U)) \tag{2.5}$$

are transported by the surviving triple. Its possible blocks form an affine linear system. The homogeneous kernel, after the diagonal maps are normalized, is exactly $\mathrm{Hom}_Q(V, U)$. Include a basis of its additive shear group. Complete class transport, one solution for each lifted generator, and these shears therefore give all pairs in $(*)$. The coefficient in (2.5) is $U \otimes V^*$, with two tensor occurrences and total underlying dimension $\dim U + \dim V = a$. Its possibly quadratic coordinate dimension is never used for vector enumeration. The next subsections construct the required H^1 transport. There are at most a module layers and exactly one recursive module call at each layer.

2.4 Explicit cohomology maps for arbitrary normal subgroups

We require affine transport on invariant sections of two kinds of cohomology spaces: $H^1(R, B)$ when B is a section on at most two tensor occurrences of specified genuine modules, and $H^2(R, A)$ when A is an actual module. All maps in this subsection are constructed on normalized cochains. Their number of coordinates is polynomial in $|R|$ and the coefficient dimension.

If $0 \to B' \to B \to B'' \to 0$ is an exact coefficient sequence, then

$$H^j(R, B') \longrightarrow H^j(R, B) \longrightarrow H^j(R, B'') \tag{2.6}$$

is exact at the middle term. To see this directly, if a cocycle maps to a coboundary $\delta u''$, lift u'' to a cochain u in B and subtract δu. The remaining cocycle has values in B'. This proves both exactness and the algorithm for its maps. Applied to a coefficient filtration $B_0 = 0 \leq \cdots \leq B_e = B$, the filtration by the images of $H^j(R, B_i)$ in $H^j(R, B)$ has each successive factor a section of $H^j(R, B_i/B_{i-1})$. Kernels caused by the map to the final coefficient are retained as additional denominators. Only the individual coefficient grades give new transport tasks; the cumulative B_i are used to compute linear maps, not as additional recursive tasks.

For a normal subgroup $S \lhd R$, restriction and inflation give

$$0 \longrightarrow H^1(R/S, B^S) \longrightarrow H^1(R, B) \longrightarrow H^1(S, B). \tag{2.7}$$

Indeed, if a cocycle restricts to a coboundary, subtract that principal cocycle so that it vanishes on S. Then $z(qs) = z(q)$ and $z(sq) = sz(q)$ imply that it descends to R/S with values in B^S. The same calculation proves injectivity of inflation. The final term in the filtration is the actual restriction image, computed linearly; no uncomputed invariant space is substituted for it.

Here is the corresponding degree-two construction, including nontrivial coefficient action. Put $W = H^2(R, B)$ and

$$F_1 = \ker(W \longrightarrow H^2(S, B)), \qquad F_2 = \operatorname{im}(H^2(R/S, B^S) \longrightarrow W).$$

For $[f] \in F_1$, solve for $s : S \to B$ with

$$s(xy) = s(x) + xs(y) + f(x, y).$$

Thus $x \mapsto (s(x), x)$ is a splitting over S in the extension $E_f = B \times R$ defined by (2.2). For $q \in R$ define

$$\zeta_q(x) = qs(q^{-1}xq) - s(x) + f(q, q^{-1}xq) - f(x, q). \tag{2.8}$$

It is the difference between this splitting and its conjugate by $(0, q)$; differences of splittings are crossed homomorphisms. Hence $\zeta_q \in Z^1(S, B)$. Comparing conjugation by $(0, q)(0, r)$ and by $(0, qr)$ yields, or direct substitution in $\delta f = 0$ verifies,

$$\zeta_q + q\zeta_r - \zeta_{qr} = -\delta_S f(q, r), \qquad (qz)(x) = qz(q^{-1}xq), \tag{2.9}$$

where $(\delta_S b)(x) = xb - b$. For $q \in S$, choosing its lift in the splitting shows $[\zeta_q] = 0$. Paired inner action on $H^1(S, B)$ is trivial: the cocycle identity gives $qz(q^{-1}xq) - z(x) = (x - 1)z(q)$ for $q \in S$. Consequently (2.9) defines a quotient cocycle.

Changing the splitting by $u \in Z^1(S, B)$ changes ζ_q by $qu - u$. Changing f to $f + \delta b$ and s to $s - b|_S$ changes ζ_q by $-\delta_S b(q)$. Changing a quotient representative has already been accounted for by the zero class on S. Thus we obtain a well-defined linear map

$$\psi : F_1 \longrightarrow H^1(R/S, H^1(S, B)). \tag{2.10}$$

Formula (2.8) commutes with compatible coefficient maps and automorphisms of R preserving S. The map on cohomology is therefore equivariant even if the chosen linear solutions for s are not.

Lemma 2.6 (Exact degree-two middle term)**.** *The kernel of* (2.10) *is* F_2*. In particular* $H^2(R,B)$ *has a natural three-step filtration whose factors are actual sections of*

$$H^2(R/S,B^S), \qquad H^1(R/S,H^1(S,B)), \qquad H^2(S,B).$$

Its spaces and all comparison maps are computable by polynomial-size linear systems.

Proof. An inflated cocycle with values in B^S has zero conjugation defect, so $F_2 \subseteq \ker\psi$. Conversely let $\psi[f]=0$. Change the splitting by a crossed homomorphism so that every $[\zeta_q]$ vanishes. Choose $b_q \in B$ with $\zeta_q = \delta_S b_q$. Replacing $(0,q)$ by $(b_q,1)(0,q)$ changes its defect by $-\delta_S b_q$, so every q has a lift normalizing the splitting $\sigma(S)$.

Let $T = N_{E_f}(\sigma(S))$. Then T projects onto R, $E_f = BT$, and $T \cap B = B^S$: conjugation by $b \in B$ preserves the splitting exactly when $b - xb = 0$ for every $x \in S$. Thus $T/\sigma(S)$ is an extension of R/S by B^S. The map

$$T \longrightarrow R \times_{R/S} (T/\sigma(S)), \qquad t \longmapsto (\bar{t}, t\sigma(S))$$

is injective and has the same order on both sides, so it is an isomorphism. Pushing out its coefficient inclusion $B^S \hookrightarrow B$ gives E_f: explicitly the map $B \rtimes T \to E_f$, $(b,t) \mapsto bt$, has exactly the diagonal kernel from $B^S = T \cap B$. Hence $[f] \in F_2$. This uses the normalizer of the splitting; the splitting need not be normal in E_f.

For computation, form the cochain spaces, restrictions, inflations, and F_1, F_2 by elimination. For each basis vector of F_1 solve for s, evaluate (2.8), and project twice through the computed coboundary spaces. This is the matrix of ψ and the explicit identification $F_1/F_2 \cong \operatorname{im}\psi$. The extension E_f is only used in the proof; its potentially enormous underlying set is not enumerated. □

If $p \nmid |S|$, then $H^j(S,B) = 0$ for $j > 0$. One direct verification uses homogeneous equivariant cochains and the contracting operator

$$(hf)(x_0,\dots,x_{j-1}) = |S|^{-1} \sum_{x \in S} f(x,x_0,\dots,x_{j-1});$$

with the alternating deletion differential, $\delta h + h\delta = 1$ in positive degrees. Equations (2.7) and Lemma 2.6 now give surjective inflation in degrees one and two. Inflation in degree two is also injective. If an inflated cocycle is δu, its restriction makes $u|_S$ a crossed homomorphism. Subtract a principal cocycle to arrange $u|_S = 0$. The equations on (q,s) and (s,q) imply that u is constant on S-cosets and takes values in B^S, proving that the original quotient cocycle is a coboundary. We may therefore replace these cohomology problems by $H^j(R/S,B^S)$.

2.5 Elementary normal subgroups

Suppose $C \lhd R$ is elementary abelian of characteristic p and dimension c. Its augmentation ideal $I \leq \mathbb{F}_p C$ is generated by $x_i - 1$ for a basis $x_1,\dots,x_c$ of C. These generators commute and have pth power zero, so $I^{c(p-1)+1} = 0$. The descending filtration $B \supseteq IB \supseteq I^2B \supseteq \cdots \supseteq 0$ is invariant under every compatible automorphism preserving C, and C acts trivially on every grade. Until zero is reached every inclusion is strict, since $I^jB = I^{j+1}B$ and nilpotence would imply $I^jB = 0$. There are thus at most $\dim B$ nonzero grades, with dimensions summing to $\dim B$. They are computed by spans of the matrices $x_i - 1$. Use (2.6) to reduce to these grades, preserving their actual subquotient descriptions.

For a C-trivial coefficient D, restriction in degree one has target $\mathrm{Hom}(C, D) = D \otimes C^*$. In degree two the natural coefficient-free identifications are

$$H^2(C, \mathbb{F}_p) \cong \begin{cases} \mathrm{Sym}^2(C^*), & p = 2, \\ \bigwedge^2(C^*) \oplus C^*, & p \text{ odd}. \end{cases} \tag{2.11}$$

Here $\mathrm{Sym}^2(C^*)$ means the quotient of $C^* \otimes C^*$ by $u \otimes v - v \otimes u$, not the invariant subspace under interchange of the two factors. This distinction is essential in characteristic two. Here are explicit reasons for these identifications and their naturality. A central extension of C by $\mathbb{F}_p$ has a commutator map $b : C \times C \to \mathbb{F}_p$, which is alternating bilinear, and a power map $q(x) = \hat{x}^p \in \mathbb{F}_p$. These do not depend on the lift. For odd p the class-two power formula makes q linear, since $\binom{p}{2} = 0$ in $\mathbb{F}_p$. For $p = 2$ it gives $q(x + y) = q(x) + q(y) + b(x, y)$, so q is a quadratic form and determines b. The space of quadratic forms over $\mathbb{F}_2$ is naturally $\mathrm{Sym}^2(C^*)$ via the polynomial q.

Conversely, on an ordered basis e_i choose the prescribed values of $q(e_i)$ and $b(e_i, e_j)$. Adjoin a central generator z of order p, with relations $[x_i, x_j] = z^{b(e_i, e_j)}$ and $x_i^p = z^{q(e_i)}$. Collecting products gives unique normal forms $z^t x_1^{a_1} \cdots x_c^{a_c}$ with $0 \le t, a_i < p$. One can verify existence without a presentation assumption. Write $a_i, a_i' \in \{0, \dots, p-1\}$ for coordinates and use the central 2-cocycle

$$f(a, a') = \sum_{i<j} a_i a_j' b(e_i, e_j) + \sum_i \left\lfloor \frac{a_i + a_i'}{p} \right\rfloor q(e_i).$$

The cross terms are bilinear modulo p, and carries satisfy the associativity identity for addition of three integers. The resulting product has order p^{c+1}, commutator b, and the specified powers of the basis lifts. Every extension admits this normal form, proving existence and uniqueness of its cohomology class from the data. The commutator and power definitions are basis-independent, so the identification is $\mathrm{GL}(C)$-equivariant. It is linear under addition of cocycles, or equivalently under addition of their commutator and power data. For general D with trivial C-action, tensor the cochain complex by D to get $H^2(C, D) = D \otimes H^2(C, \mathbb{F}_p)$.

Consequently Lemma 2.6, applied after the coefficient filtration, has three factors that are sections of

$$H^2(R/C, D), \qquad H^1(R/C, D \otimes C^*), \qquad D \otimes H^2(C, \mathbb{F}_p). \tag{2.12}$$

The last term uses at most three tensor occurrences. The middle term uses two actual R/C-modules, D and C; C is a quotient module under conjugation because it is abelian. Formula (2.11) uses a linear extra summand at odd characteristic, and never an additional p-fold tensor power.

2.6 Semisimple normal subgroups and their cochain realizations

If $R \neq 1$ has trivial solvable radical, its socle is a characteristic product $S = S_1 \times \cdots \times S_t$ of nonabelian simple groups. This standard structural assertion also has a direct elementary proof: a minimal normal subgroup is the product of the conjugates of a minimal normal subgroup of itself; distinct such subgroups intersect trivially and commute. Their product is the whole minimal normal subgroup, so its factors are simple: a normal subgroup of one factor would also be normal in the whole direct product, contradicting its minimality. Abelian factors would give a nontrivial solvable normal subgroup. Distinct minimal normal subgroups of R likewise commute and intersect trivially. Algorithmically, compute the normal closure of every element, retain the inclusion-minimal nontrivial ones, and generate their product. Repeating the normal-closure calculation inside that product gives its simple direct factors. All these operations use the supplied tables.

Every nonabelian finite simple group is generated by two elements [GK00, Theorem I and its Corollary]. A generating pair is found by enumeration. Enumerating its possible images finds all isomorphisms to another simple factor and all its automorphisms in polynomial table time. Fix a common table model and an explicit identification for each isomorphism type. The pure permutations of isomorphic factors now form an actual subgroup P of $\mathrm{Aut}(S)$. Since S is centerless,

$$J = \mathrm{Inn}(S) \rtimes P \leq \mathrm{Aut}(S), \qquad [\mathrm{Aut}(S) : J] = \prod_i |\,\mathrm{Out}(S_i)| \leq |S|.$$

The inequality follows from $|\,\mathrm{Aut}(S_i)| \leq |S_i|^2$ and $|\,\mathrm{Inn}(S_i)| = |S_i|$. No splitting of $\mathrm{Aut}(S_i) \to \mathrm{Out}(S_i)$ is required.

For clarity we specify what is retained during quotient recursion. There is a fixed table group $\widetilde{Q}$, a quotient $\pi : \widetilde{Q} \to R$ with kernel K, genuine $\widetilde{Q}$-modules $X_1, \ldots, X_s$, and a comparison group Γ acting compatibly on $\widetilde{Q}$ and these modules. A coefficient B is an invariant section of tensor products on the X_i, invariant for both $\widetilde{Q}$ and Γ, and K acts trivially on B. The X_i themselves need not descend to R. Their genuine actions are needed for the following normalization.

Restrict Γ to the preimage Γ_0 of J. Its index is at most $|S|$ and it is constructible with all coset representatives. Indeed, form the finite outer automorphism tables of the simple factors. The product of their outer-coordinate sets has size at most $|S|$; Γ acts on this set through its wreath-product restriction to S. The stabilizer of the identity tuple is Γ_0. Orbit and Schreier calculations give its full generators and representatives, without enumerating the factor-permutation group.

For $g \in \Gamma_0$ there is a unique $n_g \in S$ such that $\beta_g|_S = \mathrm{Inn}(n_g)\sigma_g$, where $\sigma_g \in P$. The elements n_g are computable by enumeration in S. Define

$$\hat{\Gamma}_0 = \{(g, q) : g \in \Gamma_0,\ \pi(q) = n_g\}, \quad (g, q)(h, r) = (gh, \beta_g(r)q). \tag{2.13}$$

Since $n_{gh} = \beta_g(n_h)n_g$, this is a group. Associativity follows from $\beta_{gh} = \beta_g\beta_h$; the inverse is $(g^{-1}, \beta_g^{-1}(q^{-1}))$. The projection onto Γ_0 is surjective with full kernel $\{(1, k) : k \in K\}$. Choose one lift for each generator of Γ_0 and add generators of this kernel to generate the entire cover. There is no need for a coherent choice for every g.

On every X_i define corrected maps

$$\alpha_i'(g, q) = \rho_i(q)^{-1}\alpha_i(g), \qquad \beta'(g, q) = \mathrm{Inn}(q)^{-1}\beta_g. \tag{2.14}$$

They are homomorphisms: for example,

$$\rho_i(q)^{-1}\alpha_i(g)\rho_i(r)^{-1}\alpha_i(h) = \rho_i(\beta_g(r)q)^{-1}\alpha_i(gh).$$

They remain compatible, preserve the actual numerator and denominator of B, and their action on S is the pure permutation σ_g. The cover is represented faithfully on the disjoint union of the old permutation domain and the set $\widetilde{Q}$, using on the latter

$$x \longmapsto q^{-1}\beta_g(x).$$

Composition gives exactly (2.13); identity on the old domain forces $g = 1$, and then identity on the new domain forces $q = 1$. Thus the added degree is $|\widetilde{Q}|$, not a Cartesian power of it.

The corrected action on $H^j(S, B)$ is exactly the original action. Indeed they differ by a paired inner action of n_g. In degree one the cocycle calculation above proves this assertion. In degree two, for a normalized cocycle f put

$$(K_x f)(y) = f(x, x^{-1}yx) - f(y, x).$$

Substitution in $\delta f = 0$ gives

$$xf(x^{-1}yx, x^{-1}zx) - f(y,z) = \delta(K_x f)(y,z).$$

Thus the discrepancy is a coboundary for arbitrary, possibly nontrivial, coefficient action. A cover-kernel element has $q \in K$; it acts literally trivially on B and on R, hence on the cohomology in question. An affine action is pulled back through the original cover projection, so that this kernel has zero affine displacement as well. A further kernel of the corrected matrices can have a nonzero displacement and is handled by Lemma 2.4.

We construct the promised tensor section rather than assert it from a cohomological dimension estimate. First remove the product S_0 of factors of order prime to p, using $H^j(S,B) \cong H^j(S/S_0, B^{S_0})$ as proved above. This product is characteristic in S, and the coefficient invariants remain a section of the same tensor products. Let t_p be the number of remaining simple factors and let Y be their permutation module. Then

$$\dim Y = t_p, \qquad p^{t_p} \le |S|. \tag{2.15}$$

For each remaining factor use its normalized free bar resolution over $\mathbb{F}_p S_i$. Tensor these resolutions over $\mathbb{F}_p$, with differential $\sum_i (-1)^{e_1+\cdots+e_{i-1}}\delta_i$ on multidegree $(e_1, \dots, e_{t_p})$. It is a free $\mathbb{F}_p(S/S_0)$-resolution of the trivial module: tensor products of the free group-algebra factors are free over the product group, and exactness follows by splitting the augmented complexes as vector-space complexes and tensoring their contractions. Applying equivariant Hom to B^{S_0} gives a cochain complex in which a degree-j term has at most j active factor indices. In degrees one and two its summands are

$$C^1 = \bigoplus_i \operatorname{Map}(S_i \setminus \{1\}, B^{S_0}),$$
$$C^2 = \bigoplus_i \operatorname{Map}((S_i \setminus \{1\})^2, B^{S_0}) \ \oplus \bigoplus_{i<k} \operatorname{Map}((S_i \setminus \{1\}) \times (S_k \setminus \{1\}), B^{S_0}).$$

The internal elements of the fixed simple-group models are fixed labels. The index i is a coordinate of Y and has its genuine factor-permutation action. The one-factor summands use one Y occurrence. The two-factor summands are obtained from the distinct-index subspace of two copies of Y by the relations $(i,u;k,v) = -(k,v;i,u)$. These are quotient relations, requiring no division by 2, and yield exactly the Koszul signs when factors are permuted. Type-specific restrictions on the internal labels are invariant because factor permutations preserve simple-group types.

If B is a section with b tensor occurrences, this gives C^j as an actual section with at most $b+j$ occurrences. The corrected compatible action makes the differential equivariant; hence its cycle numerator and boundary denominator give the same kind of section for $H^j(S,B)$. The use of C^{j+1} to impose cycle equations does not add an occurrence to the space being transported. This is the essential distinction between explicit equation size and the dimension in (2.1).

All identifications with ordinary group cochains can be computed. Between the tensor resolution and the ordinary bar resolution construct chain maps lifting the identity in degree zero. Once a map is defined through degree $j-1$, the image of the boundary of a free degree-j generator is a cycle. Solve linearly for its preimage under the other boundary and extend equivariantly. Exactness supplies that preimage. Construct the map in the other direction in the same way. The two composites are homotopic to the identity: after constructing a homotopy through degree $j-1$, solve $dH_j = (F_j - G_j) - H_{j-1}d$ on each free generator. Its right side is a cycle by the previous degree equation. This also shows that conjugating a comparison map by any compatible action changes it only by homotopy, so the resulting cohomology identifications are equivariant. Only

degrees through three are needed here. For example the comparison in degree two sends a mixed generator to $[u_i|v_k]-[v_k|u_i]$, with the same sign convention as the quotient above.

For fixed degree at most three, the free ranks are bounded by a fixed multiple of $(\sum_i |S_i|)^3 \le |S|^3$; as vector spaces the resolutions have dimension at most a fixed polynomial in $|S|$. Hence these maps, their induced cohomology maps, all corrected actions, and all section coordinates are obtained by polynomial-size linear algebra.

Finally solve every coset of Γ_0 in Γ, not a selected one. In a coset $g\Gamma_0$, normalize the target by g^{-1} and use the cover calculation. Project the complete answer to Γ_0. If t is one successful witness and t_i are the others, the branch source stabilizer together with all $t^{-1}t_i$ generates the full source stabilizer in Γ. Every such generator stabilizes the source, and every transporter element belongs to one of the branch cosets, which proves completeness. Return this reunited original group before continuing with any other cohomology factor. The cover and its corrected module actions are used only for this terminal calculation; subsequent quotient problems retain the original $\widetilde{Q}$ and its original module actions.

2.7 Closure of the two low-degree constructions

Fix in advance a coefficient-independent characteristic quotient chain

$$Q = R_0 \longrightarrow R_1 \longrightarrow \cdots \longrightarrow R_t = 1, \qquad R_{i+1} = R_i/S_i. \tag{2.16}$$

If R_i has nontrivial solvable radical, choose $S_i = A(R_i)$ as above; otherwise choose its socle. The chain length satisfies $t \le \lfloor \log_2 |Q| \rfloor$. Every operation below uses the remaining suffix of this chain, including one that begins with a new coefficient. Since the kernels are characteristic, every supplied comparison action preserves them.

Consider first $H^1(R, B)$, with the retained data $(\widetilde{Q} \twoheadrightarrow R; X_1, \dots, X_s; B)$ described above, where B has at most two tensor occurrences. If the next elementary S_i has order prime to p, replace B by its invariants and pass to the next quotient. If S_i is elementary of characteristic p, use its augmentation filtration on B. Each grade D is S_i-trivial. The positive restriction term is a section of $D \otimes S_i^*$, and is terminal: apply (2.1). Conjugation on S_i is a genuine $\widetilde{Q}$-module, so at most one occurrence is added. The inflation term continues as $H^1(R/S_i, D)$; it retains the same X_i, and the enlarged quotient kernel acts trivially on D. If S_i is semisimple, the restriction term $H^1(S_i, B)$ is terminal and has at most three occurrences by the preceding construction. The inflation term is $H^1(R/S_i, B^{S_i})$, which again retains the same genuine modules. In particular, positive normal-subgroup cohomology from this H^1 construction is never reused as a recursively continuing coefficient. Each step reduces quotient order, after a coefficient filtration whose total dimension is conserved. This proves closure and termination for all finite quotients, not just solvable ones.

For $H^2(R, A)$, the coefficient is an actual R-module of dimension at most a. The coprime and elementary steps are those in (2.12). Their inflation terms remain degree-two problems on actual modules. Their middle terms enter the proved degree-one construction with the two genuine modules D and S_i. Their last terms are terminal sections with at most three occurrences. For a semisimple S_i, Lemma 2.6 has factors in

$$H^2(R/S_i, A^{S_i}), \quad H^1(R/S_i, H^1(S_i, A)), \quad H^2(S_i, A).$$

The last is terminal, and the first continues on an actual module of dimension at most a. The middle uses the following bound.

Lemma 2.7 (A small actual middle coefficient)**.** *If S is a direct product of nonabelian finite simple groups and A is an $\mathbb{F}_p S$-module, then* $\dim H^1(S, A) \le 2 \dim A$.

Proof. Scalar extension commutes with the kernels and images of the finite cochain matrices, so extend to an algebraic closure. An irreducible module for a direct product is an outer tensor product of irreducible modules for its factors. To recall the argument, the socle of the restriction to one direct factor is nonzero and is stable under the other factors, so irreducibility makes that restriction semisimple. Its isotypic components are also stable, hence there is just one. Schur's lemma writes it as the tensor product of that simple module with its multiplicity space, on which the remaining factors act irreducibly. Induction proves the assertion.

The tensor product of the factor resolutions used above now gives

$$H^1(S, \bigotimes_i A_i) = \bigoplus_i H^1(S_i, A_i) \otimes \bigotimes_{j \ne i} A_j^{S_j}.$$

This follows directly by splitting each cochain complex into its cohomology and contractible summands over the field, and taking degree one of their tensor product. A nontrivial irreducible factor has zero invariants. Thus if at least two factors act nontrivially, the displayed space is zero. If none does, perfectness of the simple groups gives $H^1(S, \mathbb{F}_p) = \mathrm{Hom}(S, \mathbb{F}_p) = 0$. Otherwise precisely one factor can contribute. A crossed homomorphism on a two-generated group is determined by its values on the two generators, since $z(xy) = z(x) + xz(y)$ and $z(x^{-1}) = -x^{-1}z(x)$. Hence its dimension is at most twice the coefficient dimension. This proves the bound for irreducible coefficients. Exactness of (2.6) bounds the dimension for any extension of coefficients by the sum of the two dimensions. A composition series proves the stated bound for A, and scalar extension preserves it. □

The space $H^1(S_i, A)$ is therefore made into a *new actual* R/S_i-module of dimension at most $2a$. Its matrices are computed on cohomology; the paired-inner identity proves that they factor through the quotient. It is not asserted to have a linear action lifting to A. The ensuing problem has degree one, so the preceding construction never makes another positive-cohomology coefficient enlargement along this branch. This is the only place where such an enlargement is used.

In all cases intersect the filtrations with the actual supplied numerator and denominator. Solve restriction first, the degree-two middle factor second, and the inflation factor last, retaining a full coset at every step as in Subsection 2.2. If a branch section is zero, no condition is imposed. Arbitrary affine cocycles are carried through these exact linear maps and centered on each current fiber. Thus the constructions just proved apply to every invariant affine section required by earlier steps, not only to whole cohomology modules or to linear point actions.

2.8 Uniform time, complete outputs, and the single quotient recursion

Fix an extension step with $|G| = m = p^a n$ and $|Q| = n$. Every terminal tensor calculation has underlying vector-space dimension

$$d \le 2a + \log_p n, \qquad p^d \le p^{2a} n \le m^2. \tag{2.17}$$

Here is a verification for each source of dimension. A module layer uses $U \oplus V$ of dimension at most a. An elementary middle term adds a conjugation module of dimension c while passing through a quotient of index p^c. A subsequent degree-one restriction adds at most one further elementary module, or a simple-factor permutation module of dimension at most the base-p logarithm of the corresponding factor order by (2.15). These orders are distinct factors in (2.16), so their logarithms

sum to at most $\log_p n$. A semisimple middle term instead creates one actual coefficient of dimension at most $2a$; it spends the remaining cohomological degree on a degree-one problem, whose terminal dimension is bounded in the same way. A continuing coefficient filtration or invariant subspace does not increase the dimension of the specified underlying modules. Repeated occurrences of a space in a tensor product do not create new independent matrix groups. A fixed-factor realization change in Theorem 15.4, if used, multiplies this bound only by an absolute constant.

We next bound the *number* of calls; a logarithmic recursion depth alone would not suffice. In the fixed chain (2.16), put $\eta_i = \dim S_i$ when S_i is elementary of characteristic p, $\eta_i = 2$ when it is semisimple, and $\eta_i = 0$ at a coprime elementary step. Let v be the sum of the p-elementary dimensions and let s be the number of semisimple steps. Then

$$v \le \log_p n, \qquad s \le t, \qquad \sum_i \eta_i = v + 2s.$$

Let X_i be the sum of coefficient dimensions of all degree-two problems at R_i, and let Y_i be the corresponding sum for degree-one problems, including the separate roots from the module layers. Coefficient filtrations conserve the sum of dimensions. Degree-one positive restriction factors terminate. Degree-two inflation preserves at most the coefficient dimension, while its middle factor contributes at most η_i times that dimension. Therefore

$$X_0 = a, \qquad X_{i+1} \le X_i, \qquad Y_{i+1} \le Y_i + \eta_i X_i. \tag{2.18}$$

No degree-one problem creates a new degree-two problem, and every newly created problem uses the remaining suffix of the chain. It follows that

$$\sum_{i=0}^{t} (X_i + Y_i) \le (t+1)[a + Y_0 + a(v+2s)]. \tag{2.19}$$

Equivalently, $Y_i + X_i(1 + \sum_{j \ge i} \eta_j)$ is nonincreasing; substituting (2.18) proves this assertion term by term.

For the serial module construction write its successive dimensions as $a = a_0 > a_1 > \cdots > a_\ell = 0$ and put $r_j = a_j - a_{j+1}$. Its degree-one coefficient dimensions sum to

$$Y_0 = \sum_j a_{j+1} r_j = \frac{a^2 - \sum_j r_j^2}{2} \le \frac{a^2}{2}.$$

Thus the right side of (2.19) is $O((1 + \log m)^3)$. Every nonzero recursive coefficient has positive integer dimension, and a coefficient of dimension b has at most b nonzero filtration grades. This bounds recursive problems, individual grade calculations, and terminal restriction calculations by a fixed multiple of (2.19). Cumulative-coefficient maps do not create further recursive roots. The module calculations are actually performed serially under their current full groups; combining their initial masses is only an upper bound on their total work.

Put $b_* = \max(1, a^2, av, 2a)$. Every coefficient space has dimension at most b_*. Degree-two continuations never exceed a; degree-one continuations only take subquotients or invariants after their initial root. Cochains, the two nested cochain systems in Lemma 2.6, and the factor-resolution comparisons through degree three have polynomially many coordinates in $b_* n$. For example $O(b_* n^4)$ bounds all ordinary spaces and resolution vector spaces used for one comparison. There are polynomially many cumulative-coefficient systems, since at most b_* are used per problem. All maps are

stored in fixed coordinates and composed as matrices, rather than retained as a recursively growing sequence of new coefficient spaces. Thus their dense elimination and all changes of quotient coordinates have one fixed polynomial bound.

Every semisimple terminal has at most $|S_i| \le n$ coset records. Complete all of them, including failed ones, and reunite their answers before returning to a subsequent grade. They therefore multiply the *total* number of terminal calls by at most n, rather than multiplying an independent branching factor at each level of the quotient recursion. Each normalization cover adds just one $\widetilde{Q}$ domain and is projected away on return.

The original compatible-pair groups are represented on quotient table elements and the vector sets of the original module and its successive subquotients; their total degree is at most $n + O(ap^a)$. A new coefficient of dimension at most $2a$, when needed, adds at most p^{2a} points. The local normalization cover adds at most n points. No cochain space, tensor-product coefficient space, or translation space is enumerated as a permutation domain. Kernels use Lemma 2.4. Compress generators after returns and store their expressions as shared straight-line programs in earlier generators, including the accompanying matrices and cochains. A polynomial number of calls with polynomial-size compressed outputs then gives polynomial total expression size and polynomial evaluation time. This also accounts for all actual lifts, rather than only the number of abstract subgroup generators.

By (2.1) and (2.17), each terminal call has cost $m^{O(1)}$ with an absolute exponent. The bounds just proved on total coefficient mass, explicit coordinates, coset records, permutation degrees, and words give one absolute constant C such that the entire extension step costs m^C. This exponent is chosen at the root of the extension step and does not grow with cohomology depth or with p.

We can now prove Proposition 2.2. Reject unequal orders, and handle the trivial group directly. Compute the two solvable radicals. If exactly one is trivial reject; if both are trivial use the complete radical-free algorithm. Otherwise compute $A(G), A(H)$, rejecting unequal selected primes or dimensions. Make *one* recursive call on the quotient tables and retain its complete isomorphism coset. Use Subsection 2.3 to match the conjugation modules under that coset, and then Subsections 2.4–2.7 to match their extension classes under the returned full compatible-pair group. Lemma 2.3 gives the witness and entire automorphism group. Every isomorphism must preserve the characteristic subgroups, so these alternatives are exhaustive. No module or cohomology operation calls the group-isomorphism procedure.

Since $|A(G)| \ge 2$, the unique recursive quotient has order $n \le m/2$. Absorb the fixed radical-free exponent into C. The recurrence is

$$T(m) \le T(n) + O(m^C), \qquad T(N) \le O\left(\sum_{j \ge 0} (N/2^j)^C\right) = N^{O(1)}.$$

All intermediate answers were complete cosets and all projection kernels were restored, so the final generators describe the entire $\mathrm{Aut}(G)$. This proves the proposition.

3 Constructive representation theory at the original dimension

Throughout this section $k = \mathbb{F}_p$, and d is a fixed upper bound for the sum of the dimensions of the original faithful representation spaces. Additional coefficient spaces are described by bases and matrices; their size is included in the explicit input length L. The bound $p^{O(d)} \operatorname{poly}(L, \log p)$ never permits enumeration of a coefficient space of dimension d^2. A fixed number of copies, including dual spaces, changes only the absolute constant in this bound. All group elements are stored as

straight-line words in the original generators, together with their actual actions. In particular, computing an image does not discard its kernel.

3.1 The permitted group operations

We use deterministic constructive membership, point stabilizers, and point orbits for permutation groups, and the quotient-group algorithms of Kantor–Luks [KL90, Sections 3–4]. The latter supply orders, presentations, normal closures, normal-subgroup centralizers, composition series, socles, solvable radicals and prime cores in quotients of supplied permutation groups. Their running time is polynomial in the permutation degree, and their outputs include straight-line words. No unrestricted intersection or set-stabilizer operation is assumed here. A matrix group on a space of dimension at most d has a faithful permutation action on at most p^d vectors. A specified additional action can be put beside this action on a disjoint union; pointwise stabilization of the additional part then computes its kernel while preserving the original words.

For clarity, a homomorphism to an explicitly represented permutation image also permits full kernel computation from a presentation of that image. Lift its generator words and its defining relations. Adjoin the relations expressing each original generator in terms of the lifted image generators, and take their normal closure in the original group. The resulting quotient both maps onto the image and is a quotient of its presentation, so its kernel is exactly the required kernel. This observation is useful when the image has a convenient small domain but the original representation is to be kept unchanged.

Lemma 3.1 (Kernels of additive homomorphisms)**.** *Let H be a supplied finite group with constructive permutation membership, and let $\eta : H \to B_+$ be a homomorphism to an explicitly given $\mathbb{F}_p$-vector space. From values on generators one can compute* $\operatorname{im}\eta$*, word lifts of a basis, and generators of* $\ker\eta$*, in time polynomial in the permutation degree and the explicit input length. There is no enumeration of B or of* $\operatorname{im}\eta$*.*

Proof. The span of the generator values is the image. Gaussian elimination gives a basis $b_1, \dots, b_t$ and elements h_i with $\eta(h_i) = b_i$. Correct every original generator by a product of powers of the h_i so that its image becomes zero. Take the normal closure of these corrected generators, the h_i^p, and the commutators $[h_i, h_j]$. Call the result K. Then $K \le \ker\eta$, while H/K is generated by t commuting elements of order dividing p. It maps onto the t-dimensional image, so it has exactly p^t elements and $K = \ker\eta$. Normal closure and generator compression have the stated permutation complexity. □

3.2 Composition flags and primitive blocks

Lemma 3.2 (Composition flags)**.** *A supplied group $H \le \mathrm{GL}(V)$ has a constructible invariant composition flag in deterministic time $p^{O(\dim V)}\operatorname{poly}(L, \log p)$. The kernel of its action on the direct sum of the successive quotients is exactly $O_p(H)$. The same procedure refines any supplied invariant flag.*

Proof. For every nonzero $v \in V$, close kv under the supplied generators by repeated linear-span operations. This computes kHv in polynomial matrix work. If a proper nonzero submodule exists, choosing v inside it finds a proper cyclic submodule. Exhausting all vectors therefore either certifies irreducibility or finds a proper submodule. Recurse on that submodule and the quotient. There are at most $2\dim V - 1$ nodes, and the sum of the dimensions at any split is unchanged. Bases, inclusions and quotient maps are retained throughout.

The kernel R on all composition factors is block unitriangular, hence a normal p-subgroup. A finite p-group acting on a nonzero characteristic-p vector space has a nonzero fixed vector: induct on the group order using a central subgroup of order p and the kernel of its operator $z-1$. Thus a normal p-subgroup of H has a nonzero fixed space on every irreducible H-module, and normality makes that space H-invariant. It acts trivially on each factor. This proves $R = O_p(H)$. Its generators are obtained as the pointwise kernel on the graded vector action, retaining in parallel the original vector action. Both vector sets have at most $p^{\dim V}$ points. □

Lemma 3.3 (Primitive direct-sum decompositions)**.** *Let $A \le \mathrm{GL}(U)$ be irreducible. In time $p^{O(\dim U)}\,\mathrm{poly}(L, \log p)$ one can either certify linear primitivity or construct a nontrivial direct-sum decomposition permuted transitively by A. Iteration gives primitive block stabilizers and actual block transversal words.*

Proof. For each $u \neq 0$, compute the vector orbit $\Omega = u^A$. It spans U. Choose a basis $B \subseteq \Omega$ and form a graph on Ω by joining the members of each fundamental circuit of $x \in \Omega \setminus B$. The latter circuit consists of x and the basis vectors with nonzero coefficient in its unique expansion. Distinct connected components use disjoint subsets of B in these expansions, so their spans form a direct sum. Conversely every minimal linear dependency lies in one such component: projection to the direct summands would otherwise give a proper subdependency. The components are therefore intrinsic to the represented matroid, and A permutes them. It does so transitively because Ω is a single orbit.

If $U = \bigoplus_{i=1}^{t} U_i$ is a genuine imprimitivity decomposition, a seed $u \in U_1 \setminus 0$ has orbit contained in $\bigcup_i U_i$. Every circuit of that orbit lies in a single U_i, again by projection. This seed therefore gives a nontrivial direct-sum decomposition. Hence exhausting all seeds is a valid test. A block stabilizer is obtained from the action on the at most $\dim U$ blocks. Its action on its block is irreducible: inducing a proper invariant subspace over the block orbit would contradict irreducibility of U. Refine inside that stabilizer and induce by actual transversal words. The refinement tree has linearly many nodes; each orbit, circuit graph and vector search has size at most $p^{\dim U}$. Stabilizer-chain compression prevents the generator lists from accumulating exponentially. □

Apply these lemmas to the graded representation of $G = H/O_p(H)$. Its total dimension is unchanged. We obtain a G-permuted family $W_1, \dots, W_t$ with primitive irreducible stabilizer images $P_i \le \mathrm{GL}(W_i)$. Write $d_i = \dim W_i$ and let $N \trianglelefteq G$ be the kernel of the block permutation action. Thus $\sum_i d_i \le d$, and restriction embeds N in $\prod_i P_i$. All choices are made for the current supplied group. If an incoming coset representative is known, it transports these choices to the other side; no independent choice of target decompositions is required.

3.3 Large primitive components and their actual representations

The classification input is the following precise structural consequence of Halasi–Maróti–Pyber–Qiao [HMPQ19, Theorems 3.1–3.2]. There is an absolute c_0 such that an irreducible primitive $P \le \mathrm{GL}_r(\mathbb{F}_p)$ of order greater than $p^{c_0 r}$ has a normal quasisimple subgroup S with $[P : S] \le p^{5r}$. On an extension-field tensor decomposition, S acts naturally on one tensor factor and trivially on the other. The natural factor is a defining characteristic classical module or a deleted alternating module. We use this existence theorem only. The construction of the actual subgroup and its coordinates is given next. For a deleted alternating module the alternating degree is at most $r+2$; no stronger inequality is needed.

Lemma 3.4 (Extracting components)**.** *The quasisimple components of a supplied $A \le \mathrm{GL}_r(\mathbb{F}_p)$ can be constructed, with centers, simple quotients and original words, in $p^{O(r)}\,\mathrm{poly}(L, \log p)$ time.*

Proof. Compute the Fitting subgroup F as the product of the prime cores, and put $B = C_A(F)$ and $Z = Z(F)$. Since $B \trianglelefteq A$, its characteristic nilpotent subgroup $F(B)$ is normal nilpotent in A. Hence $F(B) \leq F \cap B = Z$, and equality follows because Z is central in B. A nilpotent normal subgroup of B/Z lifts to a nilpotent normal subgroup of B, since its kernel is central. Therefore $F(B/Z) = 1$. Its solvable radical is trivial as well: a nontrivial solvable normal subgroup has a nontrivial characteristic Fitting subgroup.

Compute $\mathrm{Soc}(B/Z)$, a direct product of nonabelian simple groups. Separate its factors by repeatedly finding a minimal normal subgroup inside this direct product and its centralizer there. For each simple factor S_j, take its full preimage A_j in B, and put $Q_j = A_j'$. The equalities $A_j = A_j'Z$ and $A_j' = [A_j'Z, A_j'Z] = [A_j', A_j']$ show that Q_j is quasisimple. Distinct Q_j commute: their commutators lie in Z, and a central commutator map from a perfect group is trivial. Conversely a component of A centralizes $F(A)$, and its image in the radical-free group B/Z is a simple socle factor. Thus these are exactly the components. All group operations invoked are normal-subgroup operations in the quotients specified above.

There is no integer-factorization cost depending exponentially on r^2. Every prime divisor of $|A|$ divides $p\prod_{j=1}^{r}(p^j - 1)$. Trial division of these r integers, each at most p^r, costs $p^{O(r)}$. The vector permutation domain has p^r points, and the remaining work is polynomial in its size and the input. □

Lemma 3.5 (Homogeneous coordinates)**.** *Suppose $P \leq \mathrm{GL}(W)$ is irreducible and primitive, and $S \trianglelefteq P$. Then $W{\downarrow_S}$ is semisimple and homogeneous. If S is quasisimple, one can construct a faithful simple constituent X, its actual endomorphism field $K = \mathrm{End}_S(X)$, and an isomorphism*

$$X \otimes_K M \longrightarrow W, \qquad M = \mathrm{Hom}_S(X, W), \qquad x \otimes f \longmapsto f(x), \tag{3.5.1}$$

in time $p^{O(\dim W)}\,\mathrm{poly}(L, \log p)$.

Proof. Let $A = k[S]$ be the represented image algebra, and $J = \mathrm{rad}\, A$. Conjugation by P preserves J, so JW is a P-submodule. Irreducibility and nilpotence imply $JW = 0$; faithfulness of the represented algebra gives $J = 0$. Its isotypic subspaces are permuted by P. Primitivity therefore forces just one of them. Find a simple submodule X by cyclic-span descent as in Lemma 3.2. Linear commutant equations construct $\mathrm{End}_S(X)$; it is a finite division ring and hence a field. Its elements are actual operators, so field arithmetic, bases over K, and the Hom space are effective. Semisimplicity and homogeneity prove that the displayed evaluation map is an isomorphism; this is also verified by rank. Every isomorphic summand has the same group kernel, so faithfulness on W implies faithfulness on X. A field search is allowed here because $|K| \leq p^{\dim X}$. No search through the whole image algebra is made. Finally the center of S acts faithfully by K-scalars, so $|Z(S)| < p^{\dim W}$. □

Increase c_0 to exceed 12. If $|P_i| \leq p^{c_0 d_i}$, keep its whole image and call the block bounded. Otherwise extract its components. The component guaranteed by the structural theorem has simple central quotient of order greater than $p^{(c_0-6)d_i} > p^{5d_i}$. Every other component commutes with it and has simple central quotient appearing in P_i/S, of order at most p^{5d_i}. Thus the unique component above this bound is intrinsically identifiable; denote it S_i. It is normal in P_i and transported to the corresponding component on a conjugate block. Apply Lemma 3.5.

In the classical case, the field recovered on the irreducible $\mathbb{F}_p$ constituent is its actual defining matrix field; extension scalars in the structural tensor decomposition are absorbed into M. The constituent is tested against the candidates $\mathrm{SL}_a(K)$, $\mathrm{Sp}_a(K, b)$, $\mathrm{SU}_a(K/K_0, h)$ and $\Omega_a(K, Q)$. Invariant bilinear and Hermitian forms are obtained by linear equations on the supplied generators.

Irreducibility makes a nonzero invariant form nondegenerate. In characteristic two, quadratic forms are found by the equations $Q(gx) = Q(x)$ on the coefficients of the polynomial Q; they are not reconstructed from their polar form. Polarization is injective on invariant quadratics on a nontrivial irreducible natural module: a zero-polar quadratic is the square of a linear functional, which would be invariant. Odd-dimensional quadratic natural modules in characteristic two have invariant polar radical and hence cannot be these irreducible constituents.

For reference, candidate group generators are available within the claimed dimension bound without enumerating a classical group. Elementary matrices generate SL. For formed groups use a nondegenerate initial one-space or two-plane from a standard formed basis. Enumerate its possible image frames, at most $|K|^{2a}$, test their restricted form, and extend each successful frame to a full isometry by formed-space elimination. Generate its pointwise stabilizer recursively on the orthogonal complement, once rather than once for every frame. The frame transversals and that stabilizer generate the full isometry group. Witt extension, formed-space elimination, and the stable derived-group identities used here are the classical finite-space results in Taylor [Tay92, Chapters 7–11]; each involves the supplied nondegenerate form over K. Field norm equations may be solved by enumerating K. Deriving the full orthogonal or unitary group produces Ω or SU in the stable ranks used below. Inclusion of candidate generators in the actual component, followed by an exact order comparison, certifies equality. Thus classification is used to guarantee a successful check, not to justify an unchecked recognition decision.

3.4 A genuine alternating permutation realization

Lemma 3.6 (Automorphisms of the alternating action)**.** *For $m \geq 8$, every automorphism of A_m is conjugation by a unique element of S_m.*

Proof. An element of order three is a product of r disjoint three-cycles. Its centralizer in A_m has order $3^r r!(m-3r)!/2$: when $r = 1$ there are at least five unused letters, and when $r \geq 2$ interchanging two three-cycles is an odd centralizer element. Among these orders the case $r = 1$ is uniquely largest. For $r = 2$ the ratio to its centralizer is $(m-3)(m-4)(m-5)/6 > 1$. For $r \geq 3$ the ratio is at least $(3r-3)!/(3^{r-1}r!) > 1$, starting at $r = 3$ and increasing by a factor greater than one. Hence three-cycles form a characteristic set.

Inverse pairs of three-cycles identify the three-element subsets of $[m]$. Two distinct such subsets meet in two letters exactly when some generators of their cyclic subgroups have product of order two. Indeed disjoint supports give order three, supports meeting in one letter give order five, and on supports $123, 124$ the products are checked directly. Every group automorphism thus induces an automorphism of the graph on triples adjacent when they meet in two letters.

Its maximal cliques have two kinds: all triples through one pair, or all triples inside one four-element set. To verify this, start with abc, abd. A triple adjacent to both either contains ab or is acd or bcd. If the latter occurs, no triple abe with e outside $abcd$ can occur. The pair cliques have size $m-2 > 4$, so they are intrinsically recognized. Two pair cliques intersect if and only if their pairs share a letter. The resulting graph on pairs is $L(K_m)$. Its maximal cliques are the letter stars and the triangles; the stars are distinguished by size $m-1 > 3$. We have therefore recovered an induced permutation of letters.

Compose with its inverse conjugation. The remaining automorphism fixes every cyclic three-subgroup, hence sends each three-cycle to itself or its inverse. On two adjacent supports choose generators whose product has order two; inverting exactly one changes that order to three. Thus the two signs agree. The triple adjacency graph is connected, so all signs agree. They cannot all be negative: with $a = (123)$ and $b = (124)$, the element ab^{-1} has support $\{1, 3, 4\}$, whereas $a^{-1}b$

has support $\{2,3,4\}$, contradicting the already fixed supports. All three-cycles are therefore fixed and they generate A_m. Finally a permutation centralizing A_m fixes every three-element support and consequently every letter, proving uniqueness. □

Lemma 3.7 (Alternating realization)**.** *If the dominant component on a primitive block W is alternating, its letter action and a permutation-module realization of all of W can be constructed in $p^{O(\dim W)}\operatorname{poly}(L,\log p)$ time. The realization extends to the entire actual block stabilizer, has W as a specified subquotient, and has dimension at most $4\dim W/3$.*

Proof. Enlarge the fixed bounded threshold so that the alternating degree m is at least 8. Set $P = k^m$, $P_0 = \{x : \sum x_i = 0\}$, and

$$U_m = P_0/(P_0 \cap k\mathbf{1}).$$

Its dimension is $m-1$ if $p \nmid m$ and $m-2$ otherwise. Its dot product is nondegenerate (alternating when $p = 2$), and the deleted module is absolutely irreducible. Thus Lemma 3.5 gives $W = X \otimes_k M$ with $X \cong U_m$.

We explicitly recover the letters from X. Try $m = \dim X + 1$ and $m = \dim X + 2$, retaining only the characteristic-compatible cases with $|S| = m!/2$. Compute its invariant form and enumerate projective-vector orbits. A successful orbit has size $\binom{m}{2}$ and corresponds to the lines $k(e_i - e_j)$ in U_m. These remain distinct after quotienting by $k\mathbf{1}$: an equality would equate a vector on at most four coordinates to a constant vector, impossible when $m \geq 8$ unless the constant is zero. Distinct such lines have nonzero pairing precisely when their unordered pairs meet. Their adjacency graph is therefore $L(K_m)$, in every characteristic.

The stars of K_m can be recovered directly from that graph. For adjacent vertices ij, ik, their common neighbors consist of the clique of the other $m-3$ edges through i and the isolated vertex jk. The large component together with ij, ik is the star of i. Collect all stars and check that there are exactly m, that each has $m-1$ vertices, that every vertex lies in two stars, and that graph adjacency agrees with intersection of the resulting letter pairs. Check that the supplied permutations preserve these stars. This gives action matrices for a map $S \to \operatorname{Sym}(m)$; require even generators and image order $m!/2$. The map is faithful by the checked order of S. Finally form the associated deleted permutation matrices and solve the intertwiner equations with X. Retain the candidate only if the intertwiner has full rank. A genuine model passes all these tests, and each search has at most $p^{O(\dim X)}$ objects.

For h in the full block stabilizer, conjugation induces an automorphism of A_m. By Lemma 3.6, it is induced by a unique $\pi_h \in S_m$. It can be found without enumerating S_m: for each possible image of one letter, take the orbit of the corresponding pair under the two letter actions on $[m] \times [m]$. It is the graph of an intertwining bijection exactly when it has m points and both projections are bijective. Test the resulting permutation on the component generators. Uniqueness proves $\pi_{hh'} = \pi_h \pi_{h'}$. Removing its deleted matrix from h leaves a map centralizing A_m, hence a unique matrix c_h on M. Consequently

$$h = \operatorname{del}(\pi_h) \otimes c_h, \qquad c_{hh'} = c_h c_{h'}.$$

The genuine representation $\tilde{h} = \operatorname{perm}(\pi_h) \otimes c_h$ on $k^m \otimes M$ preserves $P_0 \otimes M$ and $(P_0 \cap k\mathbf{1}) \otimes M$; their quotient is exactly W. Since $c_s = 1$ for $s \in S$, its multiplicity image has order at most $[P_W : S] \leq p^{5\dim W}$. The dimension is at most $m/(m-2)\dim W \leq 4\dim W/3$.

To induce over the block orbit, choose actual transversals t_i from a representative block. If $gW_i = W_j$, act between the corresponding copies by the constructed representation of $t_j^{-1} g t_i$. The intermediate transversals cancel in a product, proving the group law and the equivariance of the subquotient maps. The dimension bounds add over disjoint block orbits. □

3.5 The actual diagonal components

For a bounded block put $S_i = 1$ and $B_i = P_i$; otherwise put $B_i = P_i/S_i$. Then

$$|B_i| \le p^{c_0 d_i}, \qquad \prod_i |B_i| \le p^{c_0 d}.$$

Enumerate the individual quotients using coset breadth-first search and membership in S_i. Their regular actions have total degree at most $dp^{c_0 d}$. Restriction to blocks followed by these quotient actions specifies a homomorphism $\rho : N \to \prod_i B_i$. Compute its full kernel M on the disjoint union of those quotient domains. The conjugation covariance of the S_i makes $M \trianglelefteq G$.

Lemma 3.8 (Central product and diagonal links)**.** *The group $C = M'$ is a normal central product of quasisimple groups, $[N : C] \le p^{(c_0+1)d}$, and its actual diagonal links across blocks are constructible within $p^{O(d)} \operatorname{poly}(L, \log p)$ time.*

Proof. The projection $M_i \le S_i$ is normal in P_i. A normal subgroup of a quasisimple group is central or is the whole group: a subgroup surjecting onto the simple central quotient satisfies $S_i = M_i Z(S_i)$, whence perfectness gives $S_i \le M_i$. Let $Z_0 = M \cap \prod_i Z(S_i)$. It is central in M, and after removing central coordinates, M/Z_0 is subdirect in a product of nonabelian simple groups.

Such a subdirect group is a product of full diagonals. Indeed project onto all but the last factor and induct. The kernel in the last factor is normal, hence either the whole last factor, which splits off, or trivial. In the latter case the last coordinate is a surjective homomorphism from the preceding product of diagonal simple groups. Its factor images are commuting normal subgroups of a simple group, so precisely one is nontrivial and maps isomorphically. This gives the next diagonal link. In particular M/Z_0 is perfect, $M = M'Z_0$, and $C = M'$ is perfect. Hence

$$[M : C] \le |Z_0| \le \prod_i |Z(S_i)| \le p^d,$$

which proves the index bound.

For each nontrivial local simple image, compute $J_i = \ker(C \to S_i/Z(S_i))$. In a classical factor this is the pointwise kernel on the natural projective space, of degree at most p^{d_i}; in an alternating factor use Lemma 3.7. Two indices belong to the same diagonal factor exactly when $J_i = J_j$, tested by mutual generator membership. For a diagonal class I, kill all actions belonging to the other classes and derive the resulting subgroup; call it Q_I. Its central quotient is the one simple factor indexed by I, so it is quasisimple. Different Q_I commute, because their commutators are central and they are perfect. Their product maps onto $C/Z(C)$, and perfectness of C makes the residual central quotient trivial. Thus $C = \prod_I Q_I$. Every kernel is a pointwise kernel of a displayed action of degree $p^{O(d)}$, and every generator has an original word. □

3.6 A hereditary bound with a fixed threshold

A simple section means a simple quotient of a subgroup, not merely a composition factor of the original group. We use the following bound of Guralnick–Maróti–Pyber [GMP17, Proposition 6.1]: for $A \le \mathrm{Sym}(t)$ the product of the orders of its composition factors other than alternating groups of degree greater than 4 is at most $24^{(t-1)/3}$. In particular every nonalternating simple section of every subgroup of $\mathrm{Sym}(t)$ has order less than 3^t. This last application is to the subgroup in which the section occurs.

Fix once and for all

$$T = p^{C_* d}, \tag{3.1}$$

where C_* is a sufficiently large absolute constant. It is not changed when passing to a subgroup or a lower-dimensional subquotient. Let D be the product of the classical components Q_I of C for which $|Q_I/Z(Q_I)| > T$. The selection is conjugation-invariant, so $D \trianglelefteq G$. Since a simple classical natural group of degree a over $\mathbb{F}_{p^f}$ has order less than p^{fa^2}, while $fa \le d_i \le d$, choosing C_* sufficiently large forces every selected degree and Lie rank beyond any fixed stable cutoff required later. Small accidental isomorphisms and triality can therefore be excluded at this step.

Every local classical component with simple central quotient larger than T is the image of one of the selected global components. To see this, let $N_i \trianglelefteq P_i$ be the image of the block kernel. If N_i does not contain S_i, its intersection with S_i is central and the simple quotient of S_i is a section of P_i/N_i. That group is a quotient of a subgroup of $G/N \le \mathrm{Sym}(t)$, contradicting the preceding bound when $C_* > \log_2 3$. Thus N_i contains S_i. If the image of C there were central, the same simple quotient would be a section of N/C, contradicting Lemma 3.8. Exactly one diagonal component therefore maps onto S_i and is selected. A selected perfect component cannot have a nontrivial central projection, so its other projections are either full natural occurrences or trivial. Different selected components have disjoint nontrivial block supports.

Lemma 3.9 (Hereditary complementary bounds)**.** *The following actual representations, and all subgroup images and quotients subsequently obtained from them, have no nonalternating simple section of order greater than T: the bounded primitive images; the classical primitive images with unselected component; the full alternating realizations of Lemma 3.7; the quotient N/C; and the block or component permutation actions of degree $O(d)$.*

Proof. For $B \trianglelefteq A$ and $J \le A$, the group $J \cap B$ is normal in J, and $J/(J \cap B)$ embeds in A/B. Every composition factor of J therefore comes from one of those two groups. Applying this statement to any subgroup in which a simple section occurs proves that a simple section of an extension comes from a section of its normal subgroup or its quotient. The same reasoning applies to subgroups of finite direct products, by successive coordinate projections.

A bounded image has order at most $p^{c_0 d_i} < T$. For an unselected classical block use $Z(S_i) \trianglelefteq S_i \trianglelefteq P_i$: the central kernel has only cyclic sections, the simple quotient has order at most T, and P_i/S_i has order at most $p^{5d_i} < T$. The whole classical image can also be enumerated, since $|P_i| < p^{6d_i} T \le p^{(C_*+6)d}$, but no product of all such orders is needed.

In the alternating realizations the axis permutation degree is at most $4d/3$, so its nonalternating simple sections are less than $3^{4d/3}$. The axis kernel embeds in the product of the actual multiplicity images. Each has order at most p^{5d_i}, and these exponents add over disjoint original blocks. This product thus has order at most p^{5d}. The extension observation proves the assertion for their actual correlated representation. The remaining permutation actions use the same permutation bound, and $|N/C| \le p^{(c_0+1)d}$. Increasing C_* once dominates all these constants, including at $p = 2$. Finally, a subgroup or quotient of a subgroup cannot introduce a simple section not already covered by this statement. □

Corollary 3.10 (Persistence under restriction)**.** *Let B be one of the actual complementary images in Lemma 3.9, and let Q be quasisimple with nonalternating simple central quotient of order greater than T. Every homomorphism $Q \to B'$ is trivial, where B' is any subquotient of any subgroup of B. In particular the conclusion holds for every subsequent graded action of such a Q on an old complementary space.*

Proof. A nontrivial homomorphism from Q has central kernel: a proper normal subgroup of a quasisimple group is central. Its image then has $Q/Z(Q)$ as a section, contrary to Lemma 3.9. A merely central image is also trivial because Q is perfect. The assertion concerns an actual

homomorphism. If Q is a subgroup only of a graded quotient of the current group, its action on an old nonsplit representation need not itself be defined; the argument applies after descending that action, not by choosing an unproved group section. □

Theorem 3.11 (Constructive linear decomposition)**.** *Given $H \le \mathrm{GL}(V)$ with $\dim_{\mathbb{F}_p} V \le d$, one can compute in $p^{O(d)}\,\mathrm{poly}(L, \log p)$ time an invariant composition flag, its exact kernel $O_p(H)$, primitive permuted blocks in the faithful graded action, the normal central product C of Lemma 3.8, and its selected classical subgroup D. Every nontrivial local action of D has verified stable-rank natural coordinates, with its field, multiplicity, actual diagonal links and original projection maps. The complementary representations satisfy Lemma 3.9; alternating realizations have total dimension at most $4d/3$. All original representations, kernels and subquotient maps remain available.*

Proof. All constructions and bounds have been proved above. Refinement of a previous invariant flag is done within its factors. Refinement of a permuted block family is done in its stabilizers and induced over its actual orbits. These operations use the current group's invariant data; they assert no simultaneous compatibility with an unrelated choice of flags. The sum of the original dimensions remains at most d. □

3.7 Linked natural coordinates and their normalizer

For the entire graded representation, compute the algebra $k[D]$ and its isotypic spaces. It is semisimple: on each irreducible G-summand, its radical is a G-invariant nilpotent ideal and acts trivially, exactly as in Lemma 3.5. Thus all occurrences of each simple D-module can be collected together. The evaluation coordinates are now

$$V_D = \bigoplus_j X_j \otimes_{K_j} M_j, \tag{3.2}$$

where each j denotes a distinct simple type on the whole represented space, including formal dual rows if present. Copies of one type in different original rows are not counted as different types. The row labels impose further stabilizing conditions on this ambient decomposition.

Linked natural representations are compared by enumerating the field isomorphisms between their actual finite fields, trying the natural and dual alternatives, and solving their intertwiner equations. The completeness of this finite list is the stable-rank classical automorphism theorem: inner, diagonal, field and graph automorphisms exhaust the possibilities [Ste60, Sections 3.2–3.6]. In the ranks retained here graph automorphisms give the natural/dual choice, or a formed-space isometry; triality has already been excluded. No scalar character is lost in passing from projective to actual maps. Once their projective maps agree, two representations of the same perfect component differ by a scalar homomorphism, and every such homomorphism is trivial. Every proposed intertwiner is checked on the actual generators and for full rank.

Lemma 3.12 (The group of natural–multiplicity factorizations)**.** *The actual normalizer action on the spaces in* (3.2) *has a constructible covering group which records natural, multiplicity, field and type-permutation actions. Its kernel consists of the scalar identifications $(u, v) \sim (cu, c^{-1}v)$, and it has a faithful representation of dimension $O(d)$. All its projection kernels are constructible within $p^{O(d)}\,\mathrm{poly}(L, \log p)$ time.*

Proof. On $X \otimes_K M$, conjugation by a normalizer element induces an automorphism of $\mathrm{End}_K(X) \otimes 1$. First read its induced field automorphism σ on the center and remove that action. The remaining automorphism of $\mathrm{Mat}_a(K)$ is inner. An implementing matrix u is found by the linear equations

$uA = \alpha(A)u$ on a matrix-algebra basis. A nonzero solution is invertible, since it intertwines simple natural modules. Removing $u\sigma$ from the original map leaves $1 \otimes v$, read by linear equations. Two factorizations differ exactly by (c, c^{-1}), $c \in K^\times$.

Keep all factorizations, rather than choosing a possibly nonhomomorphic section. Choose one factorization of each supplied generator and add generators of all scalar kernels. These generate the full inverse image: any other factorization differs from the chosen lift word by a kernel element. For permuted isotypic factors, use the actual type permutation and the fixed intertwiners. Multiplication composes these maps and their common field automorphisms, so it is a genuine group law.

The group acts faithfully on the direct sums of the X and M factors, together with the original representation; the latter prevents loss of any invisible original action. If their K-dimensions are $a, b \geq 1$, then $a+b \leq 2ab$. The added prime-field dimension is at most twice that of the homogeneous space. Field automorphisms are actual prime-field linear maps. The scalar, field and original projection kernels are therefore specified kernels on an $O(d)$-dimensional faithful representation. Formal dual factors use inverse transpose and add at most another fixed constant factor. □

Lemma 3.13 (Whole-isotypic multiplicity bound)**.** *Suppose the selected representation, including a fixed number of copies and duals, has dimension at most κd. Put $f_j = [K_j : k]$, $a_j = \dim_{K_j} X_j$ and $b_j = \dim_{K_j} M_j$ in its whole-isotypic decomposition. Then*

$$\left|C_{\mathrm{GL}(V_D)}(D)\right| \leq p^{\sum_j f_j b_j^2} \leq p^{\kappa^2 d/C_*}. \tag{3.13.1}$$

The complete joint multiplicity image, together with field, type, graph and scalar-form records, can consequently be enumerated in $p^{O(d)}\,\mathrm{poly}(L, \log p)$ time with all original lift words and kernels. After fixing these records, the ambient normalizer of the full linked classical product L_D is L_D times its centralizer. For an actual subgroup A_0 this gives $A_0 = L_D C_{A_0}(L_D)$ only when $L_D \leq A_0$.

Proof. Schur's lemma identifies the commuting algebra with $\prod_j \mathrm{End}_{K_j}(M_j)$, whose unit group has order at most $p^{\sum_j f_j b_j^2}$. Since every selected simple central quotient has order greater than $p^{C_* d}$,

$$f_j a_j^2 > C_* d, \qquad \sum_j f_j a_j b_j \leq \kappa d, \qquad b_j/a_j < \kappa/C_*.$$

Multiplying the last inequality by $f_j a_j b_j$ and summing proves the bound. This argument requires aggregation of all equivalent copies; using a separate centralizer estimate for each original row would not bound maps mixing equivalent rows.

Each distinct type has prime-field dimension $f_j a_j \geq \sqrt{f_j C_* d} \geq \sqrt{C_* d}$, so there are $O(\sqrt{d})$ types. Their permutation group has order at most $(O(\sqrt{d}))! = p^{O(d)}$. Products of field-automorphism orders, graph orders and scalar-form orders are also $p^{O(d)}$: the field degrees sum to $O(d)$, graph orders are bounded, and scalar records lie in multiplicative groups of the original fields. Determinant records are retained for special linear or unitary groups. In odd-characteristic orthogonal groups retain determinant and the spinor quotient; in characteristic two the required record is the Dickson homomorphism $g \mapsto \mathrm{rank}(g-1) \bmod 2$, whose kernel in stable dimension is Ω [Tay92, Chapter 11]. In characteristic two determinant alone would not supply this quotient.

For the normalizer assertion, fix type, field and graph records in the factorization group of Lemma 3.12. Its natural matrix u normalizes the derived classical component. It therefore preserves the unique invariant form up to its explicitly computed scalar multiplier. On fixing that multiplier, u belongs to the full natural isometry group; for a linear factor it already belongs to $\mathrm{GL}(X)$. Use one u per diagonal component and transport it to the linked occurrences by their checked

intertwiners. The quotient of the original element by this linked natural action centralizes the component, hence its full image algebra and the associated full classical group. This proves the ambient factorization. If that full group is contained in A_0, the centralizing remainder also lies in A_0, proving the stated equality. Without containment it is only an ambient factorization, and the actual subgroup condition must be retained.

Breadth-first enumeration of a joint image whose order is bounded above costs its order times polynomial matrix and membership work. All maps are specified on the factorization group, and computing their full kernels uses the original faithful action. The assertion does not multiply the orders of unrelated complementary representations. □

3.8 Represented finite algebras without large vector searches

Lemma 3.14 (Ordinary algebra structure)**.** *Let $E \le \mathrm{End}_k(V)$ be a unital algebra represented faithfully, with $\dim_k V \le d$. Its radical, semisimple matrix-factor coordinates, actual center fields, an ordinary algebra complement to the radical, and generators of its unit group can be constructed in $p^{O(d)}\,\mathrm{poly}(L, \log p)$ time. All operations on auxiliary spaces of dimension as large as d^2 are linear algebra. The complement is not asserted to respect a supplied involution.*

Proof. Find a composition series of the E-module V by enumerating vectors and closing their spans under an algebra basis, exactly as in Lemma 3.2. The kernel of E on the sum of its simple factors is a strictly triangular ideal, hence nilpotent; conversely the radical annihilates every simple module. Thus this kernel is exactly $J = \mathrm{rad}\, E$, computed by linear equations. The action on those factors is a faithful representation of $A = E/J$ of total dimension at most d.

For each distinct simple constituent S, solve its commutant equations to obtain its finite division field K, choose a K-basis of S, and express the action as matrices over K. The semisimple Artin–Wedderburn decomposition identifies the product of these distinct images with A, and each factor image with $\mathrm{End}_K(S)$; ranks of the resulting linear maps verify these assertions. Repeated simple types are recognized by Hom equations (a nonzero map between simple modules is an isomorphism). If the factors are $\mathrm{Mat}_{a_i}(\mathbb{F}_{p^{f_i}})$, faithfulness gives $\sum_i f_i a_i \le d$. In particular $\dim_k Z(A) = \sum_i f_i \le d$. Enumeration of a center field or of this semisimple center therefore has at most p^d elements. It can supply generators and primitive central idempotents if desired. Enumeration of the center of the original nonsemisimple algebra is not used; that center can have much larger dimension.

Here is a constructive lifting proof for an ordinary complement. Suppose a unital multiplicative section of A has been obtained modulo J^i. Choose linear lifts modulo J^{i+1} and let $f(a, b)$ be their multiplication defect in $B = J^i/J^{i+1}$. The A-bimodule action on B is well defined, $B^2 = 0$, and associativity gives the Hochschild cocycle identity

$$af(b, c) - f(ab, c) + f(a, bc) - f(a, b)c = 0.$$

Finite-dimensional semisimple algebras over finite fields are separable. For completeness their separability tensor $e = \sum_r x_r \otimes y_r$ can be constructed with

$$\sum_r x_r y_r = 1, \qquad \sum_r a x_r \otimes y_r = \sum_r x_r \otimes y_r a.$$

For a field factor use a trace-dual basis; for a matrix algebra over its center use $\sum_j E_{j1} \otimes E_{1j}$, and combine over the direct factors and field extension. Alternatively solve these displayed linear equations. If $h(a) = \sum_r x_r f(y_r, a)$, the cocycle identity and the centralization identity give $ah(b) - h(ab) + h(a)b = f(a, b)$. Subtracting h from the chosen lifts eliminates their defect. Normalization gives $h(1) = 0$. Repeating lifts a unital algebra section $A \to E$. There are at most d radical layers:

a strictly descending chain J^iV has this length, and an equality $J^iV = J^{i+1}V$ forces $J^iV = 0$ by nilpotence.

Finally $E^\times \to A^\times$ is surjective using the ordinary complement. The matrix-factor unit groups are generated by elementary matrices and diagonal field units. Their lifted generators, together with $1 + J$, generate $E^\times$. For $1 + J$, choose bases of J^i/J^{i+1}, lift them, and use the elements $1 + x$; their images generate each elementary abelian quotient and hence the whole group by induction. Every search was confined to an original vector set or a center field of degree at most d. The remaining spaces and systems have polynomial explicit size. □

Proposition 3.15 (Radicals of auxiliary image algebras)**.** *Let $B \leq \mathrm{Mat}_m(\mathbb{F}_p)$ be supplied by a matrix basis; m need not be bounded by the original dimension d. Its Jacobson radical is computable deterministically in $\mathrm{poly}(m, \dim B, \log p)$ bit operations. If B is initially supplied by generators, a basis is first obtained by matrix-span closure in the same polynomial bound.*

Proof. Apply Cohen–Ivanyos–Wales [CIW97, Section 4, Algorithm 1, and Section 4.1]. Their radical algorithm uses polynomially many characteristic-polynomial coefficient evaluations, linear systems and inverse Frobenius operations. Over $\mathbb{F}_p$, Frobenius and its inverse are the identity, so its semilinear systems are ordinary linear systems. Computing characteristic polynomials and solving these systems has polynomial bit complexity in the displayed parameters. The number of successive coefficient stages is at most $1 + \lfloor \log_p m \rfloor$. Although the coefficients are interpreted in their proof as traces on exterior powers, the algorithm obtains them from characteristic polynomials and never constructs the exterior-power spaces. This is a polynomial coefficient-space computation; Lemma 3.14's vector enumeration is not applied with m in place of d. □

We also use the deterministic module-isomorphism algorithm of Chistov, Ivanyos, and Karpinski [CIK97, Corollary 3]: for two explicitly given modules over a finite-dimensional algebra over a finite field, it decides isomorphism and constructs one in polynomial time. Its module dimensions enter a polynomial factor, not the exponential parameter. This invocation is only a module-isomorphism problem, not an arbitrary matrix-space nonsingularity problem.

3.9 Affine transport on linear subquotients

Theorem 3.16 (Affine transport of degree one)**.** *Let H act faithfully on V, $\dim_k V \leq d$, and let Q be an explicit H-invariant subquotient of $V \otimes_k M$, where H acts trivially on M. Given an affine H-action on Q with this linear part and two points, one can compute its complete transporter in $p^{O(d)}\,\mathrm{poly}(L, \log p)$ deterministic time, with original witnesses and full original stabilizer. The statement also holds for a direct sum of natural, dual or Frobenius-twisted original spaces whose total prime-field dimension is $O(d)$.*

Proof. Put $E = k[H] \leq \mathrm{End}_k(V)$. Every H-invariant numerator and denominator is an E-module, so Q is an E-module. Compute its radical J and semisimple structure by Lemma 3.14. Filter Q by $Q \supseteq JQ \supseteq J^2Q \supseteq \cdots$. Each quotient is a semisimple E/J-module. Matrix units split it explicitly into copies of simple modules, with fixed multiplicity coordinates. Every such simple type occurs among the composition factors of the faithful E-module V, and hence has prime-field dimension at most d.

Process the quotient Q/JQ first and then successive fibers. Within a semisimple quotient, process its individual simple-coordinate equations in any order. The coordinate map is equivariant, so the affine action on that coordinate has at most p^d points. Breadth-first orbit search finds a transporter or proves emptiness, and Schreier generators give its full stabilizer. Retain that entire

subgroup, normalize by the witness, and process the next coordinate. After a quotient has been matched, its fiber is an affine space for the next invariant submodule, and the same argument applies. Thus every intermediate output is the complete original transporter for all equations processed so far.

There are only polynomially many layers and coordinates. Schreier lists are compressed by membership on the original $p^{\dim V}$-point vector action. Original action kernels, even if invisible on every processed coordinate, remain in each stabilizer. The cost is therefore $p^{O(d)}\,\mathrm{poly}(L, \log p)$, with no list of independent coordinate choices carried to the next stage. □

3.10 Involutions and the obstruction on a radical layer

Let $(E, *)$ be a finite-dimensional unital k-algebra with a k-linear involution, represented faithfully on a space of dimension at most d. Its unitary group is $\mathrm{U}(E) = \{u \in E^\times : u^*u = 1\}$. The radical J is $*$-stable. Its semisimple quotient $A = E/J$ acts on every J^i/J^{i+1} as a bimodule, with compatible involution $s(axb) = b^*s(x)a^*$.

Lemma 3.17 (Characteristic-two radical obstruction)**.** *Assume $p = 2$. Let A be semisimple and let B be a finite-dimensional A-bimodule with compatible k-linear involution s. Then*

$$T(B) = \ker(1 + s)/\operatorname{im}(1 + s)$$

under conjugation by $\mathrm{U}(A)$ is a direct sum of Frobenius twists of natural or dual modules of the matrix factors of A, with trivial multiplicity actions. If A has a faithful module of dimension at most d, each such simple-coordinate representation has prime-field dimension at most d. This decomposition and all maps can be constructed by polynomial linear algebra once the matrix-factor coordinates of A are given.

Proof. Decompose B by the central idempotents of A on its two sides. The involution sends the corner B_{ij} to B_{j^*,i^*}. If it exchanges two distinct simple bimodule types, every fixed vector is of the form $x + s(x)$, so their contribution is zero. A fixed corner must pair two $*$-partner algebra factors, which have the same matrix size and isomorphic center K. The tensor product of their centers decomposes into field-twist components. This is constructive without a large field search: for $K = \mathbb{F}_{p^f}$ the maps $a \otimes b \mapsto ab^{p^t}$, $0 \le t < f$, identify $K \otimes_k K$ with $\prod_{t=0}^{f-1} K$, and their inverse is obtained by linear algebra.

If the involution on the coefficient field of a fixed component is nontrivial, choose $t \in K$ with $t + s(t) = 1$. It is obtained by a linear trace equation. For a fixed vector x, one has $(1 + s)(tx) = x$, so this component has zero quotient. On a remaining component the involution is K-linear. Matrix units identify it with $\mathrm{Mat}_a(K) \otimes_K M$, and the compatible involution is transpose on the first factor, tensored with a linear involution η on M, after identifying the relevant adjoint form. Conjugation by the unitary group becomes congruence $X \mapsto gXg^t$, or its dual or field-twisted version, on the matrix coordinate, while M is fixed.

Here is the field calculation that also covers unitary centers. For a $*$-stable matrix factor write $a^* = H^{-1}a^{\sigma t}H$, with $H^{\sigma t} = H$ and $\sigma^2 = 1$. A central simple bimodule has right action twisted by $\tau \in \mathrm{Gal}(K/k)$. A fixed type requires $\tau^2 = 1$; its coefficient-field involution is $\mu = \sigma\tau$, and its matrix involution, apart from the multiplicity involution, is

$$X \longmapsto H^{-1}X^{\mu t}H^{\tau}.$$

If $\mu \neq 1$ the trace-one argument applies. Otherwise $\tau = \sigma$, and $Y = HX$ turns the involution into transpose. For $u \in \mathrm{U}(A)$ the conjugation action becomes

$$Y \longmapsto (HuH^{-1})Y(HuH^{-1})^t.$$

For an exchanged pair with $(a,b)^* = (b^t, a^t)$, the fixed cross-corners have coefficient involution τ. Only $\tau = 1$ survives, with congruence under (g, g^{-t}) on one corner and dual congruence on the other. Thus exchanged algebra factors cannot in general simply be omitted; it is the exchanged bimodule types that have zero quotient.

Over characteristic two, η is a sum of one-dimensional trivial blocks and two-dimensional Jordan blocks. A two-dimensional block has zero quotient: its fixed pairs (X, X^t) equal $(1+s)(X,0)$. A trivial block gives symmetric matrices modulo alternating matrices. The diagonal map identifies this quotient with K^a, and

$$\operatorname{diag}(gXg^t) = g^{(2)} \operatorname{diag}(X).$$

These are precisely Frobenius-natural modules (or their indicated dual and field twists). A matrix factor $\mathrm{Mat}_a(\mathbb{F}_{p^f})$ contributes prime-field degree $fa \le d$. All corner projections, trace equations, Jordan bases and quotient maps are polynomial linear algebra in the explicit bimodule dimension. □

Lemma 3.18 (Semisimple unitary groups and norms)**.** *For a semisimple algebra A faithfully represented in dimension at most d, generators of $\mathrm{U}(A)$ and a solution, when one exists, of $c^*hc = 1$ for an invertible selfadjoint h can be computed in $p^{O(d)} \operatorname{poly}(L, \log p)$ time.*

Proof. An involution permutes the simple matrix factors in orbits of size one or two. An exchanged pair contributes a general linear group and its norm equation is solved by choosing one matrix arbitrarily and reading the other from the equation. On a fixed factor the induced field automorphism σ has order at most two. Solve the linear equations $Hx^* = x^{\sigma t}H$ on a matrix basis. A nonzero solution is invertible and makes $*$ an adjoint involution. Scaling H normalizes $H^{\sigma t} = H$ when $\sigma \neq 1$; the required scalar can be found by enumeration of the original center field. For $\sigma = 1$ in odd characteristic, H is symmetric or alternating. In characteristic two it is symmetric, possibly alternating. These conclusions also follow directly by applying $*$ twice and using the scalar commutant of the matrix algebra.

The norm equation is equivalence of the nondegenerate forms with Gram matrices Hh and H. Bilinear alternating, odd-characteristic symmetric, and Hermitian forms admit standard elimination into hyperbolic planes and an anisotropic part of bounded dimension. Each needed field norm or square equation is found by a search through the center field. This constructs an isometry or detects distinct form types, and the frame-transversal construction above constructs their full isometry groups.

We spell out the remaining characteristic-two bilinear case. A nonalternating nonsingular symmetric form over a finite field is congruent to I_n. Repeatedly split off a vector of nonzero self-pairing and rescale its length using the unique square root. If an alternating remainder is reached, combine its hyperbolic pair u, v with a unit vector e. The vectors $e+u, e+v, e+u+v$ are an orthonormal basis of that three-space, which continues the diagonalization. Thus type and dimension decide equivalence.

For its group, define $\ell(v) = \sqrt{b(v,v)}$ and the canonical vector c by $b(c,v) = \ell(v)$. Every isometry fixes c. In odd dimension $b(c,c) = 1$, and $c^\perp$ is alternating and nondegenerate; the group is its symplectic group. In even dimension c is isotropic. Choose coordinates (c, E, e) with E alternating, $b(c,e) = b(e,e) = 1$ and $E \perp \langle c, e\rangle$. Every isometry has uniquely the form

$$\begin{aligned} &c \longmapsto c, \\ &v \longmapsto Av + b(Av, z)c \quad (v \in E), \qquad A \in \mathrm{Sp}(E),\ z \in E,\ a \in K. \\ &e \longmapsto e + z + ac, \end{aligned}$$

Substitution verifies all pairings and shows that the parameters are unrestricted. The kernel of projection onto $\mathrm{Sp}(E)$ has multiplication $(z,a)(w,t) = (z+w, a+t+b(w,z))$. Choosing any quadratic form with polar $b|_E$ changes coordinates to $(z, a+Q(z))$ and identifies it with an additive prime-field space. Its basis and symplectic generators give the entire group. All matrices act on the original natural dimension, which proves the bound in this last case as well. □

Theorem 3.19 (Unitary generators and norm equations)**.** *Let $(E,*) \le \mathrm{End}_k(V)$ be represented faithfully with $\dim_k V \le d$. In deterministic time $p^{O(d)}\,\mathrm{poly}(L, \log p)$ one can construct $\mathrm{U}(E)$, and solve $c^*hc = 1$ for every supplied invertible selfadjoint h. A nonempty solution set is returned as the complete coset $c_0\,\mathrm{U}(E)$ with actual matrices and full generators. No involution-stable algebra complement or surjectivity of $\mathrm{U}(E) \to \mathrm{U}(E/\operatorname{rad} E)$ is assumed.*

Proof. Compute $J = \operatorname{rad} E$, its powers and an ordinary complement by Lemma 3.14. Lift generators of $\mathrm{U}(A)$, $A = E/J$, through this algebra complement and adjoin $1+J$. The resulting group is exactly $H_1 = \{g \in E^\times : g^*g = 1 \pmod J\}$, which need not be unitary. For $i \ge 1$ define

$$H_i = \{g : g^*g = 1 \pmod{J^i}\}, \quad B_i = J^i/J^{i+1}, \quad S_i = \{x \in B_i : x^* = x\}, \quad A_i = \{x + x^* : x \in B_i\}.$$

For $g \in H_i$, the defect $n_i(g) = g^*g - 1$ lies in S_i, and multiplication gives the right cocycle identity

$$n_i(gh) = h^{-1}n_i(g)h + n_i(h). \tag{3.19.1}$$

Indeed h^* can be replaced by h^{-1} on this layer because their difference lies in J^i and $J^{2i} \subseteq J^{i+1}$. Moreover $n_i(1+x) = x + x^*$ for $x \in J^i$. Thus the actual subgroup $1+J^i$ realizes exactly the translations A_i.

In odd characteristic $A_i = S_i$, since $x = x^*/2 + x/2$ for selfadjoint x. In characteristic two, Lemma 3.17 and Theorem 3.16 compute the complete zero stabilizer of the cocycle on S_i/A_i under the actual subgroup H_i. (For left-action conventions use $n_i(g^{-1})$.) In either case this constructs exactly $K_i = \{g \in H_i : n_i(g) \in A_i\}$ with all original generators. For each generator g choose $x \in J^i$ with $x + x^* = -n_i(g)$ and replace it by $g(1+x)$, which belongs to H_{i+1}. Adjoin

$$L_i = \ker(n_i : 1 + J^i \longrightarrow A_i).$$

This is an additive homomorphism on this layer, so Lemma 3.1 supplies its entire kernel. The corrected generators map onto $K_i/(1+J^i)$, and their kernel in $1+J^i$ is exactly L_i. Since $H_{i+1} \le K_i$, this proves that they generate all of H_{i+1}. When $J^h = 0$, $H_h = \mathrm{U}(E)$.

For the inhomogeneous equation, first solve it in A by Lemma 3.18; failure there is conclusive. Lift a successful solution c to an E-unit. Suppose $c^*hc = 1 + \Delta \pmod{J^{i+1}}$ with $\Delta \in S_i$. The right affine action of $g \in H_i$ on this defect is

$$\Delta \longmapsto g^{-1}\Delta g + n_i(g).$$

Modulo A_i it is solved completely by the same degree-one procedure (or is automatically zero in odd characteristic). After choosing a witness g, replace c by cg and correct the remaining defect by $1 + x \in 1 + J^i$. The set of all next solutions is exactly cH_{i+1}, with H_{i+1} as just constructed. Induction therefore preserves the full solution coset, not a single arbitrary lift of each semisimple solution.

There are at most d radical layers. All coefficient and bimodule computations are polynomial in the explicit input, and every coordinate orbit used in the characteristic-two correction has degree at most d. Membership and normal closure use the original faithful vector action. These facts give the claimed running time, with no enumeration of J^i or of a quadratic-dimensional obstruction space. □

3.11 Exact simultaneous equations inside an algebra

Lemma 3.20 (Conductor equations)**.** *Let $(E, *)$ be as in Theorem 3.19. Suppose finitely many vectors in specified genuine E-bimodules, or subspaces in specified left E-modules, are to be transported simultaneously under $\mathrm{U}(E)$. In the subspace case the corresponding source and target dimensions are equal. There is a deterministic $p^{O(d)}\,\mathrm{poly}(L, \log p)$ procedure returning emptiness or the full transporter $u\,\mathrm{U}(F)$ for an explicitly computed $*$-subalgebra $F \le E$. In particular all solutions, including kernels invisible in individual equations, are retained.*

Proof. For subspace pairs L_j, L_j' define by linear equations

$$F = \{a \in E : aL_j \subseteq L_j, a^*L_j \subseteq L_j \text{ for all } j\},$$
$$X = \{c \in E : cL_j \subseteq L_j', c^*L_j' \subseteq L_j \text{ for all } j\}.$$

Then F is a unital $*$-subalgebra and X is a right F-module. If $c_0 \in X$ is an E-unit, equal dimensions make its inclusions and the adjoint inclusions equalities, and $X = c_0F$. For bimodule pairs v_j, w_j, use instead

$$F = \{a \in E : av_j = v_ja, a^*v_j = v_ja^* \text{ for all } j\},$$
$$X = \{c \in E : cv_j = w_jc, c^*w_j = v_jc^* \text{ for all } j\}.$$

The same right-module and principal-module identities follow by multiplying by c_0^{-1} and its adjoint. Mixed systems are obtained by imposing both sets of equations at once.

Test whether $F_F \cong X_F$ by the constructive module-isomorphism theorem [CIK97, Corollary 3]. If not, X contains no unit. If so, let c be the image of 1 and test its invertibility in the original matrix representation. This is conclusive: if a unit c_0 in X existed, every module generator would have the form c_0a with $a \in F^\times$, and hence would be invertible. If c is invertible, then $h = c^*c$ is an invertible selfadjoint element of F. The desired unitary elements in cF are exactly cf with $f^*hf = 1$. Theorem 3.19, applied to the actual subalgebra F on the same faithful space, solves this equation and returns $u\,\mathrm{U}(F)$. The defining equations show that $\mathrm{U}(F)$ is exactly the full source stabilizer. □

Remark 3.21 (Scope of the algebra operation). The preceding lemma applies to the entire unitary group of a represented algebra. It does not replace an arbitrary supplied subgroup by that unitary group. If a later argument works in a larger classical group, it must separately impose the actual subgroup condition by a previously proved complete transporter. Similarly, a quadratic refinement in characteristic two remains an explicit equation after its polar bilinear form has been matched; an alternating adjoint involution alone gives a symplectic group, not its quadratic isometry subgroup.

Corollary 3.22 (Fixed-label bilinear and quadratic systems)**.** *For two fixed-label tuples of $n \times n$ matrices over $K = \mathbb{F}_{p^f}$, simultaneous congruence under $\mathrm{GL}_n(K)$ has a complete deterministic transporter algorithm of cost $p^{O(fn)}\,\mathrm{poly}(L, \log p)$. There is no nondegeneracy assumption on the matrices or their tuple. The same bound holds for fixed-label homogeneous quadratic forms, and for additional quadratic constraints within a supplied full unitary group of an original-degree represented algebra. In characteristic two, an invariant correlated subspace of zero-polar coefficient differences may be allowed as a denominator.*

Proof. Take $E = \mathrm{Mat}_n(K) \times \mathrm{Mat}_n(K)$ in its faithful doubled representation, with $(P, R)^* = (R^t, P^t)$. Its unitary group is $\{(g, g^{-t}) : g \in \mathrm{GL}_n(K)\}$. Each matrix coordinate is a genuine E-bimodule with left action $(P, R)B = PB$ and right action $B(P, R) = BR$. Conjugation by the

unitary element is $B \mapsto gBg^t$. The conductor equations for source B_i and target C_i become

$$PB_i = C_iR, \qquad R^tC_i = B_iP^t \quad \text{for every } i.$$

No B_i is inverted. Lemma 3.20 gives the full answer on prime-field faithful dimension $2fn$, and projection to the first factor is injective on the unitary group.

More generally, if E acts faithfully by ρ on V, bilinear matrices form a genuine bimodule with left action $B \mapsto \rho(a)B$ and right action $B \mapsto B\rho(a^*)^t$. Its unitary conjugation is again congruence. Thus simultaneous polar or bilinear conditions inside $\mathrm{U}(E)$ are one conductor problem with the same original dimension bound.

In odd characteristic a homogeneous quadratic is determined by its polar form. In characteristic two, first compute the complete simultaneous polar transporter and normalize its witness. In its full stabilizer H, every remaining difference has zero polar and is uniquely $\sum_j a_j x_j^2$. Under the left action $(gQ)(x) = Q(g^{-1}x)$ the coefficient space is the Frobenius-twisted dual natural space, with the tuple labels fixed. The difference increments satisfy the affine cocycle identity because they are differences of an actual group action. Any allowed correlated square-coefficient subspace is an actual invariant denominator in this degree-one module. Theorem 3.16 therefore solves the residual affine condition under the entire H, keeping its full original kernel. Reinsert the normalized polar witness to obtain the complete congruence transporter. No freedom to permute or linearly transform the tuple labels has been introduced. □

3.12 The affine image of the flag kernel

Lemma 3.23 (Translations from a flag kernel)**.** *Let H preserve supplied flags on original spaces of total dimension at most d, let R be the kernel of its action on their successive quotients, and let H act affinely on an explicit module Q by $x \mapsto \rho(h)x + c(h)$. Suppose R acts linearly trivially on Q. Then $\eta(r) = c(r)$ is an additive homomorphism, its image is H-invariant, and conjugation equivariance on its image factors through H/R. Every composition factor of this image is a subquotient of a sum of spaces $\mathrm{Hom}_k(V_j, V_i)$ between original flag quotients. Its actual image, full kernel and word lifts are constructible without enumerating R or assuming a section $H/R \to H$.*

Proof. The cocycle identity gives $\eta(rs) = \eta(r) + \eta(s)$, and

$$c(hrh^{-1}) = c(h) + \rho(h)c(r) + \rho(hr)c(h^{-1}) = \rho(h)\eta(r).$$

This is valid for every affine origin. Let I be the nilpotent algebra of all strictly flag-lowering endomorphisms of the original direct sum. Then H normalizes I and $R \le 1 + I$. Define $R_i = R \cap (1 + I^i)$. The map $r \mapsto r - 1$ induces an equivariant injection

$$R_i/R_{i+1} \longrightarrow I^i/I^{i+1}$$

of elementary abelian groups. Since $[1 + I, 1 + I^i] \le 1 + I^{i+1}$, R acts trivially on these quotients. In a flag basis I^i/I^{i+1} is the direct sum of the Hom spaces between grades at distance i, with the natural induced action of H/R. Each quotient $\eta(R_i)/\eta(R_{i+1})$ is a quotient of R_i/R_{i+1}. This proves the composition-factor assertion for the whole image, without asserting that its extensions split.

The image of η is the span of its generator values, with lifts obtained by elimination. Its full kernel is supplied by Lemma 3.1. The groups R_i can likewise be constructed successively by the additive leading-coordinate maps, using that same lemma at each step. All original faithful representations remain in parallel. The cost is $p^{O(d)}\,\mathrm{poly}(L, \log p)$. □

4 Low-degree modules for finite classical groups

Throughout this section $k = \mathbb{F}_p$ and defining fields are finite extensions of k. A tensor word of length r is a tensor product of r natural, dual natural, or field-Frobenius-twisted natural modules. The trivial module has length zero. Direct sums may have arbitrary explicitly given multiplicities. All invariant subspaces and quotients are taken over k, unless another field is specified. In particular a subspace of a sum of fixed lines is not assumed to be a vector space over a defining field.

We use natural dimension at least 64 for special linear and special unitary factors, and Witt index at least 32 for symplectic and orthogonal factors. In characteristic two only even-dimensional orthogonal factors are used. These deliberately uniform bounds imply every local inequality below. Groups of bounded smaller natural dimension are not covered by a stable-rank assertion; their matrix groups have order $p^{O(d)}$ and can be enumerated in the algorithms where they occur. Here and below d is the sum of the prime-field dimensions of the faithful natural spaces, not the dimension of a tensor module.

4.1 Two elementary reductions

Lemma 4.1 (A computable cogenerator embedding)**.** *Let $A \subseteq \mathrm{End}_k(M)$ be a finite-dimensional algebra and let Q be an explicitly given subquotient of a direct sum of copies of M. Suppose C contains one copy of every indecomposable injective left A-module. If $f_1, \dots, f_s$ is a basis of $\mathrm{Hom}_A(Q, C)$, then*

$$\iota : Q \longrightarrow C^s, \qquad v \longmapsto (f_1(v), \dots, f_s(v))$$

is injective. It and the image algebra A can be computed by linear algebra. The target has dimension at most $\dim Q(\dim C)^2$.

Proof. The left A-module $A^* = \mathrm{Hom}_k(A, k)$, with $(a\varphi)(b) = \varphi(ba)$, is injective: the isomorphism $\mathrm{Hom}_A(X, A^*) \cong \mathrm{Hom}_k(X, k)$ identifies the former with an exact contravariant functor of X. It is a cogenerator, since for $v \neq 0$ one may choose $\lambda \in X^*$ with $\lambda(v) \neq 0$ and use the map $x \mapsto [a \mapsto \lambda(ax)]$. Decomposition of the right regular module into primitive right ideals shows that A^* is a sum of the stated injectives. Consequently maps to C separate points. A basis of their space also separates points. Intertwining equations on algebra generators compute that space. The dimension bound follows from $s \leq \dim Q \dim C$. Starting with the identity and the supplied generator matrices, repeatedly adjoining products to their linear span constructs A in at most $(\dim M)^2$ independent additions. No element enumeration of Q or A is involved. □

Lemma 4.2 (An acyclic observation and its exact residual)**.** *Let G act linearly on Q, I, E, let $\iota : Q \hookrightarrow I$ and $F : I \to E$ be equivariant, and let $z : G \to Q$ be a cocycle. Suppose $H^1(G, E) = 0$. Put $K = \ker F$. Given $x, y \in Q$, the complete affine transporter for $g * x = gx + z(g)$ is obtained by one complete linear transporter in E and one complete affine transporter in K, under the subgroup returned by the first transporter. The second cocycle is computed from the original action. This assertion needs no vanishing of $H^1(S, K)$ for that subgroup S.*

Proof. Solve $F\iota z(g) = (g-1)b$ on the supplied generators. Vanishing makes the equations consistent, and the cocycle identity then gives them on all of G. Solve $g(F\iota x + b) = F\iota y + b$. If its transporter is empty, so is the requested transporter. Otherwise write the complete answer as aS, where S fixes $F\iota x + b$, and put $y_0 = a^{-1} * y$. For $s \in S$, the displacement $c(s) = \iota(s * x - x)$ lies in K and satisfies $c(st) = c(s) + sc(t)$. The remaining equation is $c(s) = \iota(y_0 - x)$, an affine transporter from zero in K. If its answer is $s_0 S_0$, the original answer is $as_0 S_0$. All these are equalities of sets, so all stabilizer kernels are retained. Although K may contain vectors outside $\iota(Q)$, the cocycle and target belong to $K \cap \iota(Q)$, and testing their equality in K cannot introduce additional solutions. □

Lemma 4.3 (Joint scalar shifts)**.** *For matrices $A_j, B_j \in \mathrm{Mat}_n(K)$, every nonzero solution of*

$$XA_j - B_jX = \lambda_j X \quad (1 \le j \le s)$$

belongs to one of at most n^2 nonzero common scalar eigenspaces. All their full tuples $(\lambda_1, \dots, \lambda_s)$ can be found in $|K|\,\mathrm{poly}(n, s, \log|K|)$ operations. For independent matrix factors with $\sum_i [K_i : k] n_i \le d$, all joint choices number at most 4^d, and their computation costs $p^{O(d)}\,\mathrm{poly}(L)$.

Proof. On $\mathrm{Mat}_n(K)$ put $L_j(X) = XA_j - B_jX$. Starting with this whole space, replace each current space W by its nonzero intersections $W \cap \ker(L_j - \lambda)$, for $\lambda \in K$. These intersections are linearly independent for distinct λ, even if W is not L_j-invariant. Their dimensions therefore sum to at most $\dim W$. Induction bounds the final number by n^2. This also gives the stated algorithm. For products, use $n_i^2 \le 4^{n_i}$ and $|K_i| = p^{[K_i:k]} \le p^d$. □

A prescribed correlated additive subspace of allowable scalar tuples is imposed only after computing the whole tuples in Lemma 4.3. All remaining matrix, bilinear, and quadratic equations are imposed simultaneously. One must not choose independent witnesses for separate tensor coordinates.

4.2 Finite general-linear comparison, with explicit ranges

For a finite field K, let $\mathcal{F}(K)$ denote the category of ordinary functors from finite-dimensional K-spaces to K-spaces. The functors used here have finite ordinary difference degree and finite-dimensional values. We write I_a for field-Frobenius base change by p^a; as an ordinary additive functor it has degree one, regardless of a.

Lemma 4.4 (Finite general-linear computations)**.** *Let K be any finite field and $V = K^n$.*

(i) If $n \ge 8$, then $H^1(\mathrm{GL}_n(K), M) = 0$ for every whole tensor word M of length at most two and for $\mathrm{Sym}^2 V$, its dual, and their common field twists.

(ii) If $n \ge 10$, Hom spaces between whole covariant tensor words whose total length is at most four are the finite evaluations of the ordinary-functor Hom spaces. In particular words of different lengths have no maps; between two products of two additive functors the maps are the permutation matchings of equal field twists.

(iii) These assertions, and their Hom analogues with dual factors, hold for every intermediate group $\mathrm{SL}_n(K) \le G \le \mathrm{GL}_n(K)$ whenever n exceeds the coefficient weight support by two. The same vanishing holds for the trivial coefficient in degree one.

Proof. A split injection $(f, C) : V \to W$ has its specified retraction r. On a natural factor use f, and on a dual factor use r^*; apply the same rule after field base change. These are coefficient systems on the category of split injections with specified complements. Expanding a length-e word on $V \oplus K$ splits off the old word; every other term has a constant slot. Induction proves polynomial degree at most e, starting at rank zero. The decomposition $\mathrm{Sym}^2(V \oplus K) = \mathrm{Sym}^2 V \oplus V \oplus K$ proves the analogous assertion for the symmetric square, and duality gives the divided square assertion.

Putman's stability theorem [Put23, Theorem B] applies over fields with stable-rank parameter 2. Its explicit range is $n \ge 2i + 2e + 2$ for positive degree e starting at zero. Apply it to the dual coefficient system and use finite chain/cochain duality. For P, Q covariant, the homology map on $Q(V) \otimes P(V)^*$ is $Q(f) \otimes P(r)^*$; its dual is $h \mapsto P(r) h Q(f)$. This is exactly the restriction/evaluation transition in $\mathrm{Ext}_{\mathrm{GL}_n(K)}(Q(V), P(V))$. Thus the resulting stable values agree with the comparison

of ordinary finite functors in [FFSS99, Appendix, Theorem A.1]. No field-size hypothesis is used. The stated bounds are $2 + 4 + 2 = 8$ for first cohomology and $0 + 8 + 2 = 10$ for four-factor Hom spaces.

The constant functor K is both projective and injective: it is the linearized functor represented by the zero object, and also its dual corepresentable functor. Thus it has zero positive Ext to or from any reduced functor. For an extension of two additive functors, the middle functor is reduced, and its second cross-effect is zero: cross-effects are images of natural split idempotents and hence exact. It is therefore additive. Additive functors are modules over $K \otimes_{\mathbb{F}_p} K$, a product of fields, so the extension splits. This proves all first-cohomology claims for $\mathrm{GL}_n(K)$.

For clarity the Hom calculation can be made without highest weights. For products of r and s additive functors, use the sum/diagonal adjunction in $\mathcal{F}(K)^t$, with $t = \max(r, s)$, and expand the smaller product. If $r \neq s$, every term is constant in a variable where the other side is reduced, so its Hom is zero. For $r = s = 2$, the two surviving terms are the two permutations. Finally $\mathrm{Hom}(I_a, I_b)$ is K for equal Frobenius exponents modulo $[K : \mathbb{F}_p]$ and zero otherwise, as follows by evaluating at K and commuting with its scalar multiplications. Dual variances are moved across Hom by tensor duality.

We justify passage to SL, rather than inferring it from a prime-to-p quotient. Let T be the diagonal torus, $T_0 = T \cap \mathrm{SL}$, and let r bound the number of coordinates supporting any weight of M. Normalize a cocycle on T_0 to zero by averaging. Its values on the elementary root subgroups determine it, since these subgroups generate SL. On each root subgroup the action of T on its argument involves two coordinates. Thus the T-weights on the space of normalized cocycles, and on its cohomology quotient, have support at most $r + 2$. Inner automorphisms act trivially on cohomology, so any surviving character is trivial on T_0, and is consequently a constant exponent on all n coordinates modulo $|K| - 1$. If $n > r + 2$ only the zero character is possible. Hence $\mathrm{GL}\,/\,\mathrm{SL}$ acts trivially on this cohomology. Inflation–restriction and averaging identify it with the already zero GL cohomology. For invariants and Hom spaces the same argument uses r in place of $r + 2$. Intermediate determinant subgroups follow identically. For a trivial coefficient, SL is perfect in the stated rank and the determinant quotient has order prime to p. □

Lemma 4.5 (The separate trivial-coefficient input)**.** *For the stable natural groups* SL*,* SU*,* Sp*, and* Ω *in defining characteristic p, and for their products, $H^1(G, T) = H^2(G, T) = 0$ for every finite-dimensional trivial characteristic-p module T. This also holds for a prime-to-p extension of such a product.*

Proof. The stable simply connected finite groups are perfect and centrally closed by [Ste81, Theorem 1.1 and the twisted-group statement on p. 528]. All exceptional ranks in that theorem are below the bounds fixed here. The natural images have central kernels of order prime to the defining characteristic (in characteristic two the finite spin-to-Ω kernel is trivial). The low-degree inflation–restriction sequence and averaging transfer $H^1 = H^2 = 0$ to these images. Equivalently, universal coefficients identify trivial H^1 with the dual of the abelianization and trivial H^2 with the defining-characteristic part of the Schur multiplier. For products the degree-two Kunneth formula has, besides the factor terms, only products of degree-one terms; all vanish. Averaging over a prime-to-p quotient proves the last assertion. □

4.3 An exact prime-field image lemma

Lemma 4.6 (Augmentation and field descent)**.** *Let $K = \mathbb{F}_{p^e}$, let G be perfect, and let W be a $K[G]$-module with a nonzero fixed vector. Suppose distinct matrix-Frobenius twists $W^{[i]}$ and*

$W^{[j]}$, $0 \le i < j < e$, have no common nontrivial absolutely simple composition factor. Write $A_K = \operatorname{span}_K \rho(G)$ and let $\alpha : A_K \to K$ be its character on a fixed vector. Then

$$\operatorname{span}_{\mathbb{F}_p} \rho(G) = \mathbb{F}_p I \oplus \ker \alpha, \qquad \ker \alpha = \operatorname{span}_{\mathbb{F}_p} \{\rho(g) - I : g \in G\}.$$

In particular the entire augmentation ideal, including its radical, is retained over the defining field.

Proof. Put $R = K[G]$, $\Delta = \ker \epsilon$, and $J_i = \operatorname{ann}_R(W^{[i]})$. Modulo Δ^2 the map $g \mapsto g - 1$ is a homomorphism to an additive abelian group. Perfectness gives $\Delta = \Delta^2$, hence $\Delta^t = \Delta$ for all $t \ge 1$. Fixed vectors give $J_i \subseteq \Delta$. For any basis $w_1, \dots, w_s$ of $W^{[i]}$, the map $R/J_i \to (W^{[i]})^s$, $a \mapsto (aw_1, \dots, aw_s)$, is injective. Thus the composition factors of R/J_i occur in $W^{[i]}$. The quotient $R/(J_i + J_j)$ has only trivial composition factors after scalar extension. Some power of Δ annihilates it, and therefore Δ itself annihilates it. Hence $J_i + J_j = \Delta$.

Let $J = \bigcap_{i=1}^{e-1} J_i$. Expanding the ordered product $\prod_{i=1}^{e-1}(J_0 + J_i) = \Delta^{e-1} = \Delta$ shows $\Delta = J_0 + J$: every term containing J_0 lies in J_0 and the remaining product lies in every J_i. Two-sidedness, not commutativity of the ideals, is the required property. Given $a \in \ker \alpha$, choose $w \in J$ with $\rho_0(w) = a$. For coefficient Frobenius σ, put $u = \sum_{s=0}^{e-1} \sigma^s(w)$. It belongs to $\mathbb{F}_p[G]$, has augmentation zero, and

$$\rho_0(\sigma^s w) = \sigma^s \rho_{e-s}(w) = 0 \quad (0 < s < e).$$

Thus $\rho_0(u) = a$. This proves the augmentation assertion, and adjoining I proves the first. The case $e = 1$ is immediate. No division by e is used. The group algebra is only used for the proof; actual image computation uses matrix-span closure. □

Lemma 4.7 (Derived and full image algebras)**.** *On a joint sum of words of length at most r, the prime-field image algebras of $\mathrm{SL}_n(K)$ and $\mathrm{GL}_n(K)$, and of $\mathrm{SU}_n(K/K_0)$ and $\mathrm{U}_n(K/K_0)$, coincide if $n > 2r$. In odd characteristic the images of $\Omega_n(K)$ and $\mathrm{O}_n(K)$ coincide if $n \ge 4r + 3$. All occurrences, including trivial coordinates, must be included jointly.*

Proof. For a prime-to-p abelian group T and $S \le T$, restriction injective on the occurring characters gives every weight projector as $|S|^{-1} \sum_{s \in S} \chi(s)^{-1} \rho(s)$ after scalar extension. Thus every T operator belongs to the S image algebra. For general linear and unitary groups use the diagonal and diagonal norm-one tori, respectively. A character trivial on the determinant-one torus has constant exponent vector; a difference of two length-r weights has support at most $2r$, so cannot be a nonzero such vector. The full groups are their special subgroups times these tori.

In odd characteristic diagonalize a quadratic form with entries in the two square classes. Both classes can be made to occur more than $2r$ times: every nonzero element is a sum of two squares, so replacing two equal diagonal entries by entries in the other square class is possible; the interval $[2r + 1, n - 2r - 1]$ contains two consecutive integers. In the diagonal sign group the determinant and spinor norm are respectively the total parity and the parity on the nonsquare positions. The annihilator of its intersection with Ω has three nonzero supports: all positions and the two square-class sets. Each has size exceeding $2r$. The same interpolation applies. These assertions are statements on whole modules, regardless of internal extensions. Linear descent proves prime-field equality, and invariant subquotients inherit it. In characteristic two this argument gives no equality between Ω and the full quadratic orthogonal group; their Dickson quotient is retained explicitly. □

4.4 Binary tensor squares and their injectives

In this subsection K has characteristic two. For $T = V \otimes_K V$ write $\tau(v \otimes w) = w \otimes v$, $x = 1 + \tau$, $A = \operatorname{im} x$, and $\Gamma = \ker x$. Thus $A \cong \bigwedge^2 V$, $\Gamma = \Gamma^2 V$, and $T/A = \operatorname{Sym}^2 V$. The latter is the symmetric *quotient*, not the symmetric-tensor subspace.

Lemma 4.8 (The general-linear square)**.** *For $n = \dim_K V \geq 3$, the defining-field image of* $\mathrm{GL}(V)$ *on T is* $\mathrm{End}_{K[x]}(T)$*. Its indecomposable injective types are T and* $\mathrm{Sym}^2 V$*, with their respective simple socles* $\bigwedge^2 V$ *and* $V^{[1]}$*.*

Proof. First work over $\mathbb{F}_2$. A homogeneous quadratic polynomial in the n^2 matrix entries which vanishes on every invertible matrix can be made multilinear by replacing squares by their variables. For two distinct positions $(i,j),(k,l)$ choose a permutation matrix P so that $P^{-1}E_{ij}, P^{-1}E_{kl}$ are off-diagonal units with acyclic directed edges. If $i = k$, choose the common new row outside $\{j,l\}$; if $j = l$, choose distinct new rows outside $\{j\}$; otherwise choose $c \notin \{j,l\}$ and new rows c, j. In the last case the edges form the path $c \to j \to l$. Complete these choices to a permutation. All four matrices $P + sE_{ij} + tE_{kl}$ are invertible. Summing their polynomial values kills every term except the indicated mixed coefficient. Hence all mixed coefficients vanish. An invertible pair $P, P + E_{ij}$ similarly kills each remaining linear coefficient. The polynomial was zero.

The quadratic coordinate functions of $g \otimes g$ consequently have rank $n^2(n^2+1)/2$. The operator x has Jordan type $2^h 1^n$, where $h = n(n-1)/2$, so

$$\dim \mathrm{End}_{K[x]}(T) = (h+n)^2 + h^2 = n^2(n^2+1)/2.$$

The image commutes with x, giving equality over $\mathbb{F}_2$ and over any extension by taking the same binary matrices. In a basis adapted to $\operatorname{im} x \subseteq \ker x$, the algebra consists of

$$\begin{pmatrix} B & C & D \\ 0 & E & F \\ 0 & 0 & B \end{pmatrix},$$

with arbitrary blocks of the indicated dimensions. Primitive right ideals at its two diagonal types, dualized over K, have the realizations T and T/A; this can also be checked directly from their simple essential socles and dimensions $2h+n$ and $h+n$. Thus they are all the indecomposable injectives. □

For a general-linear square with no fixed constituent, its prime-field image is the same defining-field algebra: scalar matrices act by a^2, and the squares of K span K. Passage from GL to SL uses Lemma 4.7 on the whole module, including any natural summands already appearing as its Frobenius-natural constituent.

Lemma 4.9 (The finite symplectic square)**.** *Let V have dimension $2m$ with a nonsingular alternating form b, where $m \geq 5$. Write*

$$\omega = \sum_{i=1}^{m} (e_i \otimes f_i + f_i \otimes e_i), \qquad \ell(v \otimes w + w \otimes v) = b(v,w), \quad y(t) = \ell(xt)\omega.$$

Then $x^2 = y^2 = xy = yx = 0$, and the defining-field image on T is $\mathrm{End}_C(T)$*, where $C = K[x,y]/(x^2, y^2, xy)$ acts by these operators. Its prime-field image is $\mathbb{F}_2 I + \ker \alpha$, where α is its character on $K\omega$.*

Proof. Use the binary Siegel Levi on $V = E \oplus E^*$. The three whole grades EE, $EE^* \oplus E^*E$, and E^*E^* have pairwise disjoint simple types for $m \geq 5$: the first has fundamental weights ω_1, ω_2, the last their dual positions $\omega_{m-1}, \omega_{m-2}$, and the middle has only the trivial and adjoint types. The defining-characteristic restriction and tensor-product statements used here are [Ste67, Theorems 41 and 43]; all these weights are restricted, including over $\mathbb{F}_2$. Disjointness gives the three whole-grade

projections in the finite Levi algebra. Indeed for annihilator ideals J, J' a quotient by $J + J'$ would have a simple factor occurring in both modules, by the faithful evaluation embedding used in Lemma 4.6; hence $J + J'$ is the whole algebra, and the two-sided Chinese remainder theorem gives the projections.

For binary symmetric B the upper Siegel element satisfies $\rho(u(B)) = I + L_B + Q_B$, with degrees one and two relative to these grades. The projections isolate both coefficients. After scalar extension the algebra therefore contains $I + tL_B + t^2Q_B = \rho(u(tB))$ for every t. Binary symmetric matrices span all symmetric matrices, and $u(B + B') = u(B)u(B')$. The same applies to the opposite radical. These radicals generate the algebraic symplectic group (the mixed root subgroups follow by elementary commutators). Over the algebraic closure its image is the Brauer commutant by [DDH08, Proposition 1.3]; similitudes have the same linear span, since each is a scalar times a symplectic matrix. This theorem is used only after the preceding finite-density argument. Centralizers commute with scalar extension, proving the asserted finite-field identity.

The binary whole-grade projectors also lie in the prime-field image over K. Applied to $u(tB)$ they give tL_B and t^2Q_B for every $t \in K$. Thus this image contains all defining-field multiples of the root coefficients. Their unital K-algebra is $\mathrm{End}_C(T)$; they annihilate ω and the quotient by the ideal they generate is the scalar algebra. Their ideal is exactly $\ker \alpha$. Moving the coefficient of a nonempty word onto one root coefficient shows that its entire K-span lies in the prime-field image. Conversely every group element has α-value one. □

Lemma 4.10 (Symplectic injective modules)**.** *The defining-field symplectic square image has three indecomposable injective types:*

$$I_H = T/K\omega, \qquad I_W = T/A = \mathrm{Sym}^2 V,$$

and

$$I_0 = \begin{cases} T/\ker(\ell : A \to K), & m \text{ odd}, \\ (T \oplus T)/\{(a, \ell(a)\omega) : a \in A\}, & m \text{ even}. \end{cases}$$

Proof. Put $h = \dim A$. A complement of A in Γ consists of $2m$ trivial C-modules. On the remaining space, x identifies a top copy of A with its bottom copy, while y is the rank-one map $P = \omega\ell$ between them. Since $P^2 = mP$, the C-module is

$$T \cong \begin{cases} (C/(y))^{h-1} \oplus C/(x + y) \oplus K^{2m}, & m \text{ odd}, \\ (C/(y))^{h-2} \oplus R \oplus K^{2m}, & m \text{ even}, \end{cases}$$

where R has generators u, v and relations $yu = 0$, $yv = xu$. For an idempotent selecting an indecomposable summand C_i, the primitive right ideal of $\mathrm{End}_C(T)$ is $\mathrm{Hom}_C(T, C_i)$. Its dual is $T \otimes_C C_i^*$ by tensor–Hom duality. All three C_i are self-dual. For $C/(y)$ this quotient is $T/K\omega$; for K it is T/A; for $C/(x + y)$ it is $T/\operatorname{im}(x + y) = T/\ker \ell$. In the even case the presentation gives relations $(yt, 0)$ and (xt, yt). Since $\ell(\omega) = 0$, $(\omega, 0)$ already occurs in the second family, and the denominator is precisely the displayed graph. This derives all three injectives, including their nonsplit structure, from primitive right ideals. □

Lemma 4.11 (The finite unitary square)**.** *For K/K_0 quadratic in characteristic two and stable n, the ordinary square of the natural unitary module has defining-field image $\mathrm{End}_{K[x]}(T)$ and the same two injective types as Lemma 4.8. Its prime-field image contains the full scalar field. The special unitary group has the same joint image on bounded-length words in the range of Lemma 4.7.*

Proof. Write $\bar{a} = a^{|K_0|}$. In even dimension $2m$ use $V = E \oplus E^{\bar{*}}$ and the Levi $\mathrm{GL}_m(K)$. Its whole square grades EE, $EE^{\bar{*}} \oplus E^{\bar{*}}E$, and $E^{\bar{*}}E^{\bar{*}}$ have disjoint composition types: the mixed grade is the distinct-digit simple type, and the two squares have the exterior and Frobenius-natural types with their indicated variances. This remains true for $K_0 = \mathbb{F}_2$ and $m \geq 5$, by the restricted weights in [Ste67, Theorems 41 and 43]. Thus their whole projections belong to the finite Levi algebra. For Hermitian B the upper radical again has action $I + L_B + Q_B$, and projections separate these terms. Hermitian matrices span $\mathrm{Mat}_m(K)$: use the diagonal units and, for $i < j$, $E_{ij} + E_{ji}$ and $\theta E_{ij} + \bar{\theta} E_{ji}$ with $\theta \neq \bar{\theta}$. Scalar extension consequently supplies all upper and lower general-linear block unipotents. They generate SL_{2m}, whose square image is the swap centralizer; the latter also follows by scalar extension of Lemma 4.8.

In odd dimension add a one-dimensional tail Z. The six Levi summands EE, $EF \oplus FE$, FF, $EZ \oplus ZE$, $FZ \oplus ZF$, ZZ have disjoint whole type sets, and hence have projections. Choose $\theta + \bar{\theta} = 1$ and the finite unitary root element sending $z \mapsto z + e_i$ and $f_i \mapsto f_i + z + \theta e_i$. Put $a = E_{i,z}$ and $b = E_{z,f_i}$. The blocks $EZ \to EE$, $FZ \to EF$, $ZZ \to EZ$ isolate $a \otimes 1 + 1 \otimes a$, and $ZZ \to EE$ isolates $a \otimes a$. The blocks $FF \to FZ$, $FZ \to ZZ$, $EF \to EZ$ similarly isolate the two coefficients for b. The other root terms have different source or target grades. This supplies the algebraic elementary matrices $1 + ta$, $1 + tb$ in the represented image; with the even head they generate SL_{2m+1}. Linear descent proves the equality. A central norm-one scalar of order $|K_0| + 1$ acts on the square by its square, and generates K over $\mathbb{F}_2$: every proper subfield has at most $|K_0|$ elements. Thus the whole scalar field lies in the prime-field image. Finally use the joint derived transfer lemma. □

4.5 Binary first cohomology

Lemma 4.12 (Special-unitary transfer in degree one)**.** *Let $U = \mathrm{U}_n(K/K_0)$, $D = \mathrm{SU}_n(K/K_0)$, in characteristic two. If every diagonal norm-one-torus weight of M is supported on at most r coordinates and $n > r + 3$, restriction identifies $H^1(U, M)$ with $H^1(D, M)$.*

Proof. Coordinate SU_3 subgroups generate D. Here are the small-field details. A norm-one vector with two coordinates of unequal nonzero K_0-norm can have those two coordinates replaced by one using a two-dimensional unitary isometry. If all its nonzero coordinate norms coincide, their number is odd, and any three of them have nonzero total norm; a three-dimensional isometry replaces that triple by one coordinate. Witt extension in the corresponding coordinate subspace justifies each operation. Successive column elimination shows that U is generated by these coordinate unitary groups and its diagonal torus T. Each local unitary group is its special subgroup times a diagonal element. The determinant-one torus $S = T \cap D$ is generated by two-coordinate $\mathrm{diag}(t, t^{-1})$ and is inside the generated special subgroup. Thus the latter is exactly D. Normalize a D-cocycle to vanish on S. Evaluation on the generating coordinate SU_3 subgroups is injective and shows that its T-weights have support at most $r + 3$. Since $|T|$ is odd, this bound passes to the cohomology quotient. Inner D-action is trivial there, so all its characters are constant-coordinate characters trivial on S. The inequality on n makes them zero. The quotient U/D has odd order; inflation–restriction finishes the proof. □

Lemma 4.13 (Acyclic binary linear, unitary, and symplectic injectives)**.** *Every general-linear and special-unitary injective listed above, including the adjoint scalar-quotient modules described below, has zero first cohomology. All three symplectic injectives in Lemma 4.10 have zero first cohomology. The whole unmatched relative-Frobenius tensor types also have zero first cohomology. These assertions are for the full stated groups.*

Proof. For general linear groups this is Lemma 4.4; quotients of adjoint modules by fixed scalar subspaces additionally use Lemma 4.5 and the long exact sequence.

For unitary groups put $q_0 = 2^f$. On $V^{[a]} \otimes V^{[b]}$ a central norm-one scalar acts by exponent $2^a + 2^b$. This is zero modulo $q_0 + 1$ exactly when $b - a = f$ modulo $2f$. Otherwise its prime-to-two cyclic subgroup has no fixed vector, so all cohomology of U is zero by averaging; Lemma 4.12 passes degree one to D. It remains to treat the adjoint signature.

In even dimension $2m$, the Siegel parabolic of D has odd index, Levi $L = \{g \in \mathrm{GL}_m(K) : \det g \in K_0^*\}$, and additive Hermitian radical $u(Z) = \left(\begin{smallmatrix} I & Z \\ 0 & I \end{smallmatrix}\right)$. Every Levi block of $\mathrm{End}_K(V)$ is a whole word of length at most two, so normalize a cocycle to zero on L. The radical raises the usual three endomorphism grades. At its first nonzero grade the cocycle is an additive L-map from the Hermitian parameter space. Over a splitting field that parameter has types $E^{[s]} \otimes E^{[s+f]}$, $0 \le s < f$. The four-factor matching in Lemma 4.4 excludes the opposite and diagonal blocks and leaves only the top block $Z \mapsto aZ$. This is the coboundary of a times the projector onto the first half. After subtracting it no first grade remains. Restriction to an odd-index subgroup is injective in characteristic two, proving the claim.

In odd dimension write $V = E \oplus Kt \oplus E^{\bar{*}}$. The Levi is $\mathrm{GL}_m(K)$, acting as $\mathrm{diag}(g, \chi(g), g^{-\dagger})$, where $\chi = (\det g)^{q_0 - 1}$. Its radical consists of

$$u(X, Z) = \begin{pmatrix} I & X & Z \\ 0 & 1 & X^\dagger \\ 0 & 0 & I \end{pmatrix}, \qquad Z + Z^\dagger = XX^\dagger.$$

Its center is $X = 0$ and its quotient is the additive module $E \otimes \chi^{-1}$. Every adjoint Levi block has zero first cohomology. For the cross-tail blocks, normalize on the prime-to-two torus: a nonzero determinant twist has support m, whereas the possible natural or cocycle-root weight has support at most three, so the same proof as in Lemma 4.4 excludes it. Kill the center value aZ by the first-half projector, as above. A first remaining grade then factors through the radical quotient and has exactly the two possibilities

$$aX \in \mathrm{Hom}(Kt, E), \qquad bX^\dagger \in \mathrm{Hom}(E^{\bar{*}}, Kt).$$

They are the two field components, since $\chi^{[f]} = \chi^{-1}$. Commutators lie in the center, where the cocycle is now zero. The cocycle identity for two radical elements gives in the top block $(a + b)(XY^\dagger + YX^\dagger) = 0$. Distinct coordinate vectors force $a = b$. The projector onto Kt removes these values. No later first grade is allowed by the same Hom matching. Again odd-index restriction proves $H^1(D, \mathrm{End}_K(V)) = 0$.

The Hermitian K_0-space H satisfies $\mathrm{End}_K(V) = H \oplus \theta H$ for $\theta \notin K_0$. Thus $H^1(D, H) = 0$, and every fixed-scalar quotient is acyclic by Lemma 4.5.

For symplectic groups use the odd-index Siegel parabolic with Levi $\mathrm{GL}_m(K)$ and additive radical $\Gamma^2 E$, the symmetric matrices. In T its three grades are EE, $EE^* \oplus E^*E$, E^*E^*. After Levi normalization the lower and middle radical values are zero by finite Hom matching. The top maps $\Gamma^2 E \to E \otimes E$ are scalar multiples of its inclusion: this follows directly from the block algebra in Lemma 4.8. Over a proper extension field scalar squares force prime-field intertwiners to be K-linear. For $r = \sum_i e_i \otimes f_i$, $u(B)r - r = B$, so this last value is a coboundary. Hence $H^1(\mathrm{Sp}, T) = 0$. For $\mathrm{Sym}^2 V$ the same argument leaves only $B \mapsto \sum_i B_{ii} e_i^2$; it is the coboundary of $\sum_i e_i f_i$. Unequal relative Frobenius words have distinct-digit simple top type, and no top Hom from $\Gamma^2 E$, so their first cohomology is also zero. The quotient $T/K\omega$ uses Lemma 4.5.

Finally, for odd m there is the exact sequence $0 \to K \to I_0 \to \mathrm{Sym}^2 V \to 0$; for even m there is $0 \to T \to I_0 \to \mathrm{Sym}^2 V \to 0$, with first map $t \mapsto [(0, t)]$ and last map $[(s, t)] \mapsto s + A$. The displayed

denominators in Lemma 4.10 verify exactness. In either case both exterior first-cohomology terms are zero, so $H^1(\mathrm{Sp}, I_0) = 0$. □

Lemma 4.14 (Orthogonal square and natural cohomology)**.** *Let $G = \Omega(V, Q)$ in characteristic two, with either quadratic sign and stable Witt index. Then*

$$H^1(G, V^{[a]}) = 0, \qquad H^1(G, V^{[a]} \otimes V^{[b]}) = 0, \qquad H^1(G, T/K\omega) = 0.$$

Proof. In split type the odd-index parabolic has Levi $\mathrm{GL}_m(K)$ and additive alternating radical $\bigwedge^2 E$. Normalize on the Levi. Finite four-factor matching permits no radical map to a natural coefficient; for a tensor coefficient it permits only $Z \mapsto aZ$ in the top block, with equal Frobenius signatures. That map is the coboundary of the first-half projector. This proves the assertions before scalar quotient in split type.

For nonsplit type write $V = E \oplus Z \oplus E^*$, with Z an anisotropic plane. The Levi is $\mathrm{GL}_m(K) \times T_0$, where the rotation torus T_0 has order $q + 1$, $q = |K|$. The radical has center $\bigwedge^2 E$ and quotient $\mathrm{Hom}(Z, E)$; write its elements as

$$\begin{pmatrix} I & X & B \\ 0 & I & X^* \\ 0 & 0 & I \end{pmatrix}, \qquad X^* = J_Z^{-1} X^t, \quad t^t B t = Q_Z(X^* t).$$

Every X occurs. The parabolic has odd index because it contains a Sylow two-subgroup. Levi cohomology vanishes by Lemma 4.4 and the odd-order torus. Normalize there, and kill the only possible center map aB in the matched case by the first-half projector. No center map exists otherwise. Over a splitting field the tail weights are $\chi^{\pm 2^a}$. A first quotient-radical grade in a two-slot tensor requires $2^a = \pm 2^b$ modulo $q + 1$. For $0 \le a, b < \log_2 q$ this means $a = b$: a nonzero difference has absolute value below $q + 1$, and the sum is at most q. Hence unmatched signatures have no value. For a natural coefficient no matching tail character is available.

For the matched square identify T with $\mathrm{End}(V)$ by the polar form. The only possible first grade is $XD \in \mathrm{Hom}(Z, E)$ and $EX^* \in \mathrm{Hom}(E^*, Z)$, with $D, E \in \mathrm{End}_{T_0}(Z)$. Commutators lie in the center, and the cocycle identity gives $X(E + D)Y^* + Y(E + D)X^* = 0$. Choose different single nonzero head rows of X, Y and arbitrary row functionals on Z; their top matrix positions differ, so every coefficient of $E + D$ vanishes. The endomorphism supported by D on Z removes the remaining values by a coboundary. There is no other first grade, proving vanishing on the parabolic and hence on G. Finally apply the long exact sequence and Lemma 4.5 to $K\omega$. □

4.6 The binary orthogonal image, including both parities

Lemma 4.15 (Finite orthogonal density)**.** *The defining-field square image of finite $\Omega^{\pm}_{2m}(K)$, in the stated stable range, becomes the square image of the connected algebraic orthogonal group after scalar extension.*

Proof. If an image algebra contains the whole-grade projectors P_i, it contains $\rho(h(t)uh(t)^{-1}) = \sum_{i,j} t^{i-j} P_i \rho(u) P_j$ for a cocharacter having weight i on grade i and every finite element u. In split type the Levi $\mathrm{GL}_m(K)$ has three disjoint whole square grades EE, $EE^* \oplus E^*E$, E^*E^*, as in Lemma 4.9; their projections lie in its image. The cocharacter $\mathrm{diag}(tI, t^{-1}I)$ scales an upper alternating radical parameter by t^2. Thus each finite alternating matrix unit supplies its whole algebraic root subgroup. The opposite radical and commutators supply all connected type-D roots. In nonsplit type use $E \oplus Z \oplus E^*$ and Levi $\mathrm{GL}_m(K) \times T_0$, $|T_0| = q + 1$, with physical weights $2, 1, 0$. The five square grades have disjoint finite Levi types. Grades one and three are dual-natural and

natural with a nontrivial tail character; this separates them from the extreme square factors even when $K = \mathbb{F}_2$. The middle grade has adjoint/trivial head types and tail-square types. Thus all five projectors exist. The cocharacter takes $u(X, B)$ to $u(tX, t^2B)$ in the radical of Lemma 4.14. These curves contain every scalar multiple of every finite central alternating matrix, and modulo the center every scalar multiple of every finite X. The latter span $\mathrm{Hom}_K(Z, E)$. Their generated subgroup is the whole algebraic radical; similarly it contains the opposite radical. Together they generate the connected group. The reverse image containment is immediate. □

Put $h = \dim A = n(n-1)/2$, $n = 2m$. Choose $X \in T$ representing $Q \in \mathrm{Sym}^2 V$ under polar duality, so $xX = \omega$. On Γ put $f_Q(s) = \langle X, s\rangle$, for the tensor-product polar pairing. Then $f_Q(v \otimes v) = Q(v)$ and $f_Q|_A = \ell$.

Lemma 4.16 (The Dickson coordinate)**.** *For $g \in \mathrm{O}(Q)$, $gX - X \in A$ and $\ell(gX - X)$ is its Dickson invariant in $\mathbb{F}_2 \subseteq K$.*

Proof. The first assertion is preservation of the quadratic class, and invariance of ℓ makes the expression a homomorphism. After scalar extension split Q and choose $X = \sum e_i \otimes f_i$. For a nonsingular $v = u + w$ normalize $Q(v) = 1$. The symmetry $t_v = I + vb(v, -)$ satisfies $(t_v \otimes t_v)X - X = u \otimes w + w \otimes u$, of contraction one. Symmetries generate the full orthogonal group in this range, and each has Dickson invariant one [Tay92, Corollary 11.42 and Theorem 11.43]. The identity therefore holds and descends. The equality of the finite binary Ω with the Dickson kernel is [Tay92, Theorem 11.51], since every nonzero finite-field element in characteristic two is a square. □

Proposition 4.17 (Exact orthogonal square algebras and injectives)**.** *Let B be the subalgebra of the symplectic square image given by*

$$aX - \alpha(a)X \in \ker \ell, \qquad a^\dagger X - \alpha(a)X \in \ker \ell,$$

where $\dagger$ is tensor-pairing adjoint. The defining-field image of $\Omega(V, Q)$ is exactly B. Its injectives are

$$I_H = T/K\omega, \qquad I_W = T/(A + KX).$$

$$I_0 = \begin{cases} T/(\ker f_Q + KX), & m \text{ odd}, \\ (T \oplus T)/\{(s + cX, f_Q(s)\omega) : s \in \Gamma, c \in K\}, & m \text{ even}. \end{cases}$$

Its prime-field image is $\mathbb{F}_2 I + \ker \alpha$.

Proof. Containment follows from Lemma 4.16 and self-duality. Suppose first m odd and put $H = \ker \ell$, of dimension $h - 1$. The nondegenerate C-summand $KX + K\omega$ has type $C/(x + y)$. In coordinates $H_t, W, KX, H_b, K\omega$, where $\dim W = n$, B has the same arbitrary H-matrix on H_t, H_b, an arbitrary W-matrix, and a common scalar on X, ω. Its remaining blocks are precisely

$$H_t \to H_b, \quad H_t \to W, \quad W \to H_b, \quad H_t \to K\omega, \quad KX \to H_b.$$

This follows by commuting with x, y and imposing the two conditions. The radical R has $R^2 = \mathrm{Hom}(H_t, H_b)$ and $R^3 = 0$. The heart H is absolutely simple: the nonzero weight vectors of $\bigwedge^2 V$ are the root vectors, one of which generates all root spaces by Weyl transitivity and all diagonal differences by root operations. A zero-weight vector killed by all roots has equal diagonal coefficients and lies in $K\omega$, whereas $\ell(\omega) = 1$. Lemma 4.15 gives the same assertion for the finite group. The other distinct simple types are W and K, so the actual image has the full semisimple diagonal. Conjugate a separable semisimple complement by $1 + R$ to make it the displayed diagonal. A

missing arrow $W \to H_b$ or $H_t \to W$ would make W a submodule or quotient of T, contrary to Lemma 6.3. A missing $KX \to H_b$ or its dual would give a second invariant of T or T^*, whereas $T \cong \mathrm{End}(V)$ has just one by absolute irreducibility of V. Thus all four arrow bimodules occur in full. Their products through W span R^2, proving equality of the image and B.

For even m put $H = (\ker \ell)/K\omega$, $l = \dim H = h-2$. Choose $a \in A$ with $\ell(a) = 1$ and Y with $xY = a$. The nondegenerate C-summand $\langle X, Y, \omega, a\rangle$ is the four-dimensional module of Lemma 4.10. Use coordinates $H_t, W, X, Y, H_b, \omega, a$. The diagonal algebra is $\mathrm{Mat}_l(K) \times \mathrm{Mat}_n(K) \times K$, with common scalar on the four scalar positions. The radical blocks are

$$\begin{gathered} n_H : H_t \to H_b, \quad \beta : (Y \to X,\ a \to \omega), \quad p : X \to \omega, \quad q : Y \to \omega, \quad t : Y \to a, \\ a_1 : (H_t \to X,\ H_b \to \omega), \quad a_2 : H_t \to \omega, \quad a_3 : H_t \to a, \\ b_1 : (Y \to H_t,\ a \to H_b), \quad b_2 : X \to H_b, \quad b_3 : Y \to H_b, \\ H_t \to W, \quad W \to H_b, \quad W \to \omega, \quad Y \to W. \end{gathered}$$

Paired arrows have the same coefficient; all rectangular coefficients are arbitrary. To verify the list, commuting with x, y on the four-dimensional summand gives the same top/bottom block $\left(\begin{smallmatrix} \alpha & \beta \\ 0 & \alpha \end{smallmatrix}\right)$ and an arbitrary top-to-bottom block. The additional conditions remove $X \to a$, $X \to W$, and $W \to a$, exactly $2n+1$ dimensions. The quotient R/R^2 consists of $H \leftrightarrows W$ and two parallel arrows each way $H \leftrightarrows K$, represented by a_1, a_3, b_1, b_2. They generate R, since

$$a_1 b_1 = \beta, \quad b_2 a_1 = n_H, \quad a_3 b_1 = t, \quad a_1 b_2 = p, \quad a_1 n_H = a_2, \quad b_2 \beta = b_3, \quad p\beta = q,$$

and the remaining blocks are $a_1(W \to H_b)$ and $(H_t \to W)b_1$. Matrix units supply every coefficient. Levels $Y = 4, H_t = 3, W = X = a = 2, H_b = 1, \omega = 0$ show $R^5 = 0$.

The root argument proves simplicity of H in this parity too. Perfectness and $T^G = K\omega$ imply $(T/K\omega)^G = 0$, using $H^1(G, K) = 0$. The invariant module $(A + KX)/K\omega$ is an extension of K^2 by H, annihilated by R^2. Fewer than two independent $K \to H$ arrows would leave a nonzero combination of the two top coordinates killed by the radical, hence a fixed vector. Thus both occur; duality supplies both reverse arrows. The extension $\ker(f_Q : \Gamma \to K)/K\omega$ of W by H cannot split: otherwise W maps to $T/K\omega$, but $\mathrm{Hom}_G(W, T) = 0$ by Lemma 6.3 and $\mathrm{Ext}^1_G(W, K) = H^1(G, W^*) = 0$ by Lemma 4.14. This gives both H, W arrows. The displayed products and nilpotence now give the full radical.

For odd m the primitive-right-ideal dual dimensions are $2(h-1)+n+1, h-1+n, h$. The stated quotients have these dimensions and simple socles H, W, K, read directly from the five-coordinate list. For even m the dimensions are $2l+n+3, l+n+1, 3l+n+5$. The radical annihilator in $T/K\omega$ is H_b; in $T/(A+KX)$ its surviving coordinates H_t, W, Y have annihilator W. In the last quotient use first-copy coordinates H_t, Y and the whole second copy; first-copy a equals second-copy ω. The arrows a_3, b_1 eliminate the first-copy coordinates in a socle vector, and the remaining arrows eliminate all second-copy coordinates except ω. Its socle is therefore K. Every simple socle is essential; mapping into its injective envelope and comparing these dimensions proves the claimed identifications.

Finally the square has only the nontrivial algebraic factors $L(\omega_2)$ and $L(2\omega_1)$. Its coefficient-Frobenius conjugates have factors $L(2^i\omega_2)$ and $L(2^{i+1}\omega_1)$ with indices modulo $\log_2 |K|$, pairwise distinct by [Ste67, Theorem 43]. In nonsplit type the graph automorphism fixes both displayed fundamental weights; the same period applies. Lemma 4.6 now proves the prime-field equality including the radical. □

4.7 Adjoint chains and odd-characteristic image algebras

Proposition 4.18 (The joint chain algebra)**.** *In odd characteristic the only nonsemisimple same-factor components of a joint sum of words of length at most two are the following modules M, when p divides their natural dimension n:*

Group	M	*Defining field of* M
$\mathrm{SL}_n(K)$	$\mathrm{End}_K(V)$	K
$\mathrm{SU}_n(K/K_0)$	*Hermitian matrices*	K_0
$\mathrm{O}_n(K)$	*form-self-adjoint matrices*	K
$\mathrm{Sp}_n(K)$	*form-self-adjoint matrices*	K

They have the nonsplit chain $K_iI < M_{i,0} < M_i$, where $M_{i,0}$ is trace zero and $H_i = M_{i,0}/K_iI$ is simple. Its dimension is n^2-2, n^2-2, $n(n+1)/2-2$, or $n(n-1)/2-2$, respectively. If $p \nmid n$ the scalar line splits and the component is semisimple. In characteristic two the general-linear and unitary adjoint components have exactly the same description, with splitting for odd n. After identifying all isomorphic whole occurrences, the prime-field joint algebra on the nonsemisimple components is

$$kI + \bigoplus_i R_i, \qquad R_i = \left\{ \begin{pmatrix} 0 & b & c \\ 0 & D & e \\ 0 & 0 & 0 \end{pmatrix} : b, D, e, c \text{ arbitrary over } K_i \right\}.$$

The remaining nontrivial simple types give separate semisimple image factors. No independent scalar character is added for a repeated occurrence or for a second classical factor.

Proof. The degree-two constituents can be checked from the defining representations: natural squares have highest weights $2\omega_1$ and ω_2, adjoints have $\omega_1 + \omega_{n-1}$, and orthogonal and symplectic form-self-adjoint spaces have their trivial contraction and the corresponding traceless heart. The defining-characteristic square decompositions, including the single possible scalar radical, are the low-degree cases of [GT05, Section 3.2 and the proofs of Theorems 2.9 and 2.11]. All unequal digit products are simple by [Ste67, Theorems 41 and 43]. For odd p the same-digit weights are restricted even at $p = 3$. Unitary duality identifies V^* with $V^{[f]}$ when $|K_0| = p^f$; it is this fixed-field parameter, rather than $|K|$, that determines the signature. For binary adjoints the matrix commutator description of trace zero gives the same scalar and adjoint-heart chain.

The trace pairing on each M_i is nondegenerate and invariant, and its fixed space is K_iI. If the heart split off as a submodule, its dimension greater than half of $\dim M_i$ would make the restricted pairing nondegenerate (its radical is an invariant submodule of a simple module). Its two-dimensional orthogonal complement would have only trivial composition factors. On a perfect derived group its upper-unitriangular image is trivial; on the full group the remaining quotient has order prime to p, so the image is again trivial. This contradicts the one-dimensional fixed space. Thus both adjacent extensions are nonsplit, the second by duality. When $p \nmid n$, trace splits the scalar line instead.

We check the coefficient fields, including the binary case. In odd characteristic, use the diagonal matrix with entries $s, r, 1, \dots, 1$, where $r = \pm 1$. The difference of adjoint traces is $2(s + s^{-1}) + 4(n-2)$. For a unitary adjoint take norm-one s; the same expression is in K_0. In the orthogonal and symplectic cases use two hyperbolic pairs with eigenvalues s, s^{-1}, r, r^{-1}; the difference on the relevant square component is $4(s + s^{-1} + n - 4)$. Removing trivial factors does not change these differences. The set of $s+s^{-1}$ has at least $(|K|-1)/2$ values, or at least $(|K_0|+1)/2$ in the norm-one

case. These sets exceed every proper subfield when a proper subfield is possible. Thus ordinary traces generate the required defining field. For binary general linear groups use

$$C(a,b)=\begin{pmatrix}0&0&1\\1&0&b\\0&1&a\end{pmatrix},\qquad C(a,b)^{-1}=\begin{pmatrix}b&1&0\\a&0&1\\1&0&0\end{pmatrix}.$$

Both determinant and the displayed inverse are checked by multiplication. For $\epsilon=(n-3) \bmod 2$, let $g(a)=\mathrm{diag}(C(a,1+\epsilon),I_{n-3})$. Its determinant is one, its two traces are $a+\epsilon$ and 1, and its adjoint-heart trace is $a+1$. These run over all of K. For binary unitary adjoints use $\mathrm{diag}(t,t^{-1},1,\dots,1)$ with norm-one t: the heart trace, up to a prime-field constant, is $(n+t+t^{-1})^2$. The set of $t+t^{-1}$ has $|K_0|/2+1$ values, more than a proper subfield. The case $K_0=\mathbb{F}_2$ is immediate. An absolutely irreducible finite-field image is a full matrix algebra over its coefficient subfield; these trace calculations force that subfield to be exactly K_i.

Now choose a separable semisimple complement in the image. The nonsplit bottom extension forces a nonzero radical row from the heart to the scalar vertex; the full heart matrix algebra makes that row all of $K_i^{1\times h_i}$. The top extension similarly forces the entire column, and their products force every scalar top-to-bottom entry. The two endpoint diagonal entries are the same element of k, since group elements fix both pointwise. This proves the displayed single-chain algebra. For separate chains their only common absolutely simple type is trivial, after the stated identifications. The annihilator-product argument of Lemma 4.6 therefore separates their augmentation ideals, even for two chains of the same acting group. Applying it jointly to coefficient-Frobenius conjugates and taking the coefficient-Frobenius sum gives the whole defining-field ideals over k. They have zero cross-products on the joint sum, with just one common identity. Distinct nontrivial semisimple types separate by the two-sided Chinese remainder argument in Lemma 4.9. Field-conjugate, dual, and repeated whole occurrences are identified before this step; they repeat the same algebra action. This proves joint equality, including the radical. □

Lemma 4.19 (Injectives for the joint chain algebra)**.** *For the algebra of Proposition 4.18, the heart-type injectives are M_i/K_iI; its single trivial-type injective is*

$$\left(\bigoplus_i M_i\right)/P_*,\qquad P_*=\{(c_iI)_i:\sum_i \mathrm{Tr}_{K_i/k}(c_i)=0\}.$$

Each embeds in a whole matrix module modulo a fixed scalar subspace. Consequently every invariant section of length-two tensor words embeds in such a module, with polynomial auxiliary dimension.

Proof. The right ideal at the common scalar vertex consists of its common k-scalar, all heart rows, and all top-to-bottom coordinates. Pair it with $(v_i)_i$ by summing the field traces of the first coordinates of av_i. The full row and scalar entries detect all heart and top coordinates; the common scalar detects precisely the one remaining scalar quotient. Its annihilator is P_*. This identifies its dual with the first displayed quotient. The same pairing for a primitive heart-coordinate right ideal gives M_i/K_iI. Nondegeneracy of the field trace is sufficient; no assumption that $\mathrm{Tr}(1)\neq 0$ is made. Each M_i is a submodule of a whole natural matrix module with its scalar line fixed. Embed the remaining semisimple factors in their original whole tensor summands and apply Lemma 4.1. All embeddings preserve the actual numerator and denominator. □

Theorem 4.20 (Whole Hom modules in odd characteristic)**.** *Let G be a product of the stable natural derived classical groups in odd characteristic. Let B be a direct sum of whole Hom spaces between*

natural, dual, Frobenius-natural, and trivial modules. For every invariant prime-field subspace $P \le B$,

$$H^1(G, B) = H^1(G, B/P) = 0.$$

This does not assert vanishing for an arbitrary subquotient U/P *with* $U \neq B$.

Proof. General-linear whole words follow from Lemma 4.4. For symplectic and split even orthogonal groups use the parabolic with Levi $\mathrm{GL}_m(K)$ and additive symmetric or alternating radical, respectively. Its index is prime to p. Normalize a cocycle on the Levi. Its first radical grade is an additive equivariant map from the parameter module; finite four-factor matching leaves only the top map $Z \mapsto aZ^{[i]}$ when the two signatures match, and none otherwise. This value is the coboundary of the first-half projector. No first grade remains, so restriction to the parabolic is zero. For even unitary groups the parameter is antihermitian and its prime-field components are $E^{[s]} \otimes E^{[s+f]}$; the identical matching calculation gives the same conclusion. Here one may first use the full unitary parabolic. The support proof of Lemma 4.12 works in odd characteristic as well: coordinate unitary two- and three-spaces generate, and $|T|$ is prime to p. In the vector elimination proof, if all pair norm sums were zero, three nonzero coordinates would force $2a = 0$ for a nonzero norm a, impossible in odd characteristic. Thus the same conclusion holds on the special group itself.

For the remaining dimensions choose an even integer m maximal at most the Witt index and write $V = E \oplus E^{\dagger,*} \oplus Z$, $\dim Z \le 4$. The split even derived group on the first two spaces contains both its central $-I$ and a half-swap: even m gives determinant one and, in the orthogonal case, square spinor norm. Its full matrix blocks have the already proved vanishing, cross blocks have vanishing by the central involution, and tail blocks by perfectness. Normalize there. The upper radical has matrices

$$u(X, B) = \begin{pmatrix} I & B - \frac{1}{2} X H_Z^{-1} X^\dagger & X \\ 0 & I & 0 \\ 0 & -H_Z^{-1} X^\dagger & I \end{pmatrix}, \qquad B^\dagger = -B.$$

The center $u(0, B)$ belongs to the normalized even group. The cocycle commuting identities with this center force blocks $(2,1), (2,3), (3,1)$ to zero. The remaining diagonal blocks are additive in X and vanish under the central sign, since the parameter is odd and the target even in natural degree. Natural intertwiner matching leaves, for source and target twists i, j, only $X_j D$ and $E X_i^\dagger$ in positions $(1,3), (3,2)$. Comparing two upper multiplication orders gives

$$X_j (E - D H_i^{-1}) Y_i^\dagger = Y_j (E - D H_i^{-1}) X_i^\dagger.$$

Different head rows and arbitrary tail coordinates force $E = D H_i^{-1}$. The remaining top block is the corresponding quadratic block $X_j D H_i^{-1} X_i^\dagger$; after subtraction its additive remainder is again excluded by the central sign. Direct multiplication shows that these are exactly the coboundary of the constant tail matrix D. It is fixed by the even group, so subtraction preserves normalization. The half-swap handles the opposite radical, and these radicals and the even group generate the full derived group. For a natural coefficient the same equations leave $X_i v_0$, the coboundary of the tail vector. Duals follow by the form identifications. This proves every whole Hom case.

For distinct independent factors the first-cohomology product formula gives zero on $X \boxtimes Y$, since each nontrivial factor has zero fixed space. Trivial coefficients are handled by Lemma 4.5. It remains to justify the quotient assertion. Use the full chain algebra of Proposition 4.18, valid on the derived group by Lemma 4.7. For each heart type its idempotent isolates $H_i \otimes U_i$, the heart projection of P in its multiplicity space. The full row maps put the corresponding bottom

coordinates in P, and the idempotent itself puts the entire heart projection in P. Quotient these full radical submodules first. Their copies become trivial tops; the other copies remain whole chains. The residual denominator has zero heart projection, and the independent column maps force its remaining top-chain coordinates to vanish. It is therefore entirely trivial. The resulting numerator is a sum of whole chains, semisimple direct summands of whole Hom modules, and trivial modules, each with zero first cohomology. Quotienting the final trivial denominator preserves this by Lemma 4.5. All idempotents act on the actual P, not on separately enlarged projections. Scalar extension and descent of the finite cochain equations permit arbitrary prime-field-linear P. □

4.8 One common field phase

Theorem 4.21 (Compatible finite-field quotients)**.** *Let K/k be finite, $\dim_K V = a$, and let $M : C \to \mathrm{End}_k(V)^s$ be k-linear. For every subset $S \subseteq C$ of size at most three set $B_S = \mathrm{alg}_k(K, I, M_j(x) : x \in S, 1 \le j \le s)$. Suppose B_S has a field quotient $L_S = B_S/P_S$ containing the specified K, and $V_S = \mathrm{Ann}_V(P_S)$ has K-dimension at least $a - r > a/2$. There is a common field F/K and compatible embeddings of all L_S into F, unique up to one common K-Frobenius, with*

$$[F : K] \le \mathrm{lcm}(1, \dots, a) = \exp(O(a)).$$

The quotient-value map $\lambda : C \to F^s$ is k-linear and

$$\lambda(x) = 0 \quad \Longleftrightarrow \quad \mathrm{rank}_k[v \mapsto (M_j(x)v)_j] \le [K : k]r.$$

All fields, embeddings, and phases can be constructed in $|k|^{O(\dim_k C + [K:k]a)}\,\mathrm{poly}(s)$ time without enumerating F. The construction transports equivariantly under conjugation and actual changes of the coordinate space indexing the tuple.

Proof. For $S \subseteq T$, restriction of L_T to B_S is a field image, so V_T is semisimple of one B_S-type. It must be the L_S type: otherwise its intersection with V_S is zero, contrary to their dimensions. Hence $P_T \cap B_S = P_S$ and there is an actual inclusion $L_S \hookrightarrow L_T$. Singleton degrees $e_x = [L_x : K]$ are at most a. Since pair and triple algebras are generated by their singleton algebras, their field quotients are the corresponding composita; pair degree is $\mathrm{lcm}(e_x, e_y)$.

Put $E = \mathrm{lcm}_x\, e_x$ and fix a field F/K of degree E. Arbitrary singleton and pair embeddings have discrepancies $d_{xy} \pmod{\gcd(e_x, e_y)}$. Triple compatibility implies

$$d_{xy} + d_{yz} = d_{xz} \pmod{\gcd(e_x, e_y, e_z)}.$$

For each integer prime $\pi \mid E$ choose x_π maximizing $v_\pi(e_x)$, and prescribe $s_x = d_{x,x_\pi} \pmod{\pi^{v_\pi(e_x)}}$. The triangle equation gives all pair congruences at that prime; integer Chinese remaindering gives s_x modulo e_x. Twist the singleton embeddings by these phases. They now agree on every intersection and extend uniquely to the pair and triple fields. Two solutions differ, prime by prime, by one phase on the largest prime-power degree, hence by one phase modulo E.

For $z = x + y$ the triple algebra equals the pair algebra, and $M_j(z) = M_j(x) + M_j(y)$ there. Applying its common field quotient proves additivity; the scalar version proves k-linearity. If all quotient values vanish, all $M_j(x)$ kill the same V_x, so their simultaneous stack has rank at most $[K : k]r$. If one quotient value is nonzero, that operator is invertible on V_x and has rank at least $[K : k](a - r) > [K : k]r$. This proves the exact assertion for the full stack, independent of its number of labels.

For completeness, the field construction also obeys the bound. The elementary recurrence $\log \mathrm{lcm}(1, \dots, a) \le O(a) + \log \mathrm{lcm}(1, \dots, \lceil a/2 \rceil)$ follows by the central binomial coefficient, giving

$E = \exp(O(a))$. For each maximal prime-power divisor d_π of E, $d_\pi \le a$; construct a degree-d_π extension by enumerating its at most $|K|^{d_\pi}$ monic polynomials and testing irreducibility. The tensor product over K of these extensions is a field because their degrees are pairwise coprime. Its degree is E, and its arithmetic is polynomial in E; no degree-E polynomial search is needed. A required small degree-e subfield is $\ker(\mathrm{Frob}_K^e - I)$ inside F. Enumerating that subfield, of size $|K|^e \le |K|^a$, finds a root of each small input polynomial. There are at most $|k|^{3 \dim C}$ subsets to process and all represented algebras act on the original V.

Conjugation transports every actual algebra and its unique large-dimensional field-quotient subspace, hence all embeddings; the uniqueness just proved leaves one common phase. A coordinate change in tuple labels changes the spanning set of the same complete operator family. If labels are indexed by L^*, the value is the single element of $\mathrm{Hom}_k(L^*, F)$, rather than independent scalar values. If K was noncentral in B_S, decompose each operator into its K/k-semilinear components. On V_S its image belongs to a field commuting with K; independence of field embeddings forces every nonidentity component to vanish there. Retaining their untwisted maps gives an augmented K-linear stack annihilating the same V_S, with the same field values on its identity component. Its low-rank alternative has the same bound $[K : k]r$ and the same zero-field-value locus in C; equality of the kernels of the two stacks as maps on V is neither asserted nor needed. All original semilinear row actions are retained. □

4.9 Joint injectives and complete affine transport

Proposition 4.22 (The binary joint image and its cogenerator)**.** *For a product of stable binary classical groups, the image algebra on any joint sum of words of length at most two is obtained from Lemmas 4.8, 4.9, 4.11, Proposition 4.17, and the binary adjoint cases of Proposition 4.18, by identifying whole isomorphic occurrences, retaining the induced action on every natural simple module already occurring as a square subquotient, adding only composition-disjoint nontrivial semisimple factors, and identifying the single trivial character. Its injective cogenerator has polynomial dimension in d, with the explicit quotient realizations given above.*

Proof. Over a splitting field a length-two word is either a same-factor square, a same-factor adjoint, an unequal-digit simple tensor, or an external simple tensor of two independent factors. The unequal and external simple cases follow from [Ste67, Theorems 41 and 43]; scalar restriction preserves semisimplicity since extension is separable. In unitary type the relative exponents are taken modulo $2f$, with adjoint position f and ordinary square position zero. At $f = 1$ the Frobenius-natural constituent of the square is the dual natural module, so those occurrences are the same algebra module. All common twists are identified as whole prime-field modules, not as independent factors. In particular, adjoining a natural occurrence whose simple type already occurs in a square adds no new algebra factor: its action is the action induced on that square subquotient. Indeed the annihilator of the square annihilates every such simple module, so projection from the joint image onto the square image is injective as well as surjective. For symplectic and orthogonal groups polar duality makes the corresponding dual identifications. Independent factors are formed only after any diagonal identifications of the acting simple groups have been made.

Disjoint composition-type sets give independent image factors by the annihilator/Chinese remainder proof already used above. For each factor possessing the trivial character the algebra is $kI + J_i$, with J_i its full defining-field augmentation ideal. On the joint sum the exact algebra is $kI + \bigoplus_i J_i$. This remains true for different square and adjoint blocks of the same group, where a single group element need not have supported action. To prove it, use the perfect product group algebra and its augmentation ideal Δ. The annihilators of two blocks sum to Δ, because their

only common simple factor is trivial; $\Delta = \Delta^2$ kills every iterated trivial extension. The ordered product of these annihilator sums, as in Lemma 4.6, constructs an element of Δ with any specified augmentation-ideal action on one block and zero on the others. Apply the same argument to all coefficient-Frobenius conjugates and take their Frobenius sum, as in that lemma, to obtain the specified defining-field ideal action from the prime-field group algebra. Thus every supported J_i occurs, their products on different blocks vanish, and all scalar coordinates share one character. At a nontrivial primitive vertex its right ideal is unchanged by prime-field restriction; its dual is the same defining-field injective by the trace pairing. At the trivial vertex, dualizing the common-diagonal right ideal gives

$$I_{\mathrm{triv}} = \left(\bigoplus_i I_{0,i} \right) / \{ (c_i s_i)_i : \sum_i \mathrm{Tr}_{K_i/k}(c_i) = 0 \},$$

where s_i spans the scalar socle of $I_{0,i}$. This is one correlated denominator. Separate trace-zero quotients would give a different module and are not used. The semisimple factors embed by stacking all Hom maps into their actual original tensor summands. This handles smaller endomorphism fields and field-conjugate identifications without constructing or enumerating a compositum. The listed modules have dimension $O(d^2)$ up to a fixed multiplicity for each actual type, and the Hom construction of Lemma 4.1 is polynomial. □

Lemma 4.23 (The orthogonal observation maps)**.** *Every orthogonal injective has an equivariant map into one or two copies of $I_H = T/K\omega$ with a natural or trivial kernel:*

$$I_H \xrightarrow{1} I_H, \qquad I_W \longrightarrow I_H,\ [Y] \longmapsto xY + K\omega,$$

$$I_0 \longrightarrow I_H,\ [Y] \longmapsto xY + K\omega \quad (m \text{ odd}),$$

and

$$I_0 \longrightarrow I_H^2,\ [(Y,Z)] \longmapsto (xY + K\omega, Z + K\omega) \quad (m \text{ even}).$$

The second kernel is $\Gamma/A \cong V^{[1]}$; the last two kernels are a scalar line. These maps kill the full scalar socle whenever one is present.

Proof. All denominators map to zero by their displayed definitions. If $xY = c\omega$, subtract cX to place Y in Γ. For I_W the remaining denominator is exactly A, giving Γ/A. For odd I_0 it is $\ker f_Q$, giving a trivial one-dimensional quotient. For even I_0, if also $Z = b\omega$, subtract $(cX, 0)$ and then $(Y, f_Q(Y)\omega)$; only a scalar multiple of $(0,\omega)$ remains. This class is nonzero because a defining relation with zero first coordinate has $s = cX \in \Gamma$, forcing $c = s = 0$. □

Theorem 4.24 (Affine transport under the full classical product)**.** *Let D be a recognized actual product, or an explicitly represented central-product image, of groups $\mathrm{SL}_n(K)$, $\mathrm{SU}_n(K/K_0)$, $\mathrm{Sp}_{2m}(K)$, and $\Omega_n(K)$ in the stable ranges above, over finite fields of characteristic p, with total faithful natural prime-field dimension d. Let $P \le U$ be an actual invariant prime-field pair in a joint direct sum of tensor words of length at most two, with fixed multiplicity spaces. Given a cocycle $z : D \to U/P$ by its values and original words on generators, and $x, y \in U/P$, one can compute its complete affine transporter in $p^{O(d)}\,\mathrm{poly}(L, \log p)$ time. The output consists of one original witness and generators of its full source stabilizer, or emptiness. All supplied original kernels and lifts are retained.*

Proof. For odd p, Lemma 4.19 embeds the direction in a sum of whole matrix modules modulo fixed scalar subspaces. Their first cohomology is zero by Theorem 4.20 (a fixed scalar quotient is a special case). Find an origin there by generator linear equations. Thus one simultaneous linear transporter in these templates suffices.

For $p = 2$ use Proposition 4.22 and Lemma 4.1. On each nontrivial injective use identity if acyclic, and on the orthogonal injectives use Lemma 4.23. A common trivial-type injective needs care. On every nonorthogonal summand $I_{0,i}$, first quotient by its whole defining-field scalar socle, and embed that quotient once in its family's acyclic cogenerator. Its kernel is exactly the old socle. On the orthogonal summands use the maps just displayed. All maps now kill the entire sum of the original scalar socles, so descend through the one correlated trace denominator. Their total target E is a direct sum of acyclic modules, and their kernel is a section of natural, dual-natural, Frobenius-natural, and trivial rows. No new correlated quotient is introduced in E. The product cohomology formula and perfectness show $H^1(D, E) = 0$; for an external simple $X \boxtimes Y$, both terms $H^1(D_1, X) \otimes Y^{D_2}$ and $X^{D_1} \otimes H^1(D_2, Y)$ vanish. Lemma 4.2 now reduces the problem to a complete linear transporter in E and a degree-one affine problem under its actual returned stabilizer. The latter is exactly Theorem 3.16, with no renewed full-group premise.

We specify the simultaneous linear transporter to avoid a hidden intersection operation. Pair natural and dual rows when necessary so that the product of ambient full classical groups is the unitary group of one represented semisimple algebra with involution. Keep all original field-linearity constraints, baseline forms, and quadratic refinements. Whole bilinear tensors, external tensors, and fixed Frobenius signatures give simultaneous literal equations on these rows. For a coefficient block with signatures σ, τ, use the $\mathbb{F}_p$-algebra representations $a \mapsto a^{[\sigma]}$ and $a \mapsto a^{[\tau]}$ for its left and right actions. It is thereby a genuine bimodule for the same original-degree algebra; no separate faithful natural row is adjoined for each Frobenius signature. Every $T/K\omega$ or adjoint scalar quotient gives matrix equations modulo scalar. Collect all their shifts at once using Lemma 4.3, and filter the full tuples by the actual correlated scalar subspace. For a fixed tuple solve all matrix and bilinear equations at once by Lemma 3.20 and Theorem 3.19. Quadratic equations are included through their polars, followed by the complete degree-one zero-polar correction. This is Corollary 3.22, under the actual returned subgroup.

For clarity the remaining template equations are only additive. In a symplectic odd-m I_0 coordinate, first fix the quadratic class of Y; then impose $\ell(gY - Y') = 0$. In even-m I_0 coordinates (Y, Z), first fix the class of Y and the scalar shift of Z; then impose that shift equal to $\ell(gY - Y')$. For an orthogonal I_W, the scalar shift of its polar determines the same scalar multiple of Q in the quadratic equation, with Q itself included among the fixed observations. For odd orthogonal I_0 impose $x(gY - Y') = c\omega$ followed by $f_Q(gY - Y' - cX) = 0$. For even orthogonal I_0 impose $x(gY - Y') = c\omega$, $gZ - Z' = b\omega$, followed by $b = f_Q(gY - Y' - cX)$. After the preceding literal equations are fixed, each residual map on the stabilizer is additive, since ℓ or f_Q is invariant and its argument lies in A or Γ. All scalar and trace equations are imposed jointly by Lemma 3.1.

Finally return from the ambient full classical groups to the specified actual derived factors. For SL impose determinant one; for SU impose unitary determinant one; no additional character is needed for Sp. For odd-characteristic Ω impose both determinant one and spinor norm one. For binary Ω impose Dickson invariant zero. These are exactly the natural derived images, by [Tay92, Theorem 11.51]; in particular SO alone is not substituted for odd Ω.

Here is an explicit spinor computation and complete-coset filter. For an isometry g in odd characteristic, choose a basis x_i of $\mathrm{im}(1 - g)$ and solve $(1 - g)v_i = x_i$. The Wall form has matrix $W_g = (b(v_i, x_j))_{ij}$. It is well defined: adding a fixed vector to v_i changes no entry, since $\mathrm{im}(1 - g)$ is orthogonal to the fixed space. The chosen v_i, together with the fixed space, span the natural space. Hence its right radical would be orthogonal to the whole natural space and is zero. Moreover $b(x_i, v_i) = Q(x_i)$, by expanding $Q(v_i - gv_i)$. Its determinant modulo squares is the spinor norm [Tay92, Theorem 11.50 and its preceding definition]. For $g = 1$ take the empty determinant to be one. Gaussian elimination and the finite-field square test compute this character in polynomial time. Determinants and binary Dickson rank parity have the same bound.

Let χ be the product of these characters on the actual ambient factor representation. Its image has size $p^{O(d)}$: the determinant groups have order at most the product of the original field sizes, and each orthogonal factor adds at most four character values. For a complete ambient transporter aS, enumerate $\chi(S)$ while retaining words. The intersection with $D = \ker \chi$ is empty exactly when $\chi(a)^{-1} \notin \chi(S)$. Otherwise choose $s_0 \in S$ with $\chi(s_0) = \chi(a)^{-1}$ and return $as_0 \ker(\chi|_S)$. Schreier generators give the full kernel, not just a determinant-one witness. Thus every allowed element in this record is retained. For a prescribed finite extension of the product, first enumerate its actual outside cosets and apply this procedure in each normalized product record, then reunite them; one must not silently replace that extension by its derived subgroup.

This filtering is done *before* evaluating a cocycle supplied only on D. All preceding ambient equations concern translated linear observations or their tensor-derived quadratic corrections; they do not evaluate that original cocycle. An ambient orthogonal witness with a forbidden spinor or Dickson value is not an argument of the D-cocycle. For a central-product image carry out this test on its supplied product cover and project with the full central kernel retained; no choice of a section of the cover is used. For each shift retain its whole stabilizer. If a_0 is one successful witness and a_jS_j are all successful cosets, the full source stabilizer is generated by every S_j and $a_0^{-1}a_j$. Every generator fixes the source, and every original transporter belongs to the union, proving equality of the returned set.

All coefficient dimensions remain polynomial; the exponent uses only a fixed number of copies of the original natural rows. There are at most $4^{O(d)}$ shift tuples and $p^{O(d)}$ finite quotient records, all completed and reunited inside this call. No such list is passed to a further recursive call. Matrix-span and Hom calculations enumerate no coefficient vector set. For a represented central quotient, pull back the cocycle and keep its kernel; its kernel values are zero, so the constructed translated observations are fixed by that kernel. Projection and exact kernel restoration therefore preserve the entire transporter. □

5 Affine transport on tensor subquotients

This section gives the recursive passage from the classical-group algorithms of Section 4 to algorithms for a supplied group. The linear spaces on which the input groups are represented must be distinguished from spaces of tensor coefficients. Only the former occur in the exponent. All vector spaces in this section are over $\mathbb{F}_p$, unless another field is specified.

5.1 Inputs, original groups, and exact affine equations

Fix a constant r. A tensor presentation consists of specified representation spaces $V_1, \dots, V_s$, of total dimension at most d, and a direct sum Z of tensor products with at most r occurrences of these spaces, their duals, or specified Frobenius twists. Each summand may have an additional trivial multiplicity space. Repeated occurrences have the same prescribed action; writing them separately does not permit independent changes of basis. Extension-field balancing, symmetry, and other linear identifications are imposed by invariant subspaces of Z. An input section is

$$Q = N/P, \qquad P \le N \le Z.$$

There is one denominator P, which can correlate different tensor summands. The degree of a summand is its number of nontrivial representation-space occurrences, not its coefficient dimension.

The group at the initial call is denoted by Γ. We retain a faithful comparison representation of Γ throughout the computation. This may combine its given matrices with an explicitly supplied

permutation action. It may also have a specified elementary abelian normal subgroup represented symbolically by a vector space. Every group element subsequently constructed is represented by a word in original generators, stored as a directed acyclic graph. A current group is always an actual subgroup $H \le \Gamma$; changing a presentation or adjoining an auxiliary vector group does not change this meaning of H.

The affine action and the requested transporter have the form

$$\alpha_h(q) = L_h q + c_h, \qquad c_{gh} = c_g + L_g c_h, \qquad \mathrm{Trans}_H(x,y) = \{h \in H : \alpha_h(x) = y\}. \tag{5.1}$$

The cocycle is given in N/P and need not lift to a cocycle in N. It suffices to specify it on generators with the promise, or a certificate, that it is a cocycle for the supplied group. The action on N and P is induced by the specified tensor presentation. If nonempty, the answer is an element a and generators of the entire subgroup $S = \mathrm{Stab}_H(x)$, with $\mathrm{Trans}_H(x,y) = aS$. This includes elements acting trivially on some or all of the displayed representation spaces.

Every intermediate problem has an original prefix tH and an explicit comparison of its affine equation with the input equation. Its invariant is an equality of sets of original elements:

$$\{\text{input solutions remaining in this call}\} = t\,\mathrm{Trans}_H(x,y). \tag{5.2}$$

In particular, an isomorphism of two quotient vector spaces is useful only when the action, both points, and this comparison are transported with it.

We allow scalar and semilinear actions only when their support is specified. Each scalar or semilinear line action is either induced by a counted representation space, or acts on a specified one-dimensional space over a field K_j counted in d. For the latter, independently counted lines, $\sum_j [K_j : \mathbb{F}_p] \le O_r(d)$. Repeated copies use the same representation and do not introduce independent scalar or field actions. A passive multiplicity space has trivial action and introduces no new unknown scalar or field automorphism. These conventions exclude an unrepresented moving group on a large coefficient-label space.

If $d = 0$, the linear action on the presentation is trivial. Its cocycle is an additive homomorphism, so image-basis computation and Lemma 3.1 solve the equation directly, without an enumeration depending on p. We may therefore assume $d \ge 1$ in the recursion and its estimates.

Let L denote the entire explicit input length, including matrices, linear maps, subspace bases, field descriptions, the external permutation action, and any symbolic normal subgroup present at the initial call. All claims below are uniform in p. The factor $\mathrm{poly}(L, \log p)$ is permitted to depend on the fixed r, but its exponent does not depend on d.

5.2 Exact quotients, fibers, and changes of presentation

Lemma 5.1 (An affine quotient and its fiber)**.** *Let $U \le Q$ be invariant under the linear part of* (5.1)*. Suppose the complete transporter on Q/U is aK. Put*

$$y_U = \alpha_{a^{-1}}(y) - x \in U, \qquad \beta_h(u) = L_h u + \alpha_h(x) - x \quad (h \in K). \tag{5.3}$$

Then β is an affine action of K on U, and

$$\mathrm{Trans}_H(x,y) = a\,\mathrm{Trans}_{K,\beta}(0, y_U).$$

A complete answer to the latter problem therefore gives the complete answer to the former, with its full original source stabilizer.

Proof. The definition of K says that $\alpha_h(x) - x \in U$ for every $h \in K$. Furthermore,

$$\alpha_{gh}(x) - x = \alpha_g(x) - x + L_g(\alpha_h(x) - x),$$

so these increments form a K-cocycle. The quotient equation puts y_U in U. For $h \in K$, the identity $\alpha_{ah} = \alpha_a \alpha_h$ gives

$$\alpha_{ah}(x) = y \quad \Longleftrightarrow \quad \alpha_h(x) - x = \alpha_{a^{-1}}(y) - x.$$

Every element fixing x belongs to K, and within K it fixes 0 under β precisely when it fixes x. This proves both assertions, without choosing a complement to U. □

Lemma 5.2 (The kernel of an equivariant map)**.** *Let $e : Z \longrightarrow Z'$ be an equivariant linear map. Set $N_e = N \cap \ker e$ and $P_e = P \cap \ker e$. The map*

$$N_e/P_e \longrightarrow N/P, \qquad n + P_e \longmapsto n + P$$

is an isomorphism onto the kernel of $N/P \longrightarrow e(N)/e(P)$. It is computable by linear algebra.

Proof. The map is injective. A class $n + P$ lies in the displayed kernel exactly when $e(n) \in e(P)$. Choose $a \in P$ with $e(a) = e(n)$. Then $n - a \in N_e$ represents the class; a second such choice differs by an element of P_e. Solving for a is a linear system. This also specifies how to transfer every cocycle increment and every target to the kernel after applying Lemma 5.1. An uncorrected representative in N must not be used as if it belonged to N_e. □

Lemma 5.3 (Complete union of transporter cosets)**.** *Suppose a finite exhaustive search partitions a transporter into nonempty answers $g_j S_j$, each with its entire source stabilizer inside the corresponding search subgroup. Choose one witness g_0. Then the complete transporter is $g_0 S$, where*

$$S = \langle S_j,\ g_0^{-1} g_j : \textit{all successful } j \rangle.$$

Proof. Each displayed generator fixes the same source: this is immediate for S_j, and follows from $g_0 x = g_j x = y$ for $g_0^{-1} g_j$. Thus $g_0 S$ is contained in the transporter. Conversely, each $g_j S_j$ is contained in $g_0 S$. Exhaustiveness proves equality. The argument applies verbatim to affine actions and to a head quotient of an affine problem. □

Lemma 5.4 (Section presentations and all lifts)**.** *The following operations preserve* (5.2)*.*

(i) Replace N/P by an explicitly equivariantly isomorphic section and transport the affine action and the two points.

(ii) If $q : E \twoheadrightarrow Z$ is equivariant, replace N/P by $q^{-1}(N)/q^{-1}(P)$. The same assertion holds when E is an invariant subspace of a larger representation space.

(iii) For a quotient-valued cocycle $c : H \to N/P$, use the group

$$\widehat{H}_c = \{(h, a) : h \in H, a \in N, a + P = c_h\}, \qquad (g, a)(h, b) = (gh, a + L_g b). \tag{5.4}$$

Its projection onto H is surjective, with kernel P_{add}. No splitting of this extension is required.

Moreover, iterated auxiliary presentations can always be replaced by (5.4) *over the current original H.*

Proof. The first assertion is the definition of equivariant change of coordinates. For the second, the kernel of q is contained in $q^{-1}(P)$, and the first isomorphism theorem gives the asserted section isomorphism. Tensoring a factor surjection with identities is exact over a field, so the assertion also applies to factor covers and their tensor products.

For the third, closure under multiplication follows from the cocycle identity modulo P; associativity follows from the semidirect-product multiplication on $N_{\mathrm{add}} \rtimes H$. Every c_h has a representative, which proves surjectivity. Choose representatives $\tilde{x}, \tilde{y} \in N$. For $h \in H$ there is exactly one lift taking $\tilde{x}$ to $\tilde{y}$, namely

$$(h, \tilde{y} - L_h \tilde{x}).$$

It belongs to $\widehat{H}_c$ exactly when $\alpha_h(x) = y$. Thus the point transporter on representatives projects bijectively onto the quotient transporter; the corresponding point stabilizers are isomorphic. If instead the lifted action is solved on the quotient, its entire preimage, including P_{add}, must be retained before projection. These are two consistent implementations, and their kernels must not be confused.

For completeness, let $\pi : E' \twoheadrightarrow H$ be an older auxiliary presentation. The pullback of the new cocycle to E' has a lift group mapping onto (5.4) by

$$(e, a) \longmapsto (\pi(e), a).$$

Its kernel is exactly $\{(e, 0) : e \in \ker \pi\}$. All current predicates factor through this map. On an affine quotient this follows from the surjectivity of the quotient map; on the fiber in Lemma 5.1 it follows from the injectivity of the fiber inclusion and the formula there. Successive associated quotients are obtained by these two operations, so the assertion holds for them as well. An isomorphism of presentations preserves it by conjugation. Consequently the new cocycle descends to the actual original subgroup, and the older auxiliary kernel can be removed by this explicit projection. An actual representation cover whose kernel acts nontrivially on current coordinates is retained until its own equivariant comparison permits projection. □

Remark 5.5. Even very small examples require the nonsplit formulation. Over $\mathbb{F}_2$, let $H = \langle g \rangle \cong C_2$, let $N = \langle e_1, e_2 \rangle$, let $P = \langle e_1 \rangle$, and set $ge_2 = e_2 + e_1$, $ge_1 = e_1$, and $c_g = e_2 + P$. Every lift (g, a) squares to $(1, e_1)$, so $\widehat{H}_c \cong C_4$. The representative-transporter argument above still applies.

5.3 Evaluation modules and the induced filtration

In the recursive algorithm a representation-space occurrence is either kept in the value of a multilinear map or is evaluated on vectors. These are two uses of actual representation spaces, not replacements by unspecified composition factors. In a given position the evaluated spaces form a direct sum

$$R = \bigoplus_{a \in I} R_a.$$

The current group permutes the summands and supplies actual linear maps $R_a \to R_{ha}$. Individual summands need only be modules for their stabilizers; their direct sum is a module for the whole current group. We refer to I as its set of summand labels. When a covered factor occurs covariantly, evaluation uses the dual of its cover; this is simply the canonical identification of a tensor with a multilinear map.

The argument positions are the original at most r positions. Moving a factor to the arguments adds alternatives in a direct sum in its position; it does not append another tensor position. An isotypic space $X \otimes_K M$ remains one representation-space occurrence. Natural and multiplicity

coordinates are used inside a completed classical-group call and are not inserted as extra positions of the recursive presentation.

The full evaluation set is a disjoint union of tagged products of vector sets in the selected summands. A tag records both a summand and a vector, so different zero-vector tags are distinct. Tensor patterns with a value factor in position i use a single permanent symbol $\perp_i$ there. There is at most one such symbol in each position, and it is fixed by the group. A pattern exists only for its recorded tuple of selected labels and symbols. In particular, selecting a subset of the labels in position i also determines which tensor summands remain: a summand with a value factor in position i remains exactly on the side containing $\perp_i$. It is never copied to both sides. All patterns and all linear correlations among them are retained in a single numerator and denominator.

Write b_i for the number of currently selected labels in position i, including $\perp_i$ when it is present, and $b = \sum_i b_i$. The polynomial number of tensor patterns can equivalently be encoded as a direct sum of spaces $\mathrm{Hom}(W_\sigma, T_\sigma)$, with tagged evaluation domains indexed by σ. Linear operations below act on the whole direct sum. Thus it suffices to prove the following lemmas for one Hom space; applying their maps to the direct sum retains all correlations.

Lemma 5.6 (Evaluation-kernel saturation)**.** *Let $Z = \mathrm{Hom}(W, T)$ and let A be a finite H-set with an equivariant map $\epsilon : A \to W$. Put*

$$e_A(f)(a) = f(\epsilon(a)), \qquad K_A = \ker e_A.$$

Suppose that the section N/P is faithfully evaluated on A, meaning $N \cap K_A \le P$. Then

$$N/P \;\cong\; (N + K_A)/(P + K_A)$$

equivariantly. This change preserves the observed image on every subset $B \subseteq A$, whether or not B is invariant. If $T' \le T$, then

$$e_A^{-1}(\mathrm{Fun}(A, T')) = \mathrm{Hom}(W, T') + K_A. \tag{5.5}$$

Proof. The natural map is surjective and has kernel $(P + (N \cap K_A))/P = 0$. Since $K_A \le \ker e_B$ for $B \subseteq A$, adding K_A changes neither $e_B(N)$ nor $e_B(P)$.

Let $S = \mathrm{span}\, \epsilon(A)$. If $f(\epsilon(A)) \subseteq T'$, then $f(S) \subseteq T'$. Extend the linear map $f|_S : S \to T'$ to a linear map $f' : W \to T'$ by extending a basis of S. The difference $f - f'$ lies in K_A. This proves the nontrivial inclusion in (5.5); the other is immediate. The linear extension is not asserted to be equivariant. It proves an equality of subspaces, which are invariant whenever T' is invariant. □

Lemma 5.7 (Filtration of the current section)**.** *Under the hypotheses of Lemma 5.6, put $N^+ = N + K_A$ and $P^+ = P + K_A$. For an invariant flag $0 = T_0 \le \cdots \le T_m = T$, set $Z_i = \mathrm{Hom}(W, T_i)$ and $U_i = N^+ \cap Z_i$. The induced filtration of Q is*

$$Q_i = (P^+ + U_i)/P^+,$$

and its ith quotient has the exact presentation

$$Q_i/Q_{i-1} \cong \frac{U_i}{(P^+ \cap Z_i) + U_{i-1}} \cong \frac{q_i(U_i)}{q_i(P^+ \cap Z_i)}, \tag{5.6}$$

where $q_i : Z_i \to \mathrm{Hom}(W, T_i/T_{i-1})$ is the actual quotient map. This quotient is faithfully evaluated on A. The positive quotient dimensions sum to $\dim Q$.

Proof. The kernel of q_i is Z_{i-1}, whose intersection with U_i is U_{i-1}. Also

$$U_i \cap (P^+ + U_{i-1}) = (P^+ \cap Z_i) + U_{i-1}.$$

The two isomorphisms in (5.6) follow.

To check evaluation faithfulness, first suppose $u \in U_i$ evaluates to zero modulo T_{i-1}. Equation (5.5) gives $u = w + k$ with $w \in Z_{i-1}$ and $k \in K_A$. Because $u, k \in N^+$, we have $w \in U_{i-1}$; because $u, w \in Z_i$, $k \in P^+ \cap Z_i$. Thus the class of $q_i(u)$ is zero. If its evaluation lies in the evaluated denominator instead, first subtract a representative from $q_i(P^+ \cap Z_i)$ and apply the same argument. This proves faithfulness of the quotient, not merely injectivity on a chosen list of representatives. Dimension additivity follows from the filtration $0 = Q_0 \leq \cdots \leq Q_m = Q$. □

For several value factors, take the tensor-product filtration of their actual invariant flags, refined in a fixed order to successive tensor products of the actual quotients. A factor that is already used as an evaluation argument is unchanged. Equation (5.6) applies to the whole correlated section. This is a filtration of the current numerator and denominator; an older presentation of their composition factors cannot be substituted for it.

The affine processing order is determined by Lemma 5.1: solve the top quotient completely, then its exact affine fiber, and continue. The nominal associated quotients do not carry independent affine actions before these normalizations. When a fiber class represented by $n \in N^+$ belongs to $(P^+ + U_i)/P^+$, solve for $a \in P^+$ with $n - a \in U_i$. Two choices differ by $P^+ \cap Z_i$. This specifies the requisite representative correction at each filtration step.

Lemma 5.8 (Restriction to selected evaluation summands)**.** *Suppose $A = B \sqcup C$ is an invariant partition for the current subgroup and N/P is faithfully evaluated on A. After completing the transporter in $e_B(N)/e_B(P)$, use its exact fiber presentation*

$$N_B = N \cap \ker e_B, \qquad P_B = P \cap \ker e_B.$$

Let r_C be the equivariant tensor projection onto exactly the summands and patterns selected by C, so that $e_C r_C = e_C$. Put

$$U_C = r_C(N_B), \qquad D_C = r_C(P_B) + (U_C \cap \ker e_C). \tag{5.7}$$

Then $N_B/P_B \cong U_C/D_C$. Replacing the subsequent evaluation set by the full vector domains in these same selected summands preserves the entire original predicate.

Proof. The map $n + P_B \mapsto r_C(n) + D_C$ is well defined and surjective. If its value is zero, choose $a \in P_B$ for which $e_C(n-a) = 0$. Both n and a vanish under e_B, so $n-a$ vanishes on A. Faithfulness gives $n - a \in P$, and hence $n \in P_B$. The map is injective.

Full multilinear evaluation is injective on the tensor space: evaluate on tuples of basis vectors to recover every coefficient. It remains injective on a direct sum with separate tags. Therefore U_C/D_C is faithfully evaluated on the full domains of its selected summands. The isomorphism just proved transports the action, points, and original comparison. Future restrictions of the evaluation map may change, but the current predicate does not. The projection r_C first removes excluded pattern blocks, including every tensor summand with a value factor in position i when its symbol $\perp_i$ was excluded. Thus enlarging the vector domains inside surviving summands cannot restore a discarded summand or pattern. □

The image of the evaluation map on B is represented similarly by $r_B(N)/(r_B(P) + (r_B(N) \cap \ker e_B))$. On a full selected domain its evaluation is faithful by construction. Consequently both children of an evaluation split have presentations on exactly their selected summands. All these assertions concern actual subquotients and computed linear maps, not just their dimensions.

5.4 Moving complementary factors to evaluation arguments

Apply Theorem 3.11 only to the current value representation spaces after the filtration above. Use one fixed threshold

$$T = p^{C_* d}$$

chosen from the initial dimension, with $C_* \geq 128$ and larger if the structural constants require it. It is not decreased when the current value dimension decreases. We take the stable ranges of Section 4: natural dimension at least 64 for special linear and unitary factors, and Witt index at least 32 for symplectic and even orthogonal factors. In particular the nonsplit even orthogonal condition is a Witt-index condition. A fixed-rank exception has group order $p^{O(d)}$ with an absolute constant; choosing C_* above all these constants leaves it in the complementary part. The theorem gives invariant direct sums of spaces supporting the selected large classical components and complementary spaces. Its component links are the actual diagonal, dual, and Frobenius links. A whole isotypic summand is kept intact.

Each complementary space has an actual section presentation E/F in a module C. For a bounded-image space we take its actual representation. For a deleted-permutation representation we use the actual permutation cover of Lemma 3.7, including the action on its multiplicity space and the inter-block identifications. This is a section of a module, not an abstract replacement of a composition factor. The cover has dimension at most a fixed multiple of the original space; in the stable deleted-permutation case the multiple is at most 4/3.

Replace the factor by this section, tensor its surjection $E \to E/F$ with identities on the other factors, and pull back the entire numerator and denominator. In particular the new denominator includes the full kernel of this tensor map. Then regard the covered factor as an evaluation argument, using its dual if necessary. Lemma 5.4 proves exactness. The cover module is a direct sum of actual bounded-image summands, or of the coordinate summands of an actual permutation cover, with its full group action. Its summand labels are added in the same original position. Existing evaluation modules and their tags are never replaced by their composition factors or covered a second time.

Let v be the sum of the dimensions of spaces still used as value factors, counted with the fixed occurrence convention. Converting total dimension $c > 0$ gives, for a constant a_r,

$$v' = v - c, \qquad b' \leq b + a_r c. \tag{5.8}$$

Only a currently present alternative can be converted. A surviving value alternative retains its old $\perp_i$; if none survives, that symbol is removed. No new symbol is introduced. Pattern blocks excluded by an ancestor remain excluded in the pullback. Thus (5.8) has no additive term for newly created null symbols.

Along a path, conversions consume disjoint portions of the remaining representation dimension. Passing to an actual subquotient can decrease this dimension; decomposing a direct sum preserves its sum. Consequently

$$\text{current value dimension} + \text{total dimension of all evaluation covers} \leq O_r(d). \tag{5.9}$$

The constant cover inflation is summed over the disjoint converted spaces, not multiplied over successive stages. Direct-sum alternatives preserve the original number of tensor positions.

Later restrictions to a subgroup can destroy a previously useful permuted-block description of the value spaces. New composition flags are then computed on their actual subquotients. The old value-block names are not retained as additional constraints. In contrast, evaluation summands remain their actual modules with their existing tags. Nothing in this operation makes a discarded evaluation summand available again.

Lemma 5.9 (Hereditary exclusion on the evaluation representation)**.** *Choose C_* sufficiently large in terms of r and the constants in Theorem 3.11. The joint image on the stored evaluation modules, their labels, and the specified scalar and semilinear lines has no nonalternating nonabelian simple section of a subgroup of order greater than T. The assertion persists after every subgroup restriction.*

Proof. The individual assertions are those of Lemma 3.9 for bounded complementary modules and Lemma 3.7 for permutation covers. In the latter case the kernel of the action on cover coordinates has order $p^{O_r(d)}$: its local multiplicity images have exponents bounded by their original disjoint dimensions. Its permutation degree is $O_r(d)$ by (5.9), so the hereditary simple-section bound of Lemma 3.9 applies. Supported scalar and semilinear line images are solvable.

Here is the elementary closure argument needed to combine these facts. If $K \le G$ and $M \trianglelefteq G$, then $K \cap M \trianglelefteq K$ and $K/(K \cap M)$ embeds in G/M. Apply this fact to a subgroup whose quotient is the proposed simple section, and then use a composition series of its intersection and image. Every nonabelian simple section obtained is a section of a subgroup of M or of G/M. Induction gives the same conclusion for products and nonsplit extensions. Thus the exclusion holds for the actual joint representation, including any nonsplit linear action left in a stored module. Since it quantifies over all subgroups, further restrictions preserve it. □

5.5 Recursion on invariant summands

Lemma 5.10 (Splitting the evaluation summands)**.** *If b exceeds a constant depending on r, an affine problem can be partitioned into at most 2^b coset cases. In each case it is solved by two successive exact affine problems whose positive direction dimensions sum to the incoming dimension. Both problems use at most $\gamma_r b$ selected labels, where $\gamma_r = 1 - 1/(3r) < 1$. The construction and all coset representatives are explicit.*

Proof. Choose a position i with $b_i \ge b/r$, and choose a subset S of its selected labels of size $\lfloor b_i/2 \rfloor$. Enumerate its orbit under H and compute its set stabilizer H_S by the orbit–stabilizer algorithm, retaining actual words. The orbit has at most $2^{b_i} \le 2^b$ elements. Let $H = \bigsqcup_j a_j H_S$ be the resulting left transversal. For the representative a_j, replace y by $\alpha_{a_j^{-1}}(y)$ and solve under H_S.

The subset S partitions the tagged evaluation domain into the patterns whose label in position i belongs to S and their complement. The permanent symbol $\perp_i$, if present, belongs to exactly one side. First solve the transporter on the evaluated image of the first part. Under its complete returned stabilizer, solve its exact fiber on the complement, using Lemmas 5.2 and 5.8. The first direction is the image of an equivariant linear map and the second is its kernel, so their dimensions sum to $\dim Q$.

Each projected presentation uses at most half, rounded upward, of the labels in position i, and at most the previous labels elsewhere. Thus

$$b_{\text{child}} \le b - \lfloor b_i/2 \rfloor \le b - \frac{b}{2r} + \frac{1}{2} \le \left(1 - \frac{1}{3r}\right) b$$

when $b \ge 3r$. An empty domain or a zero direction is tested directly. Every coset case is completed before its answer is used in Lemma 5.3. The cosets exhaust H, so that lemma recovers the entire original transporter. Subsequent full evaluation on surviving summands cannot undo the displayed decrease of b. □

5.6 The joint classical subgroup at a terminal

After the value-space preparation and conversions above, all remaining value spaces support the selected classical components. Let A be the actual image on their associated quotient spaces, let B

be the actual image on the selected evaluation modules and the independent supported scalar and semilinear line actions, and let

$$J \le A \times B$$

be the joint image of H. Denote by $C \trianglelefteq A$ the selected central product of perfect quasisimple classical components. Its simple central quotients have order greater than T and lie outside the bounded exceptional identifications with alternating groups.

Lemma 5.11 (An actual subgroup and a bounded quotient)**.** *One has $C \times 1 \le J$. If the number of selected evaluation summands is bounded in terms of r, then*

$$[J : C] \le p^{O_r(d)}. \tag{5.10}$$

Generators of $C \times 1$, their original preimages, and a complete list of actual coset representatives for J/C can be computed in $p^{O_r(d)} \operatorname{poly}(L, \log p)$ time.

Proof. Put $A_0 = \operatorname{pr}_A \ker(J \to B)$. Then $A_0 \trianglelefteq A$, and the rule sending b to aA_0 for $(a, b) \in J$ is a well-defined surjection $B \twoheadrightarrow A/A_0$. If a selected quasisimple component had nontrivial image in this quotient, its kernel would be a proper normal subgroup, hence central. Its image would therefore have the same large simple central quotient as a section. This contradicts Lemma 5.9. A perfect component has no nontrivial abelian quotient either. Thus every component is contained in A_0, which proves $C \times 1 \le J$.

This argument is constructive using homomorphism kernels and membership, not a general intersection algorithm. Compute the kernel of the actual map $J \to B$, project it onto A_0, and express each supplied generator of C in those projected generators. Evaluate the same word in the kernel. Its image is exactly $(c, 1)$. Original preimages need not obey a splitting of the original group.

Let s be the number of selected evaluation summands. Once their labels are fixed, their joint image embeds in the product of the previously constructed images of their summand stabilizers. Each such image has order $p^{O_r(d)}$ by the structural construction. Between-summand maps are the fixed actual transversal identifications; they introduce no independently chosen maps. For an axis of a permutation cover the relevant image is its bounded multiplicity image, not the entire alternating group. Therefore

$$|B| \le s! \prod_{a \text{ selected}} |B_a|\,|B_{\mathrm{scal}}| = p^{O_r(d)}.$$

Indeed, if $f_j = [K_j : \mathbb{F}_p]$ for the independently counted semilinear lines, then

$$|B_{\mathrm{scal}}| \le \prod_j f_j(p^{f_j} - 1) \le p^{2\sum_j f_j} = p^{O_r(d)}.$$

The induced scalar and field-automorphism tuples have the same bound by Theorem 3.11; correlations only reduce the image. This is the step that uses $s = O_r(1)$.

We include the multiplicity estimate because the whole commutant, rather than separate block commutants, is needed. Aggregate every equivalent irreducible C-type across all value spaces, including required dual copies and actual diagonal links. Write an isotypic space as $X_i \otimes_{K_i} M_i$, where $f_i = [K_i : \mathbb{F}_p]$, $a_i = \dim_{K_i} X_i$, and $m_i = \dim_{K_i} M_i$. Increasing the constant in the permanent threshold if necessary gives

$$f_i a_i^2 > C_* d, \qquad \sum_i f_i a_i m_i \le \kappa_r d.$$

Consequently

$$\frac{m_i}{a_i} < \frac{\kappa_r}{C_*}, \qquad \sum_i f_i m_i^2 \le \frac{\kappa_r^2}{C_*} d. \tag{5.11}$$

The commutant embeds in $\prod_i \mathrm{GL}_{m_i}(K_i)$ and hence has order $p^{O_r(d)}$. This calculation includes intertwiners between equivalent copies in different original spaces.

Each distinct type uses at least $\sqrt{C_* d}$ prime-field dimensions, so there are $O_r(\sqrt{d})$ types. Their permutations have order $p^{O_r(d)}$. The natural-normalizer calculation of Lemmas 3.12 and 3.13 gives at most $p^{O_r(d)}$ joint tuples of field and graph automorphisms, form multipliers, determinants, spinor norms, and Dickson invariants. Fixing one such tuple leaves the derived classical product times this commutant. Since C is contained in the actual image, it follows that $[A : C] \le p^{O_r(d)}$. Thus $[J : C] \le [A : C]|B|$, proving (5.10).

Finally, breadth-first coset enumeration in the actual quotient J/C, using constructive membership in C, lists all representatives within the stated bound. It enumerates actual joint elements, so separate marginal scalar, permutation, and field choices are never assumed compatible without verification. □

Lemma 5.12 (Removing translations of a linear kernel)**.** *Let $\pi : H \twoheadrightarrow J$ be the actual joint action and let $K = \ker \pi$. Center the source at 0, with cocycle $c : H \to Q$ and target t. Then $\tau = c|_K$ is an additive homomorphism, $E = \tau(K)$ is H-invariant, and the affine action on Q/E factors through J. A complete transporter on this quotient lifts to the full original transporter by correcting its witness and every stabilizer generator and adjoining* $\ker \tau$.

Proof. Elements of K act linearly as the identity on the entire tensor presentation, and hence on Q. The cocycle identity therefore says $c(kk') = c(k) + c(k')$. Another application gives $c(hkh^{-1}) = L_h c(k)$, proving invariance of E. Two lifts of the same joint element have increments differing by an element of E, so the quotient action is well defined.

Compute E, preimage words for a basis, and $\ker \tau$ using Lemma 3.1. This uses linear algebra and group operations, not enumeration of the additive group E. If the quotient answer is $a\overline{S}$, lift a to $h_0 \in H$. Its discrepancy $t - c(h_0)$ lies in E. Choose $k_0 \in K$ with $\tau(k_0) = t - c(h_0)$; then $k_0 h_0$ is a witness. For every generator $\bar{s}$ of $\overline{S}$, lift it to h_s and choose $k_s \in K$ with $\tau(k_s) = -c(h_s)$. The elements $k_s h_s$ fix 0. Together with $\ker \tau$, they generate a subgroup projecting onto $\overline{S}$ with the full kernel of the source stabilizer. Equality with that stabilizer follows. If $Q/E = 0$, the same construction applies with $\overline{S} = J$, or directly to the translation equation. □

Proposition 5.13 (The complete quadratic terminal)**.** *For a degree-at-most-two problem with a bounded number of selected evaluation summands, the preceding constructions and Theorem 4.24 compute its complete original affine transporter in $p^{O_r(d)} \operatorname{poly}(L, \log p)$ time.*

Proof. First form E as in Lemma 5.12. Replacing P by the entire preimage of E in N gives an explicit presentation of $\overline{Q} = Q/E$ in the same tensor space. This retains every correlation in the denominator. If $\overline{Q} = 0$, the additive construction finishes the problem.

Otherwise enumerate the actual cosets uC of Lemma 5.11. Normalize the target using the true descended affine action of u^{-1} and restrict the source action to C. On C, every evaluation space and independently specified scalar or semilinear line is fixed pointwise. A whole isotypic space $X \otimes_K M$ has C acting on X and fixing M. Expand only in fixed bases of these multiplicity and evaluation spaces. A tensor word with at most two remaining occurrences becomes a direct sum of whole Hom spaces between the specified natural, dual, or Frobenius-natural representations; one or zero occurrences use a common trivial row. Symmetry, field balancing, and the original equations

remain one invariant numerator and denominator inside that whole direct sum. Passive copies affect L, not the total natural degree.

Thus each coset case satisfies precisely Theorem 4.24: an actual stable central product with its links, an arbitrary invariant prime-field section of the indicated quadratic Hom presentation, and its actual quotient cocycle. A supplied natural cover is used by inflating these data and retaining its full kernel on projection. A diagonal link is not replaced by independent classical factors. If $C = 1$, each case is the direct equality test. All coset cases are solved completely and combined by Lemma 5.3; only then is the result lifted by Lemma 5.12.

We spell out two scope points in the invocation of the pure theorem. In characteristic two its cogenerator argument acts on the actual section, pushes the cocycle to its specified acyclic target, solves the joint target equations inside the actual derived group, and then solves the remaining degree-one affine equation under the entire returned subgroup. It does not assert that the original arbitrary section has vanishing first cohomology. In odd characteristic, Lemma 4.19 embeds the actual section into a direct sum of whole matrix modules modulo fixed scalar subspaces, using its exact image algebra and all Hom maps. Theorem 4.20 gives vanishing first cohomology in that larger target, so the projected cocycle has an origin there. It need not have an origin in the original section. The resulting simultaneous template equations are solved by the represented-algebra, scalar-shift, form-tuple, and additive-kernel operations proved in Theorem 4.24; the actual derived-group conditions are imposed before its cocycle is evaluated. The original group H is not replaced by a full classical group or a split extension. These are the constructions proved in the pure theorem, so neither case requires a general subgroup-intersection algorithm or any cubic solver.

There are $p^{O_r(d)}$ coset cases, each has natural degree $O_r(d)$, and each complete pure call has the stated bound. The maps, kernel corrections, and reunion use the same bound. Their product is still $p^{O_r(d)} \operatorname{poly}(L, \log p)$. □

5.7 The recursive construction and its full cost

We first give the common recursion and its accounting. For quadratic problems its terminal is already provided by Proposition 5.13. The cubic terminal, constructed below using earlier completed algorithms, returns a complete positive-dimensional quotient before one remaining affine fiber.

At an entrance, retain the current original group, its affine predicate, the actual value spaces, and the selected evaluation summands. Apply the following finite sequence of operations.

1. Present the equation on the selected summands by Lemma 5.8 and use their full vector domains. If the direction is zero, test the equality and return the whole current group or the empty set.

2. Compute composition flags only for value spaces not already prepared for this current original subgroup and their actual subquotient maps. Use Lemma 5.7. If there are at least two positive induced quotients, solve the top quotient and then its affine fiber, with the original comparison of Lemma 5.1. If there is just one positive quotient of the full direction dimension, install its exact presentation in place and continue. A zero quotient is omitted.

3. Apply the constructive structural decomposition to those actual value quotients and convert every complementary summand into evaluation arguments as above. Reuse this decomposition when the original subgroup and the actual representation have not changed. A different numerator or denominator requires new linear maps, not a new structural decomposition of an unchanged representation.

4. If the selected-label count is large, use Lemma 5.10, complete every coset case, and return their complete union. Otherwise apply the appropriate complete terminal.

There is no instruction to repeat this sequence merely because a basis, a denominator, or an auxiliary lift group has changed. A single full-dimensional induced quotient proceeds to the next operation in the same sequence. After a strict original-subgroup restriction, new flags and block descriptions may be needed, but previously stored evaluation modules remain unchanged.

Every completed child returns an original transporter. Its temporary coordinate description is not imposed on the next fiber. The fiber uses the caller's exact presentation and its returned original subgroup, with the denominator correction already proved. If the completed quotient had positive dimension, that fiber has smaller direction dimension. A zero quotient cannot authorize a new same-dimensional invocation.

Lemma 5.14 (Bounded equal-parameter sequences)**.** *For the preceding construction, the number of successive operations on a path that leave b, v, and the positive direction dimension m unchanged is bounded by a fixed polynomial in the initial explicit input length and* $\log p$.

Proof. First fix the actual subgroup $H \leq \Gamma$. On each actual value subquotient compute one composition flag, with deterministic tie-breaking in its present coordinates. Transport the flag under a coordinate isomorphism instead of computing a new unrelated flag. Mark each interval as it is used. A composition flag has at most the dimension many proper intervals, and the fixed-arity tensor filtration has polynomially many interval tuples. Once a full associated value quotient has been reached, it is semisimple for this fixed H, and further subquotients are semisimple. The finite sequence above is performed once per such interval tuple.

On a path keeping b, v, m unchanged, no complementary summand is converted, no child from a balanced evaluation split occurs, no split into two positive directions occurs, and no positive terminal quotient is removed. A change of presentation transports previously computed structural data. A restored caller presentation following a positive quotient has smaller m and therefore does not lie on the path under consideration. Structural preparation without any of these changes proceeds directly to its bounded terminal. Consequently, at fixed H only polynomially many marked flag intervals and stages can occur.

The only event that invalidates these flags without decreasing the three parameters is a strict decrease of the actual subgroup of Γ. If the original external permutation degree is n_0 and an elementary abelian subgroup specified at the initial call has dimension t_0, then

$$\log_2 |\Gamma| \leq O(d^2 \log p + n_0 \log(n_0 + 1) + t_0 \log p). \tag{5.12}$$

Every proper subgroup has index at least two, so this bounds the number of such changes on the path. New flags need not refine obsolete permuted-block descriptions: their actual invariant subquotient ancestry and dimension bound suffice. A new presentation of the same subgroup is not a strict decrease. In particular, auxiliary denominator groups in (5.4) introduce no term in (5.12).

Multiplying the polynomial number of stages in one fixed-subgroup segment by the right side of (5.12) gives the asserted polynomial, denoted by P_0. One may take, for example, a constant multiple of $(1 + \log_2 |\Gamma|)(d + 1)^{r+2} \operatorname{poly}(L, \log p)$. The assertion does not license a repeated call whose stipulated structural or terminal input has not been constructed; those inputs are supplied by the preceding lemmas and the specified terminal theorem. □

Lemma 5.15 (A common bound for local operations)**.** *All presentations and local operations in the construction have a common bound*

$$p^{O_r(d)} \operatorname{poly}(L, \log p), \tag{5.13}$$

measured from the initial call, provided each completed earlier terminal algorithm has that bound at degree $O_r(d)$. Auxiliary affine lifts do not create an increasing tower of additive spaces.

Proof. The factor spaces have total dimension $O_r(d)$ by (5.9). At fixed arity their tensor coefficient dimension is polynomial in d times the initial passive multiplicity size. Each numerator, denominator, and comparison is an image, kernel, intersection, or inverse image in such a space. No repeated inverse image introduces a new coefficient space: the ambient is the current actual tensor cover, and its entire kernel is already part of the denominator. Spaces created inside a completed pure or lower-degree call are bounded by that call's stated polynomial and are not added to the caller's presentation on return.

Full evaluation domains have size $p^{O_r(d)}$. Indeed a product of vector sets has cardinality p raised to the sum of the factor dimensions; the sum over the polynomial number of tagged summand choices has the same bound. Enumeration of these domains and exact linear algebra on the evaluated maps therefore satisfy (5.13). The algorithm does not recurse on vector orbits or introduce a factor equal to domain size at every level.

For clarity, symbolic group operations can be implemented without the action on an auxiliary coefficient-vector space. Let E_0 be a generated subgroup of $M_{\mathrm{add}} \rtimes G$. Compute its image G_0, retaining actual generator lifts. The faithful physical action of G_0 can be regarded as a permutation action on vectors, of degree at most $p^{O_r(d)}$, together with the original explicit permutation action. The constructive permutation algorithms of Lemma 3.1 give a membership procedure and a presentation of G_0 on its generators. Evaluate the relators on their actual lifts; they are translations in M. Take their linear span and close it under conjugation by the lifted generators. At most $\dim M$ strict span enlargements occur.

The resulting space is exactly $E_0 \cap M$. To see this, the relators normally generate the kernel of the map from the free group on the chosen generators onto G_0. Evaluation in E_0 sends this normal closure to the kernel of $E_0 \to G_0$. Since the kernel is an elementary abelian group, taking its additive span and its conjugates is precisely that normal closure. Retain a word for each basis vector. Membership of (g, a) is now tested by a membership word for g, lifting it to (g, a_0), and testing the appropriate translation difference in that space by linear algebra. A generating set consists of one lift of each compressed image generator and a basis of the kernel.

By Lemma 5.4, each new artificial lift is formed directly over the current original subgroup. If the initial group has its own symbolic elementary abelian kernel, first perform the preceding construction at that fixed layer and then at the single current artificial layer. There is no unbounded sequence of new extension bases. Group-cover kernels with a genuine current action are treated by their separate equivariant comparison and not discarded as artificial.

All words are shared directed acyclic graphs. Each new local word stores pointers to previous words, rather than their expanded strings. Generator compression can retain an element only when it enlarges the represented subgroup; (5.12) bounds the number retained in the original comparison representation. The full number of graph nodes is charged to the local work summed over the recursion tree. These facts prove (5.13), including exact witness and kernel operations. □

Proposition 5.16 (The complete recursion-tree bound)**.** *Suppose each bounded-label terminal either completes its affine equation or computes a complete positive-dimensional quotient, reunites all of its coset cases, and then makes exactly one call on the original remaining fiber. Suppose the terminal work, apart from that fiber, has the bound* (5.13) *using only previously established complete algorithms. Then the construction returns a complete original transporter in $p^{O_r(d)}\,\mathrm{poly}(L, \log p)$ time. It counts all failed coset cases and all stabilizer and lift computations.*

Proof. Correctness follows inductively from (5.2), Lemmas 5.1, 5.3, 5.4, 5.7, and 5.8. In particular, every coset case is an exact search inside an original coset and is finished before reunion. The terminal hypothesis gives the same invariant for its single remaining fiber. We give a bound on the entire tree rather than multiplying local costs along its depth.

Choose fixed integers $\lambda > a_r$ and k sufficiently large. For a call of positive direction dimension m, assign the positive integer

$$\Phi = p^{k(b+\lambda v)}m. \tag{5.14}$$

No evaluation-domain cardinality occurs in this expression. A conversion of dimension c satisfies

$$(b' + \lambda v') - (b + \lambda v) \le -(\lambda - a_r)c,$$

so it cannot increase Φ. The positive dimensions in an exact filtration or an image–fiber decomposition add to the original m; their inherited exponents do not increase. A quotient of dimension $h > 0$ computed in a complete terminal leaves a single fiber of dimension $m-h$. It therefore strictly decreases the integer weight.

For a balanced family, every positive child has at most $\gamma_r b$ labels before any subsequent separately charged conversion. In each coset case the two positive dimensions add to at most m, including failures of the first child. Thus the sum of the weights over all coset cases and both children is at most

$$2^b p^{-k(1-\gamma_r)b}\Phi \le \tfrac{1}{2}\Phi$$

outside a fixed cutoff, after choosing $k(1-\gamma_r) > 1$ and enlarging that cutoff if necessary. Here $2^b \le p^b$ for every prime p. A later conversion decreases its own child's weight independently. The selected-summand isomorphism preserves b; it cannot restore a label lost at a still unfinished ancestor. The permanent-symbol convention ensures the same assertion for tensor summands with no evaluated factor.

Contract maximal unary chains whose weight is unchanged. By Lemma 5.14, each has length at most P_0. The resulting tree has positive integer weights, the sum of child weights at most the parent weight, and every unary edge strictly decreases the weight. Assign Φ_0 distinct units to the initial call. Partition them among children in accordance with their weights. Charge a strict unary vertex to one unit not forwarded, and leave one unit at every leaf. No unit is charged twice. Hence

$$\#\{\text{leaves}\} + \#\{\text{strict unary vertices}\} \le \Phi_0.$$

After deleting unary vertices, the number of branching vertices is at most the number of leaves minus one. Restoring the contracted chains shows that the full tree has $O((P_0+1)\Phi_0)$ vertices. Initially $b + v = O_r(d)$, and m is bounded by the explicit coefficient size. The tree size is therefore $p^{O_r(d)}\,\mathrm{poly}(L, \log p)$.

A failed positive child is a leaf and uses its assigned weight. A zero-dimensional child is a direct check; all such checks and the construction of the complete list of at most 2^b coset cases belong to the local bound. Thus all-failing families are counted as well. Complete reunion includes all branch stabilizers and witness ratios, followed by compression in the original representation. Every projected generator, not just a witness, is corrected when a translation kernel is restored. These operations are included in Lemma 5.15.

Finally, multiply the common local bound by this tree size once. Earlier completed algorithms in a terminal enlarge its fixed constant in the exponent; they do not introduce another factor per level of this tree. The dependency order used below has fixed length. Hence the product has the claimed bound. The exponent $b + \lambda v$ and the direction m are nonincreasing along a path, and a strict-weight edge decreases one of these integer parameters. Together with P_0, this also bounds the

depth by a polynomial in the initial explicit input length. Keeping one symbolic affine presentation per open call therefore has polynomial space per path, apart from the already counted explicit evaluation data. □

Theorem 5.17 (Complete quadratic affine transport)**.** *For every fixed arity r, a supplied-group affine transporter on an arbitrary invariant subquotient of a tensor presentation of degree at most two can be computed deterministically in*

$$p^{O_r(d)}\,\mathrm{poly}(L,\log p)$$

time. The output is empty, or an actual original witness and generators of its entire original source stabilizer.

The same conclusion holds if the presentation has at most two value-space occurrences and additional evaluation factors of fixed total arity, provided their actual summand action, section maps, local stabilizer-image bounds, and hereditary exclusion of the selected large classical simple sections are supplied as above. Their total cover dimension is included in the $O_r(d)$ bound.

Proof. Start with the given representation spaces as value factors and no evaluation modules, except any explicitly supplied factors in the second version. The filtration and conversion constructions preserve the stated presentation class and produce all hypotheses of Proposition 5.13 whenever the selected-label count is bounded. That proposition completes the full direction and calls no cubic algorithm. Apply Proposition 5.16. The quotient and fiber formulas and complete unions prove the output claim at the original-group level. □

5.8 Intersection and frames after quadratic transport

Corollary 5.18 (Actual matrix-group and coset intersection)**.** *Let $G_1, G_2 \le \mathrm{GL}(W)$ be supplied by generators, with retained original comparison representations, and let $u, v \in \mathrm{GL}(W)$. One can compute $uG_1 \cap vG_2$, including a witness and its entire subgroup of differences, in $p^{O(d)}\,\mathrm{poly}(L,\log p)$ time, where d is the total degree of the two original representations and the displayed copies of W. Invisible kernels in original preimages are retained.*

Proof. Use the action of $G_1 \times G_2$ on $\mathrm{Hom}(W_2, W_1)$ given by

$$(g_1, g_2)X = g_1 X g_2^{-1}.$$

Transport I to $u^{-1}v$ by Theorem 5.17. The equation $g_1 g_2^{-1} = u^{-1}v$ is equivalent to $ug_1 = vg_2$. Thus the first projection of the returned transporter, multiplied by u, is the requested intersection. The stabilizer of I consists exactly of pairs (g, g) with $g \in G_1 \cap G_2$; its projection gives the full subgroup of differences. Original groups mapping onto G_i are handled in their original product comparison representation, so elements invisible on $\mathrm{Hom}(W_2, W_1)$ are not lost. The coefficient dimension $\dim \mathrm{Hom}(W_2, W_1)$ is not used as a faithful representation degree. □

Corollary 5.19 (Frames and subspaces)**.** *Let H act on $V \otimes L_0$, and let $f, g : E \hookrightarrow V \otimes L_0$ be injective maps with $\dim E = O(d)$. Suppose L_0 is trivial, or has an actual evaluation-factor presentation satisfying the second version of Theorem 5.17. Then the entire H-transporter from $\mathrm{im}\, f$ to $\mathrm{im}\, g$ is computable with the same bound, including original kernels.*

Proof. Under $H \times \mathrm{GL}(E)$, the action on frames is $(h, a)f = hfa^{-1}$. This has two value occurrences, V and E^*; the specified L_0 is an evaluated or trivial factor. Apply Theorem 5.17 to transport f to g. For a given h carrying $\mathrm{im}\, f$ to $\mathrm{im}\, g$, there is exactly one a satisfying $hf = ga$, because both

maps are injective. Therefore projecting the complete frame transporter onto H gives exactly the full subspace transporter. The same argument applies to its stabilizer and invisible original kernels. An arbitrary moving label module does not satisfy the hypotheses merely because its dimension is written in the input. □

These corollaries occur after Theorem 5.17 in the proof order. The pure quadratic theorem used in its proof must therefore not invoke them. The ordinary three-tensor algorithm of Theorem 7.1 uses only their completed outputs; it does not receive their temporary filtrations, denominators, or evaluation domains.

6 Cubic representations of the natural classical groups

We prove the representation statements used below before giving any cubic transporter algorithm. Thus none of these statements uses the algorithm it will subsequently justify. All ranks in this section are in the stable range: natural dimension at least 36, and Witt index at least 12 where a parabolic calculation is used. Smaller dimensions belong to the bounded-rank part of Theorem 3.11.

A *natural occurrence* is a natural module, its contragredient, or a specified coefficient-Frobenius twist of either. Tensor products are formed with the actual defining-field structures. Restriction to the prime field is performed only with the resulting field equations retained. Let D be a product of recognized derived classical groups, with diagonal identifications already imposed. A simple D-module is called *quadratic* if it occurs in a tensor word of length at most two in these occurrences, including the trivial module. This terminology describes its occurrence, not its dimension. For a cubic word T, let $J(T)$ be the sum of its natural pairing insertions: an invariant tensor occupies two positions and a natural parameter occupies the remaining position. All eligible dualities and twists are included.

6.1 Submodules of quadratic composition type

Lemma 6.1 (An elementary trilinear calculation)**.** *Let b be a nondegenerate alternating form on a vector space of dimension at least six. A trilinear form vanishing whenever its three arguments are pairwise b-orthogonal is a sum*

$$\ell_1(x)b(y,z) + \ell_2(y)b(x,z) + \ell_3(z)b(x,y).$$

In characteristic two the analogous assertion holds for the nonalternating dot product in dimension at least six.

Proof. Choose a hyperbolic basis e_i, f_i. A coefficient on three basis vectors containing no paired e_i, f_i is zero. For a free basis vector w in plane j, compare $(e_i + e_k, f_i + f_k, w)$ in characteristic two, and $(e_i + e_k, f_i - f_k, w)$ otherwise, for $i, k \neq j$. This shows that the coefficient with its paired positions in plane i is independent of i. The triple $(e_i + f_i, e_i + f_i, w)$ identifies the two orders of the pair, with the sign prescribed by b. Repeat in each free position and subtract the three resulting insertions. The remainder has support only on the six mixed triples inside one hyperbolic plane. For distinct i, j, k, the triple

$$e_i + e_j, \quad e_i + e_k, \quad f_i + f_j + f_k$$

is pairwise orthogonal in characteristic two and isolates the coefficient (e_i, e_i, f_i). In other characteristics change the two spare coefficients in the last vector to their negatives. Permuting positions and interchanging e_i, f_i kills the other five coefficients.

For the dot product in characteristic two, three distinct basis indices give zero coefficients. The orthogonal triple $(e_i + e_k, e_i + e_k, e_j)$ shows that the coefficient of (e_i, e_i, e_j) is independent of $i \neq j$. Subtract the three insertions. Only diagonal coefficients d_i remain. Evaluating on three copies of $e_i + e_j$ gives $d_i = d_j$. Finally the three vectors

$$(1,1,1,1,0,0), \quad (1,1,0,0,1,1), \quad (1,0,1,0,1,0)$$

are pairwise orthogonal and have common support of size one. Their value is the common d_i, which is therefore zero. □

Lemma 6.2 (Cubic submodules in characteristic two)**.** *Suppose the characteristic is two. If $P \leq T$ is a submodule of a cubic word and every composition factor of P is quadratic, then $P \leq J(T)$. This holds for special linear, special unitary, symplectic, and both even orthogonal derived groups, with arbitrary twists and fixed multiplicities.*

Proof. First consider $\mathrm{Sp}_{2m}(K)$. On each hyperbolic plane take the order-three subgroup of $\mathrm{SL}_2(\mathbb{F}_2)$. Their product is an odd-order torus. After extending the coefficient field, a natural module has characters supported on one plane, whereas any quadratic word has characters supported on at most two planes. Taking invariants and weight spaces for this torus is exact. Thus P has no component whose three positions lie in three different planes. This remains true after every group conjugation. Witt extension therefore implies that every member of P, as a trilinear form using the polar identification, vanishes on independent totally isotropic triples. The independent ordered triples span the third tensor power of a three-dimensional space, even over $\mathbb{F}_2$: with two independent arguments fixed, the vectors outside their span span the space; apply this successively in all three positions. Extending an isotropic subspace to dimension three removes the independence condition. Lemma 6.1 now gives the three insertion diagrams.

Every Frobenius twist has the same matrices on the subgroup over $\mathbb{F}_2$, so the preceding argument applies to each fixed twist signature. Isolate a free coordinate in one plane and a pairing in another. Scaling e_i by t and f_i by t^{-1} makes the two orders of that pair acquire factors $t^{2^a - 2^b}$ and $t^{2^b - 2^a}$. Since the insertion relation holds for every t, their equality forces $2^a = 2^b$ modulo $|K| - 1$ or the coefficient is zero. The oddness of $|K| - 1$ justifies cancelling the factor two. This is exactly the eligible-twist condition.

For an even quadratic orthogonal group, use hyperbolic four-spaces $A_i \oplus A_i^*$ with $\dim A_i = 2$ and the subgroup $g \oplus g^{-t}$ for $g \in \mathrm{SL}_2(\mathbb{F}_2)$. Its order-three subgroup has no fixed natural vector and lies in Ω, since the rank of $g \oplus g^{-t} - I$ is even. The preceding character-support argument excludes three different four-blocks. Any independent triple with zero mutual polar products can be moved into these three blocks: each block represents each prescribed quadratic value, and Witt extension applies to the isometry of their spans. A symmetry in an unused hyperbolic plane corrects the Dickson invariant without changing the three images. For twisted positions apply this to $F(x, y, z) = T(x^{[a]}, y^{[b]}, z^{[c]})$. Additive replacements with spare orthogonal vectors remove dependent triples. Choose a symplectic basis for the polar form, also in the nonsplit case. The coordinate proof of Lemma 6.1 uses only addition and coefficients zero and one, so the Frobenius twists do not alter it. Hyperbolic scalings prune the same ineligible pairings; their plane can be chosen disjoint from a free coordinate in the anisotropic two-space. This proves both signs.

For a unitary group use the same hyperbolic four-block subgroups. Each represents every Hermitian norm, since $t + t^\sigma$ maps the defining field onto its fixed field. Hermitian Witt extension moves independent mutually orthogonal triples into the blocks; a spare anisotropic line corrects the determinant. If a repetition occurs, its repeated vector must be isotropic. For example choose $w \in x^\perp \cap z^\perp$ outside $\langle x, z \rangle$ and subtract the values on $(x, x + w, z)$ and (x, w, z). Three repeated

arguments are handled in a three-dimensional totally isotropic subspace. In an orthonormal basis the zero-one coordinate calculation is the dot-product part of Lemma 6.1. A norm-one diagonal element in a paired coordinate, with determinant compensation on a spare coordinate, forces

$$2^a + 2^b \equiv 0 \pmod{2^f + 1}, \qquad |K| = 2^{2f}.$$

The order of 2 modulo $2^f + 1$ is $2f$: a smaller period would divide $2f$ and be at most f, which is impossible. The unique power equal to -1 is therefore the fth. The displayed condition is exactly the Hermitian dual-twist condition.

For $\mathrm{SL}_n(K)$ in even dimension restrict first to the symplectic subgroup over $\mathbb{F}_2$, identifying duals by its form. An elementary determinant-one transvection outside that symplectic subgroup takes a natural–natural pairing coefficient, with its free coordinate in a third plane, to a three-distinct-plane coefficient. Its free-position origin is unique, so it cannot cancel with another diagram. Consequently only natural–dual evaluation pairings survive. Diagonal elements with determinant compensation on spare coordinates impose matching twists. In odd dimension apply this argument in every even-dimensional coordinate subspace. For a mixed word written as $B : V \times V \to V$, it gives $B(x, y) \in \langle x, y \rangle$ for the zero-one vectors after every zero-one change of basis. Write $B(e_i, e_j) = a_{ij}e_i + b_{ij}e_j$. Evaluating on $(e_i, e_j + e_k)$ and $(e_i + e_k, e_j)$ makes the respective coefficients depend only on the free index. Evaluating on $(e_i + e_j, e_i + e_k)$ determines the diagonal coefficient. Thus $B(x, y) = \alpha(x)y + \beta(y)x$, exactly the two evaluation insertions. The pure words vanish, since every three coordinates lie in an even coordinate subspace. Full-field diagonal elements again impose the twists.

For occurrence pattern $2+1$ in two independent factors, apply the product of the same tori. The repeated-factor bilinear form must vanish on all orthogonal pairs, hence be a multiple of the pairing: for fixed $x \neq 0$ its second functional is proportional to $b(x, -)$, and comparison at independent x makes the scalar constant. The free factor has no fixed natural vector; conjugating its torus tests every free coordinate. Three independent factors give zero immediately by support three. The arguments are coefficientwise in passive multiplicities. They identify only the ambient insertion span and do not replace P by that span.

Finally extend scalars from the prime field, decompose its natural rows into their Frobenius signatures, and apply the proved containment to each projection of P. All insertion maps are defined over the prime field using trace pairings. Containment descends because it is a linear-system rank statement. No denominator is replaced by a defining-field span. □

Corollary 6.3 (Literal cubic invariants)**.** *In the setting of Lemma 6.2, $T^D = 0$. In particular, for every stable even orthogonal derived group and arbitrary a,b,c,*

$$\mathrm{Hom}_D(V^{[a]}, V^{[b]} \otimes V^{[c]}) = 0.$$

Proof. A fixed vector generates a trivial, hence quadratic, submodule and lies in $J(T)$. The eligible insertion maps form a direct sum of natural modules. To check injectivity, put the free basis coordinate in one block and a nonzero pairing coefficient in another. This isolates each insertion position, so every parameter of a zero insertion is zero. The natural parameters have no fixed vectors. The Hom assertion follows by the actual polar self-duality of the twisted natural module. This proof is independent of any square-image algebra calculation. □

Lemma 6.4 (Insertion support in all characteristics)**.** *For every characteristic, every cubic submodule of quadratic composition type lies in $J(T)$. In particular $\mathrm{Soc}_D(T/J(T))$ has no quadratic simple constituent. The eligible insertions are exactly the positionwise natural intertwiners and invariant pairings, with their actual endomorphism fields.*

Proof. Characteristic two is Lemma 6.2. In odd characteristic use the joint derived-to-full image comparison in Theorem 4.7 on the sum of the words being compared. The full-classical central signs separate their parity signatures. A quadratic constituent in a cubic word must therefore be a degree-one natural, dual, or twisted-natural constituent. For two such constituents S, S', the whole-Hom vanishing in Theorem 4.20 gives $\mathrm{Ext}^1_D(S, S') = H^1(D, \mathrm{Hom}(S, S')) = 0$. Hence the submodule in question is semisimple. Every map $S \to T_1 \otimes T_2 \otimes T_3$ is, by currying, a map

$$T_3^* \otimes S \longrightarrow T_1 \otimes T_2.$$

The full degree-two intertwiner calculation gives three pairing diagrams: the straight identification, its switch, and contraction followed by insertion. For completeness, on a nonsplit bottom–simple–top component, commutation with its simple-block idempotent and full simple matrices forces a scalar on the simple block. Subtract the corresponding whole component map. Commutation with all simple-to-bottom and top-to-simple matrices then forces the remainder to factor from trivial top to trivial bottom. Thus there is no additional isolated simple-block map. On squares the two position projections are $(1 \pm \tau)/2$, and different twists allow only the actual natural intertwiners. Uncurrying gives precisely the three insertion positions. This argument divides by two only, so it includes characteristic three.

Since $J(T)$ itself has quadratic composition type, the inverse image of a quadratic simple submodule of $T/J(T)$ would be a larger submodule of quadratic composition type, contradicting the proved containment. □

6.2 First cohomology of the canonical cubic quotients

We use left cocycles $c(gh) = c(g) + gc(h)$. Existence of a coboundary can be implemented by solving $(g - I)b = c(g)$ on any supplied generating set. Every cohomology assertion below is for the indicated full finite group, not for its arbitrary subgroups.

Lemma 6.5 (Odd-degree sign argument)**.** *In odd characteristic, a section of a sum of tensor words of odd degree at most three for a stable natural derived classical group has zero fixed space and zero first cohomology. The same holds for each nonzero parity summand for a product of such groups.*

Proof. For $\mathrm{SL}_n(K)$ use the determinant-one diagonal sign group S. Average to make a cocycle vanish on S. The elementary matrix $u_{ij}(t)$ commutes with $S_{ij} = \{s \in S : s_i = s_j\}$, so $c(u_{ij}(t))$ is fixed by S_{ij}. An occurring sign character has odd support at most three. The annihilator of S_{ij} has representatives of supports $0, 2, n, n-2$, none of the required odd size when $n \geq 6$. Thus this fixed space is zero, also on every section, because $|S_{ij}|$ is prime to the characteristic. Elementary matrices generate SL, proving the assertion. The same sign argument already gives the absence of fixed vectors.

For Sp its central $z = -I$ acts as -1 on the section. Comparing $c(zg)$ and $c(gz)$ gives $2c(g) = c(z) - gc(z)$, an explicit coboundary. For SU choose an orthonormal basis. The coordinate SU_2 groups generate: if $N = \bar{a}a + \bar{b}b \neq 0$ and $\bar{d}d = N$, then

$$\begin{pmatrix} \bar{a}/\bar{d} & \bar{b}/\bar{d} \\ -b/d & a/d \end{pmatrix}$$

has determinant one, is unitary, and sends (a, b) to $(d, 0)$. For a vector with at least three nonzero coordinates, some pair of coordinate norms has nonzero sum, since the characteristic is odd. These transformations reduce a unit vector to one coordinate; induction on its stabilizer proves generation.

Each coordinate group commutes with the corresponding S_{ij}, so the special-linear sign proof applies unchanged.

For $\Omega(Q)$ diagonalize the quadratic form with coordinate norms 1 and a fixed nonsquare ξ. Arrange at least nine coordinates of each norm class I_0, I_1. This is possible in dimension at least twenty: every $\xi \neq 0$ is a sum of two squares, and the change of basis with rows $(x, y), (-y, x)$ changes two square coordinates to two nonsquare coordinates. The parity prescribed by the discriminant is preserved. The orthogonal group is generated by reflections supported on at most three coordinates. Indeed an anisotropic vector with coordinates from both norm classes has either a cross-class pair of nonzero norm or a three-coordinate block with nonzero norm. A vector supported on one norm class has a pair of nonzero total norm, unless its support is one; include a zero coordinate of the other class. In each case the block has an axis in the norm class of that total and its support can be reduced. Vectors v, w of equal nonzero norm are carried to each other by reflection in $v - w$ when that vector is anisotropic; otherwise reflection in $v + w$ followed by reflection in w suffices, since $Q(v + w) = 4Q(v)$. Induction on an orthogonal complement proves the generation claim. The last complement may be ordered to retain both norm classes until dimension two.

The determinant and spinor-norm kernel is $\Omega(Q)$ [Tay92, Theorem 11.51]. Choose coordinate reflections r_0, r_1 of the two norm classes. The four elements $1, r_0, r_1, r_0 r_1$ are a transversal of this kernel. Their Schreier generators relative to the three-coordinate reflections are supported on at most five coordinates. Take $S \leq \Omega$ to be the signs with an even number of minus signs in each I_j. For a generator supported on B, $|B| \leq 5$, let S_B be its signs constant on B. The annihilator is

$$\langle 1_{I_0}, 1_{I_1} \rangle + \{v : \operatorname{supp}(v) \subseteq B,\ |\operatorname{supp}(v)| \text{ is even}\}.$$

A nonzero class-indicator coefficient leaves at least four coordinates outside B; without one, the weight is even. Thus no odd weight of size at most three is trivial on S_B. Normalize on S, and the commuting cocycle equation kills every generator. This proves both orthogonal signs and odd-dimensional orthogonal groups.

For a product, the parity projectors preserve all sections by joint image comparison, Theorem 4.7. Choose a factor with odd occurrence number. Normalize the cocycle to zero there by the proved one-factor result. All other factors commute with it, so their cocycle values lie in its zero fixed space. A central product is treated by pullback to its product cover; inflation is injective in degree one. □

Lemma 6.6 (Finite general-linear coefficient calculations)**.** *Let K have characteristic two and $G = \mathrm{GL}_n(K)$. In stable rank $n \geq 12$: (i) every odd-length natural/dual/Frobenius word of length at most five has no invariants; (ii) every cubic word has zero H^1; (iii) every natural word has zero H^1 and H^2. These assertions, when applicable to a coefficient or its indicated quotient, also hold for intermediate groups $\mathrm{SL}_n(K) \leq G' \leq G$ in the ranges used below. A nontrivial determinant character times a word of length r has zero H^0 for $n > r$ and zero H^1 for $n > r + 2$.*

Proof. Work in the category of ordinary functors on finite-dimensional K-spaces and all linear maps. The natural Frobenius functors are additive, regardless of their strict-polynomial degree. The zero-object maps split every functor into its constant part and reduced part; these form orthogonal exact factors. Exact sum–diagonal adjunction, followed by the decomposition of an additive functor on a direct sum, shows that a tensor product of two reduced functors has zero Ext to or from an additive functor. More generally unequal numbers of additive tensor factors have zero Ext: split the variables on one side, expand the other, and note that a nonzero term must have exactly one additive factor in each variable. This is also the exponential-functor calculation in [FFSS99, Theorem 1.7].

A pure cubic is reduced and therefore has zero Ext from the constant functor. A mixed cubic is the finite-group Ext between one additive factor and two, and the preceding argument applies in both directions. The same constant/reduced argument applies to the natural word in degrees one and two. The comparison in [FFSS99, Appendix A.1] identifies these ordinary functor Ext groups with the stable finite-general-linear Ext groups. The dual coefficient system on split injections has cross-effect degree equal to the number of occurrences: adding one direct summand replaces each natural, dual, or Frobenius occurrence by its old value plus a constant one-dimensional summand. Repeated differences lower that degree. [Put23, Theorem B], with stable rank parameter two and starting index zero, gives the range $n \geq 2i + 2r + 2$. Apply it to the dual homology system. Finite-dimensional duality changes its stabilization to cohomology restriction followed by coefficient projection, exactly the maps in the finite-functor comparison. This gives rank ten for cubic H^1, rank eight for natural H^2, and rank twelve for degree-five H^0.

To check the determinant assertion, normalize a cocycle on the full odd-order diagonal torus. Evaluation on elementary root groups embeds the normalized cocycle space in the direct sum of their function spaces. Each weight is a coefficient weight plus a character supported on the two root coordinates. It therefore has support at most $r + 2$, whereas a nontrivial determinant character is a nonzero constant exponent vector. These cannot cancel when $n > r + 2$. The invariant-vector version omits the two root coordinates. For transfer from SL normalize on its diagonal torus. The same evaluation argument bounds the support of characters in $H^1(\mathrm{SL}, A)$ by $r+2$. Inner conjugation is trivial on cohomology, so every character factors through determinant and is a constant vector. It must be zero. Since the determinant quotient has odd order, $H^1(\mathrm{GL}, A) = H^1(\mathrm{SL}, A)$ and the same holds for intermediate determinant subgroups. The invariant-vector assertion uses the same torus separation. These support arguments pass to the indicated invariant quotients because the torus has odd order. Finally the stable special-linear group is perfect and its Schur multiplier has no characteristic-two part [Ste81, Theorem 1.1]. The low-degree extension sequence and the odd determinant quotient give $H^1(G, K_\theta) = H^2(G, K_\theta) = 0$ for every determinant character θ. □

Lemma 6.7 (Symplectic cubic quotients in characteristic two)**.** *For $G = \mathrm{Sp}_{2m}(K)$, $m \geq 12$, and any cubic twist word T,*

$$(T/J(T))^G = 0, \qquad H^1(G, T/J(T)) = 0.$$

Proof. Let I_a be the additive Frobenius functor and $A(U) = K[\bigwedge^2 U]$. Write $E_a = \mathrm{Ext}^1(A, I_a)$ in the ordinary functor category. The symplectic comparison [DV10, Corollaries 3.32–3.33] applies over every finite field to ordinary polynomial degree r, in rank $m \geq 2i + r + 6$. We compute

$$\mathrm{Ext}^1(A, I_a \otimes I_b \otimes I_c) = \bigoplus_{i:\text{the other two twists agree}} E_{a_i}. \tag{6.1}$$

Here is the full degree-one argument. Sum–diagonal adjunction in three variables replaces the source by

$$\bigotimes_i K[\bigwedge^2 U_i] \otimes \bigotimes_{i<j} K[U_i \otimes U_j].$$

Split each factor into its constant and augmentation summands. A summand missing a variable has zero Ext to the reduced target in that variable. A variable in two reduced factors also gives zero Ext: duplicate that variable, retain all other shared parameters, apply exact sum–diagonal adjunction, and use additivity of the target. Thus the only survivors are a singleton augmentation in one variable and a pair augmentation in the other two, or the three singleton augmentations.

Put $\bar{A} = \ker(A \to K)$. Then $\mathrm{Hom}(\bar{A}, I_a) = 0$, since $\bar{A}(K) = 0$ and all linear functionals to K separate vectors. For pointwise finite-dimensional parameter functors R, Q, this gives

$$\mathrm{Ext}^1(P \boxtimes R, I_a \boxtimes Q) = \mathrm{Hom}(R, \mathrm{Ext}^1(P, I_a) \otimes Q), \qquad P = \bar{A}.$$

To justify it without a finite-resolution hypothesis, restrict an extension to each parameter object. A morphism between its source and target extensions exists precisely when the associated Ext classes agree under pushout and pullback. It is unique because its possible difference belongs to $\mathrm{Hom}(P, I_a)$ tensored with finite-dimensional multiplicities, which is zero. These unique morphisms satisfy composition, so the parameterwise classes assemble uniquely to the required functor extension.

For $B(U, W) = K[U \otimes W]$ its augmentation has

$$\mathrm{Hom}(\bar{B}, N \otimes I_b \boxtimes I_c) = \begin{cases} N & b = c \pmod{[K : \mathbb{F}_2]}, \\ 0 & b \neq c. \end{cases}$$

Indeed at (K, K) scalar multiplication on either variable gives $(s^{2^b} - s^{2^c})f(t) = 0$. Matching twists force $f(t) = t^{2^b} f(1)$; coordinate functionals determine its unique extension. The triple-singleton term vanishes by the same one-dimensional test in one of its remaining variables. This proves (6.1).

The eligible insertions are an injective sum of natural twists, as checked in Corollary 6.3. Their first cohomology therefore has the same dimension as $H^1(G, T)$ by the comparison and (6.1). To prove injectivity also in degree two, write $V = W \perp U$ with U a hyperbolic plane and restrict to $\mathrm{Sp}(W)$. For insertion position i, project that position to W and the other two to U, and apply any linear functional taking the invariant tensor of U to one. The resulting map D_i satisfies $D_i \iota_j = \delta_{ij} \mathrm{pr}_W$. No invariant contraction on U is required. The induced composite in degrees one and two is diagonal with the natural-coefficient restriction/projection stabilization maps. The smaller rank $m - 1 \geq 11$ is in the comparison range, so these maps are isomorphisms. Thus $H^i(G, J) \to H^i(G, T)$ is injective for $i = 1, 2$, and is surjective for $i = 1$ by the dimension equality. The cohomology exact sequence proves $H^1(G, T/J) = 0$. Its earlier terms, using $T^G = 0$ and this injectivity in degree one, prove $(T/J)^G = 0$. □

Lemma 6.8 (Unitary cubic quotients in characteristic two)**.** *For $G = \mathrm{SU}_n(2^f)$, $n \geq 36$, and every cubic word T,*

$$(T/J(T))^G = 0, \qquad H^1(G, T/J(T)) = 0.$$

Proof. Put $K = \mathbb{F}_{2^{2f}}$, $\sigma : x \mapsto x^{2^f}$ and $m = \lfloor n/2 \rfloor$. The parabolic P stabilizing a maximal isotropic subspace has odd index, so restriction on cohomology is injective. We give its radical calculation explicitly; all coefficient scalars may be extended for this calculation.

For $n = 2m$ write $V = A \oplus A^{\sigma,*}$. The Levi is $L' = \{g \in \mathrm{GL}_m(K) : \det(g) \in \mathbb{F}_{2^f}^*\}$, and its abelian radical consists of

$$u(Z) = \begin{pmatrix} I & Z \\ 0 & I \end{pmatrix}, \qquad Z + Z^\dagger = 0.$$

Its additive module, after prime-field scalar extension, is a direct sum of whole products $A^{[a]} \otimes A^{[a+f]}$. The Levi restriction of T is a sum of cubic general-linear words; that of J is a sum of natural words. Lemma 6.6 and the exact sequence give $H^1(L', T/J) = 0$. Normalize a cocycle on L'. Give the lower summand weight zero and A weight one. The radical strictly raises weight and J is homogeneous. A first nonzero weight of the radical cocycle is an additive Levi map from one of the whole quadratic products S to $(T/J)_d$. But

$$\mathrm{Hom}_{L'}(S, T_d) = 0, \qquad \mathrm{Ext}^1_{L'}(S, J_d) = 0$$

by odd-length-five invariants and cubic H^1, respectively. The Hom exact sequence forbids such a map. Thus the radical cocycle is zero.

For $n = 2m + 1$, order $V = A \oplus Kt \oplus A^{\sigma,*}$ and put $\chi(g) = (\det g)^{2^f-1}$. The Levi is the full $\mathrm{GL}_m(K)$, embedded as $\mathrm{diag}(g, \chi(g), g^{-\dagger})$. The radical is

$$u(X,Z) = \begin{pmatrix} I & X & Z \\ 0 & 1 & X^\dagger \\ 0 & 0 & I \end{pmatrix}, \qquad Z + Z^\dagger = XX^\dagger,$$

with product $u(X,Z)u(Y,W) = u(X+Y, Z+W+XY^\dagger)$. Its center C is the Hermitian $X = 0$ subgroup, and $U/C = A \otimes \chi^{-1}$. Give the three summands weights two, one, zero. For twist r_i their Levi constituents are $A^{[r_i]}, K_{\chi^{2^{r_i}}}, (A^{[r_i+f]})^*$.

Levi H^1 of T vanishes blockwise. Three head occurrences use cubic vanishing; two head occurrences and one tail have a nontrivial determinant twist; one head occurrence is either a nontrivial determinant twist or a natural word; zero head occurrences are determinant characters. The same lemma gives $H^2(L, J) = 0$, including the one-dimensional characters. Thus normalize on L. A first central-radical weight would be a map from a Hermitian quadratic product S to $(T/J)_d$. For three head occurrences $\mathrm{Hom}(S, T_d)$ is an odd word of length five; for two it has a nontrivial determinant character; for one it is either such a character or an odd word of length three; for zero it is a determinant twist of a pure dual quadratic word. Each has zero invariants. Also $\mathrm{Ext}^1(S, J_d) = 0$: a head constituent gives cubic H^1, a tail gives a nontrivial determinant twist of a quadratic word. Therefore $c(C) = 0$.

A first nonzero weight of $c(U)$ now factors through U/C. For its scalar component $W_a = A^{[a]} \otimes \chi^{-2^a}$ one has $\mathrm{Hom}_L(W_a, J_d) = \mathrm{Ext}^1_L(W_a, J_d) = 0$. Head targets have a nontrivial determinant twist, and tail targets reduce to natural/dual coefficients with or without such a twist. Consequently every map to $(T/J)_d$ lifts uniquely to T_d. The possible literal maps have exactly one head occurrence, that occurrence is the matching $A^{[a]}$, and the other two are tails. Such maps occur only if

$$\chi^{2^{r_1}+2^{r_2}+2^{r_3}} = 1,$$

and only in weight four. Their value has the form

$$\gamma_1 X^{[r_1]} \otimes t \otimes t + \gamma_2 t \otimes X^{[r_2]} \otimes t + \gamma_3 t \otimes t \otimes X^{[r_3]}.$$

Here γ_i are scalar parameters and r_i are the specified twist indices. Commutators of radical elements lie in C. Since $c(C) = 0$, the cocycle identity gives $(u(X) - I)c(u(Y)) = (u(Y) - I)c(u(X))$. In weight five, project to the summand with heads in positions i, j. Its value is

$$(\gamma_i - \gamma_j)(Y^{[r_i]} \otimes X^{[r_j]} + X^{[r_i]} \otimes Y^{[r_j]}) = 0.$$

Two different coordinate vectors make the summands independent, hence all three parameters agree. There is no quotient ambiguity: J has weights only two, three, four. Subtract the coboundary of the corresponding tail cube, which is Levi-fixed by the character condition. It removes the only possible first weight and preserves $c(L) = c(C) = 0$. Thus $c(U) = 0$. Odd-index restriction proves the claimed H^1 in both parities.

The same argument on natural coefficients proves $H^1(G, V^{[a]}) = 0$. In even dimension a quadratic radical has no natural target. In odd dimension first kill the center similarly; the quotient radical $A \otimes \chi^{-1}$ has no map to a natural, dual, or tail constituent, by the determinant-character test. Since J is a direct sum of natural twists and $T^G = 0$ by Corollary 6.3, the exact sequence gives $(T/J)^G = 0$. □

Lemma 6.9 (Orthogonal cubic quotients in characteristic two)**.** *For either even orthogonal derived group with Witt index at least twelve, and every cubic word T, the quotient $T/J(T)$ has zero invariants and zero first cohomology.*

Proof. Use the polar form to identify dual occurrences. Write $V = A \oplus Z \oplus A^*$, $\dim A = m$, where $Z = 0$ in split type and Z is an anisotropic quadratic plane otherwise. The maximal-singular-space parabolic has odd index. Its Levi is $\mathrm{GL}_m(K)$ in split type and $\mathrm{GL}_m(K) \times S$, with $S = \mathrm{SO}(Z)$ cyclic of order $|K|+1$, in nonsplit type. In split type the radical is the additive $\bigwedge^2 A$, with its alternating matrix action. In nonsplit type it is

$$u(X,B) = \begin{pmatrix} I & X & B \\ 0 & I & J_Z^{-1}X^t \\ 0 & 0 & I \end{pmatrix}, \quad b^tBb = Q_Z(J_Z^{-1}X^tb) \quad (b \in A^*),$$

where J_Z is the polar matrix. Every X occurs; the remaining freedom in B is an alternating matrix. Multiplication adds $XJ_Z^{-1}Y^t$ to the upper-right block. Thus the central subgroup $C = \bigwedge^2 A$ has quotient $U/C = \mathrm{Hom}(Z, A)$. These displayed elements lie in Ω: if $r = \operatorname{rank} X$, block elimination gives $\operatorname{rank}(u-I) = 2r + \operatorname{rank}\beta$, where the induced form β on $\ker X^t$ is alternating. Its rank is even. The radical orders, respectively $|K|^{m(m-1)/2}$ and $|K|^{m(m-1)/2+2m}$, multiplied by the two-part of the Levi order, are $|K|^{m(m-1)}$ and $|K|^{m(m+1)}$, exactly the two-parts of the corresponding orthogonal derived-group orders [Tay92, Chapter 11]. This also verifies odd index.

Let $Q = T/J$. Lemma 6.6 gives $H^1(L, Q) = 0$ in split type. In nonsplit type diagonalize the odd cyclic S after extending scalars. Each twisted tail has two nontrivial characters. A block with three heads uses cubic H^1; two heads and one tail is killed by its nontrivial S-character; one head uses natural H^1 when its tail character is trivial; zero heads uses the absence of homomorphisms from $\mathrm{GL}_m(K)$ to an additive characteristic-two group. Also $H^2(L, J) = 0$: head insertions are natural words and tail insertions have nontrivial S-character. Normalize $c(L) = 0$.

In split type give A weight one and A^* weight zero. A first nonzero radical value is a map $\bigwedge^2 A^{[a]} \to Q_d$. Precompose with $A^{[a]} \otimes A^{[a]}$. Its Hom into T_d is an odd invariant problem of length five, and its Ext into J_d is cubic H^1. Both vanish. Therefore $c(U) = 0$.

In nonsplit type give A, Z, A^* weights two, one, zero. First kill $c(C)$ by the same precomposition. In the Hom calculation, three heads give length-five odd invariants; two heads give a nontrivial tail character; one head gives length-three odd invariants if its tail character is trivial; zero heads give a pure dual quadratic word. For Ext into J, a head gives cubic H^1 and a tail has nontrivial S-character. All vanish.

Now $c(U)$ is constant on C-cosets and is C-fixed. A first nonzero weight is a map from a scalar component W_a of $\mathrm{Hom}(Z, A)$ to Q_d. Its Hom and Ext into J_d vanish: a head target leaves the nontrivial S-character of W_a; a tail target leaves a natural dual coefficient. Hence it lifts uniquely to T_d. The only possible literal maps have one matching A head and two tails, in weight four. Write them

$$F(X) = X_1(b_1) + X_2(b_2) + X_3(b_3), \qquad b_i \in (Z^{[a_1]} \otimes Z^{[a_2]} \otimes Z^{[a_3]})^S.$$

The notation X_i applies X with the specified twist in position i. This description includes every tail character; an incompatible position has $b_i = 0$. Uniqueness follows by varying every coordinate functional $Z \to K$ in the corresponding output coordinate of A. The commutator identity, projected to weight five with heads in positions i, j, gives

$$(Y_iX_j + X_iY_j)(b_i + b_j) = 0.$$

Choose X, Y with images on different coordinate axes of A. One output coordinate isolates Y_iX_j. Varying the two maps $Z \to K$ separates both tail tensor positions and forces $b_i = b_j$. Thus all are

one S-fixed tail tensor b. Subtract its coboundary. It is Levi- and C-fixed, has weight three, and removes exactly F. There is no weight-five insertion, because a pairing has weight two and the free occurrence has weight at most two. The preceding classification forbids any later first weight. This kills $c(U)$ and proves $H^1(G, Q) = 0$ by odd-index restriction.

On natural coefficients the identical argument is shorter: neither the exterior-square center nor $A \otimes Z$ has an equivariant map to a natural head or tail, by odd-word invariants and the nontrivial tail characters. Thus $H^1(G, V^{[a]}) = 0$. The exact sequence, the direct-sum insertion parameters, and Corollary 6.3 give $Q^G = 0$. Inflation–restriction also proves these conclusions for the full orthogonal group, since its normal Ω has zero coefficient fixed space. □

Theorem 6.10 (Cohomology of canonical cubic quotients)**.** *For every stable product of natural classical derived cores and every cubic word T,*

$$H^1(D, T/J(T)) = 0, \qquad (T/J(T))^D = 0.$$

The assertion holds after prime-field descent, with fixed multiplicities, and after passage from a product cover to the actual central product.

Proof. Odd characteristic follows from Lemma 6.5. For one factor in characteristic two use the three preceding lemmas. For the special-linear case Lemma 6.6 gives $H^1(T) = 0$ and $H^2(J) = 0$ since J is an injective sum of natural twists; its transfer argument applies to T/J. Natural H^1 and $T^D = 0$ also give the fixed-space assertion. For pattern $1 + 1 + 1$, the degree-one product formula has a fixed-space factor from another nontrivial natural module in every summand and is zero. For $2 + 1$, write the quotient as $(W/I) \otimes C$. The natural factor C has no invariants. Also $(W/I)^{D_1} = 0$: for a matching pair identify W with an endomorphism module and I with its scalar field. A fixed coset $X + I$ gives an additive homomorphism $g \mapsto gXg^{-1} - X$ from the perfect core to the scalar field. It vanishes, and the actual natural commutant is the scalar field. For a nonmatching pair its ordinary invariant space is zero. The product formula proves $H^1 = 0$. These product arguments apply to each recognized factor identification, not to an enlargement of an arbitrary supplied subgroup. Scalar extension is exact and cohomology is the kernel modulo image of finite-dimensional linear systems, so all assertions descend. Inflation is injective in degree one for the product-to-central-product map, and fixed vectors are unchanged. □

6.3 Exhaustion of the simple cubic types

Lemma 6.11 (A useful socle criterion)**.** *Let $P \leq T$ have at least one nonquadratic composition factor but at most one isomorphism type of such factors. Then that simple type embeds in $T/J(T)$.*

Proof. The module $P/(P \cap J(T))$ is nonzero and embeds in $T/J(T)$. Its nonzero socle has no quadratic constituent by Lemma 6.4. It therefore contains the unique nonquadratic type. No complement to P or $J(T)$ is asserted. □

Theorem 6.12 (Canonical cubic targets)**.** *From the actual natural types and twists one can form a polynomial-size family of quotients $C_j = T_j/J(T_j)$ such that every nonquadratic simple constituent of the given cubic cover embeds in some C_j, and no quadratic simple embeds in any C_j. Each C_j has zero fixed space and zero first cohomology for the full recognized core. All maps and Hom spaces are computed over the actual prime field. For a semisimple module Q, the kernel of the joint map formed from a basis of all $\mathrm{Hom}_D(Q, C_j)$ is exactly its sum of quadratic simple summands.*

Proof. We give the complete degree-three type check over an algebraic closure. Write ω_i for fundamental weights, normalize Frobenius digits, and identify duals by forms where appropriate. The finite restriction theorem and its twisted version are used only after this check [Ste67, Theorems 41 and 43].

In odd characteristic, for an equal-digit pure cube the possible cubic highest weights are $3\omega_1$, $\omega_1+\omega_2$, and ω_3. For symplectic and orthogonal restrictions a natural factor may also occur. In odd orthogonal type the extra zero weight introduces only lower-degree possibilities; the full-group parity comparison excludes the even ones. For characteristic greater than three use the symmetric, alternating, and standard idempotents of the position group S_3. In the symmetric part the weights $3\epsilon_1$, $2\epsilon_1+\epsilon_2$, and $\epsilon_1+\epsilon_2+\epsilon_3$ have multiplicity one. The highest simple $3\omega_1$ supplies all three: restriction to the SL_3 on three positive coordinates contains its simple symmetric cube, since all raising/lowering coefficients one, two, three are nonzero. Thus no other high type occurs in this part. The alternating part has only high type ω_3. In the standard part the first weight is absent, the second has multiplicity two and the third multiplicity four. The two copies of $\omega_1+\omega_2$ each supply a two-dimensional middle weight by the simple SL_3 adjoint module; these exhaust the third weight. This part has only that high type. Apply Lemma 6.11.

In characteristic three, $3\omega_1$ is Frobenius-natural and quadratic. Use the actual submodules $\Gamma^3V=(V^{\otimes 3})^{S_3}$ and the image of the alternating tensor map $\bigwedge^3 V\to V^{\otimes 3}$. The latter map is injective, because its distinct basis triples have disjoint six-term supports with nonzero signs. It has only high type ω_3. In Γ^3V, both $2\epsilon_1+\epsilon_2$ and $\epsilon_1+\epsilon_2+\epsilon_3$ have multiplicity one. The first forces one copy of $\omega_1+\omega_2$. Its SL_3 highest-vector submodule is a quotient of the adjoint Weyl module; its simple central quotient has one-dimensional middle weight in characteristic three. Thus this copy exhausts the second weight, excluding ω_3. Again the socle criterion applies, without splitting the position-group action.

For an odd-characteristic mixed general-linear word $V\otimes V\otimes V^*$, the possible weights are $2\omega_1+\omega_{n-1}$, $\omega_2+\omega_{n-1}$, and ω_1. Split the first two positions symmetrically and alternatingly. The symmetric highest simple supplies the one-dimensional weight $\epsilon_1+\epsilon_2-\epsilon_n$, by the simple degree-two SL_2 representation, so the other high type is absent there. The alternating part has exactly the second high type. These statements hold also at characteristic three. The dual cases are obtained by duality.

We next give the two equal-digit exceptions in characteristic two, including their required actual submodules. For the pure general-linear cube, put $c=(123)$, $e=1+c+c^2$, and $\varepsilon=(1+(12))e$. The principal position block is a module for $R=K[\varepsilon]/\varepsilon^2$ of the form $R^{\binom{n}{3}}\oplus K^{n^2}$. Its filtration

$$\operatorname{im}\varepsilon\subseteq\ker\varepsilon\subseteq eT$$

has successive modules $\bigwedge^3 V$, $V^{[1]}\otimes V$, $\bigwedge^3 V$. For the middle map send $x^{[1]}\otimes y$ to the cyclic sum of $x\otimes x\otimes y$ modulo $\operatorname{im}\varepsilon$; its additive cross term is exactly the alternating six-term tensor. Basis words verify the identification. The complementary standard position block has the form $W\oplus W$, where W is the simple highest-weight module $\omega_1+\omega_2$. Here is an actual image proof, including the field of order two, rather than an inference from a commutant.

A reduced polynomial in matrix-entry variables, each of whose monomials has support at most s, cannot vanish on all invertible matrices when $n\geq 2s$, unless it is zero. Choose a monomial of maximal support S, with row and column sets R_0, C_0. A permutation matrix A can be chosen so that $A^{-1}X$ has disjoint row and column supports for every X supported on S. Then $(A^{-1}X)^2=0$ and every $A+X$ is invertible. On this affine box the coefficient of a monomial with support exactly S is its original coefficient: another contributing monomial would strictly contain S. A reduced polynomial vanishing on the box has all coefficients zero. Remove the maximal supports successively.

In characteristic two these permutation and square-zero matrices also have determinant one. With $s = 3$ this proves independence of all reduced cube coefficient functions on the finite special-linear group.

Let $u = \binom{n}{3}$, $v = n^2$, and $w = n(n^2-1)/3$. The full commuting algebra of the position group has principal block

$$\begin{pmatrix} A & 0 & 0 \\ D & E & 0 \\ B & C & A \end{pmatrix}$$

with all displayed blocks free, together with $\mathrm{Mat}_w(K)$. Its dimension is $2u^2+2uv+v^2+w^2 = \binom{n^2+2}{3}$. For $|K| \geq 4$, all cubic matrix-entry monomials are reduced, and the preceding independence gives exactly this image dimension. For $K = \mathbb{F}_2$, with $N = n^2$, the coefficient dimension is $N+\binom{N}{2}+\binom{N}{3}$. The middle block is the actual square image, of dimension $N + \binom{N}{2}$, and

$$2u^2 + 2uv + w^2 + N + \binom{N}{2} = N + \binom{N}{2} + \binom{N}{3}.$$

Thus every other block is still full, in particular the standard block. The only high types are consequently ω_3 in the principal part and $\omega_1 + \omega_2$ in the standard part. The middle is literally quadratic, including finite-field Frobenius wrap. The socle criterion covers both.

For an equal-digit symplectic or even orthogonal cube in characteristic two, the possible highest weights are $3\omega_1, \omega_1+\omega_2, \omega_3, \omega_1$. The first is $V \otimes V^{[1]}$ and is quadratic. Let j_i, c_i be the three polar insertions and contractions; write j, c for their joint maps. Their scalar matrix is $cj = G \otimes I$, with $G = I_3 + 11^t$ and $G^2 = G$. Set $e_{\mathrm{st}} = (123) + (132)$, $E = jGc$, and $f = e_{\mathrm{st}} - E$. Then $E^2 = E$, $e_{\mathrm{st}}E = Ee_{\mathrm{st}} = E$, $f^2 = f$, and $fj_i = c_i f = 0$. This follows directly from $cj = G$ and the action of e_{st} on the three insertion positions. The tensor pairing makes f self-adjoint. In degree three the algebraic-group commutant is the span of the fifteen pairing diagrams. For the symplectic group this is symplectic Schur–Weyl duality [DDH08]; for even orthogonal type in characteristic two it is the multilinear degree-six consequence of [DF04, Theorem 4.9], with the natural dimension strictly greater than six. Only this commutant statement is used. Write $\mathbf{G}$ for this connected algebraic group; the following simplicity argument is initially an argument for its rational modules. Since f kills the contraction ideal, $f\,\mathrm{End}_{\mathbf{G}}(T)f = \mathrm{Mat}_2(K)$, the standard position block. It is nonzero on weight $2\epsilon_1 + \epsilon_2$, whereas E has only natural weights. Thus $fT = W \oplus W$ with $\mathrm{End}_{\mathbf{G}}(W) = K$. Self-adjointness and Krull–Schmidt make W self-dual. All the possible four simple types are self-dual. Such a Schurian module is simple: a simple socle and its dual quotient would otherwise give a nonzero noninvertible endomorphism. The highest weight is $\omega_1 + \omega_2$. It is restricted, so the finite restriction theorem makes the same W simple for the actual finite derived group. No equality with a finite-group commutant is used here.

The weight $3\epsilon_1$ has multiplicity one in T, hence its simple occurs once and contributes one dimension at $2\epsilon_1 + \epsilon_2$. The other two dimensions at that weight are supplied by fT. Therefore $(1-f)T$ has no further $\omega_1 + \omega_2$. Its exterior highest vector has weight ω_3 and belongs to the principal position block. Neither remaining low type supplies that weight. The socle criterion on $(1-f)T$ proves that ω_3 also embeds in T/J.

All remaining signatures follow from the square list, which we record explicitly to exhaust the cases. In characteristic two an equal-variance equal-digit square has exterior-square, Frobenius-natural, and possibly trivial types; an opposite-variance square has adjoint and trivial types. In odd characteristic split an equal-variance square by $(1 \pm (12))/2$, obtaining its two separate quadratic types, and possible trivial types for form-preserving groups. An opposite-variance general-linear square again has adjoint and trivial types. With exactly two equal digits, tensor each resulting

quadratic type with the natural occurrence at the other digit. Steinberg's tensor-product theorem makes it simple because the digits are distinct. Frobenius-natural square factors give literal words of length two, and trivial square factors give natural words. Each selected submodule therefore has at most one nonquadratic type, and the socle criterion applies. Three distinct digits give a simple tensor product, already its own target.

For a mixed equal-digit characteristic-two general-linear or unitary word $V \otimes V \otimes V^*$, the possible weights are $2\omega_1 + \omega_{n-1}, \omega_2 + \omega_{n-1}, \omega_1$. The first is $V^{[1]} \otimes V^*$, hence quadratic. The second is the sole high type, so the same criterion applies. For unitary groups normalize digits modulo the fixed-field degree, reversing natural/dual variance at each wrap; all carries remain quadratic. For two independent factors with counts $2 + 1$ use the same square decomposition, now with an external simple product. Three independent factors give an external product of three natural simples. These exhaust all partitions of three occurrences.

Each displayed high weight is restricted at its digit. In characteristic three the only offending weight $3\omega_1$ was explicitly removed as Frobenius-natural. Finite restriction preserves and distinguishes the remaining simple weights. The unitary graph action reverses the recorded variance, and the stable type-D graph action fixes the low-index weights used here. No low and high type fuse, since the low catalog consists only of trivial, one natural digit, one square type, or two natural digits. Prime-field Galois descent preserves this catalog. Rather than choosing independent splitting-field labels, compute the joint Hom spaces over the prime field; rank and existence of embeddings descend by linear algebra.

There are only constantly many position partitions for each input word; thus the constructed family has polynomial explicit length. Its absence of quadratic socle is Lemma 6.4, and its cohomology is Theorem 6.10. Finally a semisimple module is a direct sum of simples. All maps from its quadratic simples to the targets are zero, whereas each high simple is separated by some map into a target. Stacking a basis of every Hom space therefore has exactly the claimed kernel, including repeated copies and their endomorphism fields. □

7 Complete isomorphism of ordinary three-tensors

Let $q = p^e$, and let n_1, n_2, n_3 be the three argument dimensions over $\mathbb{F}_q$. A tensor is supplied by its coefficients. Its transporter is taken in the product of the three argument general-linear groups, unless a subgroup is explicitly supplied. Our output convention is one witness and generators of its entire source stabilizer, in the original argument matrices. We work over $k = \mathbb{F}_p$ and retain the original $\mathbb{F}_q$-linearity equations. Write d for the total dimension of the original faithful argument representation, with formal duals if used; thus $d = O(e(n_1 + n_2 + n_3))$. Auxiliary coefficient dimensions are not included in d.

Theorem 7.1 (Complete ordinary tensor isomorphism)**.** *There is a deterministic algorithm which, for two three-tensors of format (n_1, n_2, n_3) over $\mathbb{F}_q$, returns their complete transporter in*

$$q^{O(n_1+n_2+n_3)} \operatorname{poly}(L)$$

time. The same bound holds in an explicitly supplied subgroup on the original argument spaces, with full original witnesses and stabilizer kernels. The polynomial depends on the explicit coefficient and generator input length L, not exponentially on the number of coefficients.

We prove the theorem below. Only the completed quadratic transporter, Theorem 5.17, is used as a transporter subroutine. The representation lemmas already proved above do not use a cubic transporter, and the more general supported cubic algorithm occurs later.

7.1 Exact coefficient fibers

Lemma 7.2 (Sequential affine fibers)**.** *For a linear observation $\epsilon : U \to E$ and $P \le U$, put $U_0 = U \cap \ker \epsilon$ and $P_0 = P \cap \ker \epsilon$. Then*

$$0 \longrightarrow U_0/P_0 \longrightarrow U/P \longrightarrow \epsilon(U)/\epsilon(P) \longrightarrow 0$$

is exact. If P is the image of an actual translation homomorphism, normalization of the observed value lifts through that homomorphism.

Proof. The last map sends $u + P$ to $\epsilon(u) + \epsilon(P)$. If its value is zero choose $p \in P$ with $\epsilon(p) = \epsilon(u)$; then $u - p \in U_0$. Two choices differ in P_0. The recorded preimage of p supplies the actual normalization. Both numerator and denominator are restricted; keeping the old numerator would give the wrong fiber. □

Lemma 7.3 (Translations of a coefficient layer)**.** *Let R be the kernel of the current action on the quotients of retained argument flags. On a coefficient layer, after all previous coefficients have been fixed, R has identity linear action and its affine increment $\eta : R \to M_+$ is a homomorphism. Its image P is invariant under the current group and all its simple factors occur in Hom spaces between original argument-flag quotients. Its image basis, complete kernel, and original preimages are computable within the original representation bound.*

Proof. The cocycle equation becomes addition on R, and conjugation gives $\eta(hrh^{-1}) = h\eta(r)$. Let I be the nilpotent algebra of operators strictly lowering the argument flags. Filter R by $R \cap (1 + I^i)$. The map $r \mapsto r - I$ embeds each consecutive quotient in I^i/I^{i+1}, a sum of the indicated Hom spaces. Its image filtration on P proves the composition-factor statement. Evaluate η on constructive group generators and use the exact additive-kernel procedure of Lemma 3.1; retain preimages of an additive basis. All operations use prime-field linear algebra. No defining-field closure of P is taken. □

Lemma 7.4 (Normal-core affine descent)**.** *On a coefficient layer with three large natural occurrences, let J be the full insertion submodule and let P be its actual translation image. After a computable coordinate-origin change, the entire cubic cocycle is insertion-valued. The actual denominator P is unchanged. No subgroup section of the flag-kernel extension is required.*

Proof. By Lemmas 7.3 and 6.4, $P \le J$ on the cubic projection. Hence the affine action on $A = M/J$ factors through the actual associated-graded group H/R. Let D be its recognized normal product of derived cores. Theorem 6.10 gives $H^1(D, A) = 0$ and $A^D = 0$. Solve $(\rho(g) - I)b = c(g)$ on its generators. Any actual H-preimage can be used, since changing that preimage adds an element of J. Vanishing gives existence and the zero fixed space gives uniqueness. After this origin change the cocycle is zero on D. For $h \in H/R$ and $a \in D$, normality and the cocycle equation imply $ac(h) = c(h)$. Therefore $c(h) = 0$ in A. A lift of b in M makes the original cocycle insertion-valued. Since R has identity linear action, the coordinate shift leaves $\eta(R)$ unchanged. The chosen lift need not be an available group translation. □

Here is the exact form of the residual used throughout the algorithm. Let $J_0 : S_0 \hookrightarrow M$ be the faithful insertion parametrization, so $S_0 = S/\ker J$; let $R_0 = J^{-1}(P)/\ker J$. On the scalar-extraction class constructed below there is an equivariant map $s : M \to S_0$ with $s(t + J_0 r) = s(t) + r$ whenever both tensors are in that class. The two values are

$$C(t) = t - J_0 s(t), \qquad s(t) + R_0. \tag{7.1}$$

If the current action is $h * t = \rho(h)t + c(h)$, with $c(h) = J_0 a(h)$, then

$$C(h * t) = \rho(h)C(t), \qquad s(h * t) = \rho_S(h)s(t) + a(h).$$

Thus the first entry is an ordinary linear tensor observation and the second is a quadratic-or-lower affine observation. Their common transporter is exactly the original quotient transporter. If a lower-degree part shares the translations, use the single map

$$F : M_{\le 2} \oplus S_0 \longrightarrow M_{\le 2} \oplus M, \qquad F(x, s) = (x, J_0 s)$$

and its one actual denominator $F^{-1}(P)$. This prevents losing correlations between lower-degree coordinates and insertion parameters.

Definition 7.5 (Recursive input)**.** A recursive input consists of an actual original subgroup or its coset; three active argument subquotients and one simultaneous tensor family; its exact coefficient fiber $t_0 + U/P$ with actual translation lifts and kernel; retained argument flags and every previously imposed condition; and the structural data of Theorem 3.11. All field, duality, diagonal, and label actions are retained. New flag refinements occur inside old intervals. An entire completed family retains its upper triangular action under subsequent coordinate refinement. During a sequence without a constant-factor argument decrease we retain its initial argument dimensions, natural-type identifications, and whether its present insertion denominator has already been replaced by (7.1). Labels are either fixed multiplicities or have an actual evaluation-module presentation on total cover dimension $O(d)$, with the hereditary bound on every subgroup image and bounded label-stabilizer images required by Theorem 5.19; their realizations have bounded tensor arity on the original rows; no auxiliary space is silently declared to have an independent general-linear action.

Lemma 7.6 (Separating a large core from old complementary actions)**.** *Let $\alpha : H \to A$ be the action on all quotients of the retained flags in every original faithful row, including inactive rows, with kernel R. If a natural–multiplicity cover is currently retained, include every grade of its faithful $O(d)$-dimensional factor-cover realization, including scalar-kernel coordinates, in this action. Let $\beta : H \to B$ be the simultaneous action on retained labels, fields, and other complementary records. Assume every subgroup image of B has no simple section isomorphic to $Q/Z(Q)$, where $Q \le A$ is a newly selected quasisimple subgroup. Then the actual joint image $(\alpha, \beta)(H)$ contains $Q \times 1$. Its full preimage is constructible with original kernel words. The same assertion holds simultaneously for the whole perfect central product of selected components, provided each of its nonabelian simple quotients is excluded. This assertion does not assert a section of $H \to A$.*

Proof. All degree and composition-factor statements are made in the actual physical matrix image; any extra invisible comparison kernel is retained by its full preimage for the final return. In that faithful image the kernel R strictly lowers the complete original flags, so it is a normal p-group. Put $H_Q = \alpha^{-1}(Q)$. The quotient $\beta(H_Q)/\beta(R)$ is a homomorphic image of Q and a section of a subgroup image of B. A nontrivial image of a quasisimple group has its nonabelian simple quotient as a section: every proper normal subgroup of Q is central, and a perfect group has no nontrivial abelian image. The hypothesis therefore makes this quotient trivial. For a central product of components, every nontrivial perfect quotient has a simple section from one of those components, giving the simultaneous assertion as well. For each $h \in H_Q$, choose $r \in R$ with $\beta(r) = \beta(h)$; then $hr^{-1} \in \ker \beta$ has the same α-image as h. Hence $H_Q \cap \ker \beta \to Q$ is onto. Its image in the simultaneous representation is exactly $Q \times 1$. Kernels and preimages are computed in the original represented group by the constructive quotient operations. The hypothesis follows at every use here

from the uniform hereditary bound in Theorem 3.9, applied to the retained actual complementary actions. For several such actions use their joint image; its subgroup sections are controlled by successively taking coordinate kernels, not by multiplying their cardinalities. □

Algorithm 7.7 (Complete ordinary tensor transporter)**.**

1. Normalize an incoming coset and prepare its actual subgroup, refining only the retained flags.
2. Process consecutive coefficient layers with their exact translation images. Complete a layer with at most two large occurrences by Theorem 5.17. On a cubic layer use Lemma 7.4 and retain its one correlated insertion residual.
3. For a group-permuted direct sum of an argument compute each component pair transporter once, in the same original group, and combine them by Theorem 8.1. Keep the unique possible pair exceeding half the argument inside the present restriction sequence.
4. For a new natural multiplicity greater than one enumerate its compatible-frame orbit, retaining the scalar kernel, and put its coordinates in one value family as in Lemma 8.2.
5. For three remaining large occurrences choose a smallest argument and apply Proposition 12.2. Retain every marked-vector condition until the original tensor is complete or a corresponding argument decreases by a factor $1 - 1/128$. Field-value and insertion-only cases are completed before another case is considered.
6. Repeat component and multiplicity preparation after every restriction. A common zero block gives the two intrinsic smaller arms and the exact affine top of Theorem 11.1. A scalar insertion case is replaced once by (7.1), solving its full affine parameter condition before the ordinary tensor condition.
7. Lift all complete child answers, correct through the actual translation map, restore its entire kernel, and generate the full source stabilizer from the complete successful branches. Recompute structure on this actual returned subgroup.

The following subsections prove every reduction and the complete tree bound.

8 Permuted components and multiplicities

An invariant direct-sum decomposition of one argument decomposes the tensor into a family indexed by those summands. The current group may permute the summands. We must therefore match two multisets of component tensors, while using the same transformations on the other arguments and on the value space. This section gives a subset dynamic program for that task. Its important cost property is that each component-pair problem is solved once, before the subset calculation.

A second decomposition, $A \otimes_K M$, has a different role. It describes several isomorphic copies of one natural module. Fixing a compatible basis of the multiplicity space M turns those copies into an ordered tensor family on the smaller argument A. We bound the cost of choosing this basis by the decrease in the argument dimension.

Theorem 8.1 (Fixed-source-prefix multiset transport)**.** *Let Γ act on a set Ω, let $b \geq 1$ be an integer, and let $x_i, y_i \in \Omega$ for $i \in [b]$, allowing repetitions. Suppose every pair transporter $C_{ij} = \mathrm{Trans}_\Gamma(x_i, y_j)$ is supplied as empty or a complete coset, and exact coset intersection is available. The full multiset transporter can be computed using 2^b states, at most $b2^{b-1} - b$ intersections, and $O(b2^b)$ additional group operations.*

Proof. For $J \subseteq [b]$ define

$$D(J) = \mathrm{Trans}_\Gamma(\{x_1, \dots, x_{|J|}\}, \{y_j : j \in J\}),$$

where braces denote multisets. Set $D(\emptyset) = \Gamma$ and $D(\{j\}) = C_{1j}$. For $k = |J| \geq 2$,

$$D(J) = \bigcup_{j \in J} (C_{kj} \cap D(J \setminus \{j\})). \tag{8.1}$$

A solution maps the last source occurrence to some target occurrence. Removing that occurrence proves one inclusion; adding it proves the other, including repeated values.

Let H_k stabilize the source prefix multiset and let G_k stabilize x_k. Every nonempty branch in (8.1) is $g_j K_k$ with $g_j \in \Gamma$, where

$$K_k = G_k \cap H_{k-1}$$

is independent of j and J. Choose a nonempty branch representative $g_0 \in \Gamma$ and let

$$L = \langle K_k,\ g_0^{-1} g_j : \text{nonempty branches} \rangle.$$

Every generator fixes the source prefix, so $L \leq H_k$. Conversely $g_0 h \in D(J)$ for every $h \in H_k$, hence $g_0 h = g_j u$ for some $u \in K_k$, and $h \in L$. Thus $L = H_k$ and the answer is the full coset $g_0 L$. This generation is justified for the entire successful union, not a partial union of unrelated probes.

The number of intersections is at most $\sum_{k=2}^{b} k\binom{b}{k} = b2^{b-1} - b$. Only one branch subgroup list is needed at a state. All nonempty level-k states have the same H_k, so its generators may be reused. □

For example, with two source occurrences x_1, x_2 and two targets y_1, y_2, the last state joins the two complete sets $C_{21} \cap C_{12}$ and $C_{22} \cap C_{11}$. Repeated values cause no difficulty because target occurrences, rather than only distinct target values, are indexed. The common action of Γ is essential to the conclusion that the complete union is a coset.

8.1 Pair calls in the original group

Let I be the finite set of invariant blocks in an active argument and write $Q_1 = \bigoplus_{i \in I} A_i$. Let Γ be the current subgroup permuting these blocks. For $i \in I$, write M_i for the component coefficient space and $t_i \in M_i$ for its value. If $b : \Gamma \to \bigoplus_i M_i$ is the actual affine cocycle, with component $b_i(g)$, write $\rho_i(g) : M_i \to M_{gi}$ for the induced linear map. The action on indexed fibers is

$$g(i, t_i) = (gi, \rho_i(g) t_i + b_{gi}(g)).$$

Normalize an incoming coset once, so both source and target use this same action of Γ. For a pair (i, j), first compute an index transporter r_{ij} and the full index stabilizer Γ_i. Solve the whole fiber equation

$$h(i, t_i) = r_{ij}^{-1}(j, t'_j), \qquad h \in \Gamma_i.$$

The child retains the full ancestor representation with original group words, the two unchanged arguments, all labels, and every invisible evaluation kernel. If its complete result is $h_{ij} S_i$, return $r_{ij} h_{ij} S_i$ in the original Γ.

For example, a global translation space $P = \langle (1, 1) \rangle$ has a full one-dimensional image on either coordinate. The two pair calls must still return cosets in that same diagonal group. Replacing them by independent marginal shift groups would permit incompatible translations.

The exact group-coset intersection used by (8.1) is a consequence of the degree-two subroutine, not a permutation-group primitive. For an integer $d \geq 1$, subgroups $G_1, G_2 \leq \mathrm{GL}_d(k)$, and a matrix $t \in \mathrm{GL}_d(k)$, the product action on $\mathrm{Hom}(k^d, k^d)$ is

$$(g_1, g_2)X = g_1 X g_2^{-1}.$$

The complete transporter of I to t projects onto $G_1 \cap tG_2$. Its source stabilizer is the diagonal copy of $G_1 \cap G_2$. Preparation of the actual product group and the degree-two solver therefore give exact intersection in $p^{O(d)}$; Theorem 5.17 establishes the required noncircular dependency order.

There are at most b^2 pair calls, each computed once. Their costs add. The 2^b dynamic-program states use these completed results and add local work; they do not multiply every child cost. There is at most one source block larger than half its argument, and at most one dimension-compatible target block of that size. Hence there is at most one heavy pair. Keep it in-node with the inherited marks and ties. All other pair calls shrink that argument by at least half.

8.2 Fixing a multiplicity basis

A basis and its evaluation-dual basis must be moved together. This preserves the natural–dual identifications and the scalar kernel in the group of tensor factorizations. The next lemma gives the exact orbit to enumerate and explains why it is small enough.

Lemma 8.2 (The cost of fixing a multiplicity basis)**.** *Suppose a large isotypic original row is $A \otimes_K M$, where $K = \mathbb{F}_{p^f}$, $\dim_K A = a$, $\dim_K M = d \geq 2$. Supply the actual natural–multiplicity group of factorizations, its scalar kernel, field and duality records, and original projection words. A compatible-frame orbit has at most p^{fd^2} elements. Within each frame branch the active row becomes A, with all multiplicity coordinates retained in one label family. If the fixed threshold for selecting large factors implies $a \geq cd$ for an absolute $c > 0$, then*

$$fd^2 = O(fa(d-1)),$$

so the branching exponent is dimension-charged by the actual row loss. The row at least halves.

Proof. Choose a reference basis e of M and its evaluation-dual basis e^*. Compatible frames are

$$\mathcal{F} = \{(eB, e^*B^{-T}) : B \in \mathrm{GL}_d(K)\}.$$

The formal contragredient is used even if an original dual row carries an additional character; that character stays separate. A factorization-group element acts on frame coordinates by

$$B \longmapsto m\theta(B),$$

where $m \in \mathrm{GL}_d(K)$ is the multiplicity matrix and θ is a field map or a field map followed by inverse transpose. The semidirect multiplication makes this a genuine action and $\theta(I) = I$. It is not a homomorphism obtained by forgetting the outer record.

Breadth-first orbit search and Schreier generators on the orbit of I use at most $|\mathrm{GL}_d(K)| \leq p^{fd^2}$ points. The stabilizer has multiplicity matrix part I, while compatible field and exchange records remain. Keep the full central factorization kernel until complete original returns are projected.

Expand M in the fixed frame. Its d coordinates are one entire tensor family on A, with common natural maps, field-balancing equations, old frames, and explicitly realized label action. Conversely these retained factorization equations are exactly the lifting conditions to the original row. The new natural row is realized as $A \otimes e_1$ inside the current grade, with its paired dual component index when an exchange occurs. Its ancestry is therefore still an original invariant subquotient under the frame stabilizer. The new degree is fa, the old degree fad, and their difference is $fa(d-1)$. Since $d/(d-1) \leq 2$ and $a \geq cd$, the displayed bound follows. □

Selection of large factors is measured against original degree: $fa^2 > C_* d$ and original mass $fad_{\mathrm{total}} = O(d)$, where d_{total} is the original total multiplicity of the linked type, imply $a > C'd$, for $d \le d_{\mathrm{total}}$. Only the new local multiplicity is enumerated. Prior frame conditions remain imposed. When $d = 1$, no matrix-frame list is opened; scalar, field, duality and similitude records remain in the current subgroup.

9 Canonical common zero blocks for low-rank matrix spaces

The purpose of this section is to turn low rank into a large common zero block, in a way preserved by every equivalence of the matrix space. A pair of subspaces $U \le X$ and $B \le Y$ with $\mathcal{S}(U) \le B$ has *boundary size* $\operatorname{codim}_X U + \dim B$. In bases adapted to this pair, every matrix of $\mathcal{S}$ is zero from U to Y/B. We first prove a boundary bound over the base field, then construct an invariantly defined pair by noncommutative rank.

In this section, k is the stated base field, $q_k = |k|$, and N denotes the faithful degree of the represented algebra or original rows over k. In the main algorithm $k = \mathbb{F}_p$, so $q_k = p$ and this degree is $O(d)$. Auxiliary module dimensions are not included in that faithful degree.

Lemma 9.1. *Let X, Y be finite-dimensional k-spaces and $R \ge 0$ an integer. Let $\mathcal{S} \le \operatorname{Hom}_k(X, Y)$ be a matrix space whose base-field elements have rank at most R. There are subspaces $U \subseteq X$ and $B \subseteq Y$ such that*

$$\mathcal{S}U \subseteq B, \qquad \operatorname{codim}_X U + \dim B \le 4R.$$

No lower bound on q_k is required.

Proof. Choose a base-field $A \in \mathcal{S}$ of maximum rank a, and use coordinates in which A has an a-dimensional identity block and zeros elsewhere. For $B \in \mathcal{S}$, let b be the rank of its complementary lower-right block. Reduce that block to an identity of size b. Its row and column pivots can clear the adjoining blocks of B without changing A, since the corresponding rows and columns of A vanish. Thus the two matrices B and $A + B$ have a common separate identity block of size b, and their remaining parts differ by a rank-a matrix. The rank triangle inequality gives

$$\operatorname{rank} B + \operatorname{rank}(A + B) \ge 2b + a.$$

Both matrices are base-field elements of $\mathcal{S}$, so $b \le a/2$. In particular the complementary projected matrix space has maximum base-field rank at most $a/2$.

Delete the a pivot-domain directions and allow the a pivot-image directions in the boundary pair, then repeat on that projected space. The successive maximum ranks are at most $R, R/2, R/4, \ldots$. At termination the remaining projected space is zero. The sum of deleted domain and allowed image dimensions is at most $2\sum_{j \ge 0} R/2^j = 4R$. This argument only uses A, B, and $A + B$ over the original field. □

The preceding proof establishes existence; the algorithm does not need to find its maximum-rank matrix. The next proposition constructs the canonical pair by noncommutative rank and a Wong sequence, using only explicit linear-algebra operations and the cited deterministic algorithm.

For a positive integer e, the blow-up of $\mathcal{S} \le \operatorname{Hom}_k(X, Y)$ is

$$\mathcal{S}^{[e]} = \mathcal{S} \otimes_k \operatorname{Mat}(e, k) \le \operatorname{Hom}_k(X \otimes_k k^e, Y \otimes_k k^e).$$

Its noncommutative rank is $\operatorname{ncrk}(\mathcal{S}) = \sup_{e \ge 1} \operatorname{mrk}(\mathcal{S}^{[e]})/e$. For a subspace $U \le X$, its deficiency is $\dim U - \dim \mathcal{S}U$. After padding to a common square dimension d, a subspace of maximum deficiency has minimum boundary size $d - \max_{U \le k^d}(\dim U - \dim \mathcal{S}U)$. The noncommutative-rank theorem identifies this minimum with $\operatorname{ncrk}(\mathcal{S})$ and computes the necessary blow-up.

Proposition 9.2. *There is a deterministic canonical choice of a pair $(U, \mathcal{S}U)$ with*

$$\operatorname{codim}_X U + \dim \mathcal{S}U \le 4R.$$

For the original pencils here its time bound is $q_k^{O(N)} \operatorname{poly}(\log q_k)$. More generally the construction has only polynomial dependence on matrix dimensions, apart from a conservative $q_k^{O(1)}$ finite-field factor.

Proof. Write ρ for noncommutative rank. Every boundary pair in Lemma 9.1 bounds the rank of every blow-up $\mathcal{S} \otimes \operatorname{Mat}(e, k)$ by $4Re$: on $U \otimes k^e$ its image lies in $B \otimes k^e$, and the complementary domain has dimension $e \operatorname{codim} U$. Hence $\rho \le 4R$.

The deterministic noncommutative-rank algorithm [IQS18, Theorem 1.5 and Corollary 1.7(2)] supplies an optimal-rank base-field blow-up

$$B \in \mathcal{S} \otimes \operatorname{Mat}(e, k), \qquad \operatorname{rank} B = e\rho.$$

Its small-field construction passes through an extension and then uses the regular representation to return a base-field blow-up; an arbitrary extension-field subspace is not used as the output here. The standard shrunk-space characterization gives minimum boundary size ρ.

For completeness, canonicity follows by the following Wong sequence in the blow-up domain. After zero-padding rectangular spaces to a common size d, put $\mathcal{S}_e = \mathcal{S} \otimes \operatorname{Mat}(e, k)$ and

$$U_0 = \ker B, \qquad U_{j+1} = B^{-1}(\mathcal{S}_e U_j).$$

If U has maximum deficiency $e(d - \rho)$, then

$$e(d - \rho) \le \dim(U \cap \ker B) \le \dim \ker B = e(d - \rho).$$

Thus $\ker B \subseteq U$ and $BU = \mathcal{S}_e U$, whence $B^{-1}(\mathcal{S}_e U) = U$. Every sequence term is consequently contained in every maximum-deficiency subspace. The increasing sequence stabilizes in polynomially many steps. At its limit its image lies in $\operatorname{im} B$, because it is contained in a maximum-deficiency subspace, and $BU_\infty = \mathcal{S}_e U_\infty$. Its deficiency is therefore $\dim \ker B$, so it is itself maximum and is the unique smallest such space.

Right changes of basis on the blow-up index preserve $\mathcal{S}_e$. Uniqueness makes U_∞ invariant under all $I_X \otimes \operatorname{GL}(e, q_k)$. These matrices span $I_X \otimes \operatorname{Mat}(e, k)$ in every characteristic. For $e \ge 2$, the difference between I and $I + E_{ij}$ gives every off-diagonal matrix unit. Multiplying the permutation matrix interchanging i, j by such a transvection and taking another difference gives E_{ii} or E_{jj}. The assertion for $e = 1$ is immediate. Thus $U_\infty = U_* \otimes k^e$. Recover U_* by the linear equations $x \otimes e_1 \in U_\infty$. It is the unique smallest original maximum-deficiency subspace. If padding added a zero domain summand, that summand belongs to every maximum-deficiency space and is discarded after descent. The image $\mathcal{S}U_*$ is computed by linear algebra. The resulting original pair is therefore covariant under every left-right equivalence, independently of the blow-up witness chosen.

The finite-field algorithms in the cited theorem are used with their deterministic finite-field implementation. It suffices here to retain a possible polynomial factor in p or q_k, rather than assert an unjustified polynomial dependence on $\log q_k$ alone. A small field is extended only to size polynomial in the matrix dimension, and direct deterministic field construction has that same bound. These costs are absorbed in $q_k^{O(N)}$ on the original modules. The rank R in the statement is the maximum rank over the original field; no large-field commutative-rank identity is assumed. □

10 Compatible field quotients of contraction algebras

In the same-type contraction case, each small tuple of operators has a field quotient acting on most of the argument. These fields need not all be generated by one short tuple. Instead, their pair and triple inclusions determine a consistent system of embeddings. We construct their common field and show that the quotient values form a linear map, including when the previously known scalar field is noncentral in the operator algebra.

A short tuple need not generate a single large field. The following construction uses all contraction triples, while keeping the total work singly exponential in original degree.

Theorem 10.1 (Direct field gluing)**.** *Let k be a finite field and let K/k have degree f. Let $a \ge 1$, $m \ge 1$, and $r \ge 0$ be integers with $2r < a$. Put $A = K^a$, let C be a finite-dimensional k-space, and let $M : C \to \mathrm{End}_k(A)^m$ a simultaneous k-linear family of operators. The constant field K is distinguished in every contraction algebra*

$$B_S = \mathrm{alg}_k(K, I, \{M_j(x) : x \in S, 1 \le j \le m\}).$$

Assume for every $|S| \le 3$ that B_S has a field quotient $L_S = B_S/P_S$ with K embedded, and $A_S = \mathrm{Ann}_A(P_S)$ has K-dimension at least $a - r > a/2$. Then there is a field F/K, with

$$[F : K] = E \le \mathrm{lcm}(1, \dots, a) = \exp(O(a)),$$

and compatible K-embeddings of all the quotient fields L_S into F, with F generated by their images. These embeddings, with the quotient maps $B_S \to L_S$ fixed, are unique up to one common K-Frobenius. Define $\lambda_j(x)$ to be the image in F of the quotient class of $M_j(x)$. The resulting quotient-value map $\lambda : C \to F^m$ is k-linear and satisfies

$$\lambda(x) = 0 \quad \Longleftrightarrow \quad \mathrm{rank}_k[v \mapsto (M_j(x)v)_j] \le fr.$$

The diagram, field, map and all embeddings are computable in $|k|^{O(\dim_k C + fa)}$ times polynomial label work. Conjugation and actual label changes transport the whole construction equivariantly.

Proof. For $S \subseteq T$, $|T| \le 3$, the image of B_S on A_T is a subfield of L_T. Thus A_T is semisimple of one B_S-type. It must be the L_S type, because otherwise A_T and A_S have zero intersection and dimensions summing to more than $\dim A$. Hence

$$P_T \cap B_S = P_S, \qquad i_{ST} : L_S \hookrightarrow L_T$$

are actual compatible K-field embeddings, and $A_T \subseteq A_S$. Write $e_x = [L_x : K] \le a$. The contraction pair algebra is generated by its two singleton algebras, so its field quotient is their compositum and $e_{xy} = \mathrm{lcm}(e_x, e_y)$; the same holds for triples.

Let $F = \mathbb{F}_{|K|^E}$ with $E = \mathrm{lcm}_x\, e_x$. Choose arbitrary singleton and pair embeddings into F. Their pairwise discrepancies are phases

$$d_{xy} \pmod{\gcd(e_x, e_y)}.$$

An embedding of a triple field shows

$$d_{xy} + d_{yz} = d_{xz} \pmod{\gcd(e_x, e_y, e_z)}.$$

For each rational prime $\pi \mid E$, choose a reference x_π with maximal π-adic valuation of e_x and prescribe

$$s_x = d_{x,x_\pi} \pmod{\pi^{v_\pi(e_x)}}.$$

The triangle relation proves every pair congruence at this prime. Integer CRT gives s_x modulo e_x. Twisting singleton embeddings by these phases aligns every pair and triple diagram. Any two solutions differ by one common phase modulo E, since at each prime their differences are restrictions of one constant at maximal valuation. Composita give unique extensions to pairs and triples.

Define $\lambda_j(x)$ as the coherently embedded quotient value of $M_j(x)$. For $z = x + y$, ordinary pencil linearity gives $B_{x,y,z} = B_{x,y}$ and $M_j(z) = M_j(x) + M_j(y)$. Applying the common quotient proves additivity. The same argument proves k-homogeneity. These entries generate all singleton fields and hence F over K.

If $\lambda(x) = 0$, every $M_j(x)$ kills the same A_x, so the simultaneous stack has k-rank at most fr. If some value is nonzero, that operator acts invertibly on A_x, and its rank is at least $f(a - r) > fr$. This proves the exact kernel assertion without a label factor. In the noncentral-old-field case, decompose each original k-linear operator into its K/k Frobenius-semilinear components. On the K-stable field-quotient subspace it commutes with K; independence of field embeddings makes every nonidentity component zero there. Untwisting outputs gives one whole K-linear stack with the same core and rank bound, and with the actual twisted row actions retained.

A genuine tensor transporter conjugates all contraction algebras, unique dominant annihilator subspaces and quotient embeddings. The uniqueness of the gluing gives one common field isomorphism, hence a common Frobenius after source and target are placed in one F. An invertible label change merely replaces a spanning set of the same complete operator image. If the scalar observations are indexed by the dual L^* of a k-space L of dimension m, the construction with moving labels is the value in $\mathrm{Hom}_k(L^*, F)$, not an independent value for each chosen coordinate. □

10.1 Constructing the large field without enumerating its elements

The common extension degree can be exponential in the argument dimension. Arithmetic in that many coordinates is still affordable, but exhaustive enumeration of the field elements would not be. We construct it as a tensor product of small prime-power-degree extensions and find each required small subfield by linear equations.

For $n \geq 2$, put $L(n) = \mathrm{lcm}(1, \ldots, n)$. The quotient $L(n)/L(\lceil n/2 \rceil)$ divides $\binom{n}{\lfloor n/2 \rfloor}$: each new prime power lies above $n/2$ and its extra prime exponent appears in that binomial coefficient. Therefore $\log_2 L(n) \leq n + \log_2 L(\lceil n/2 \rceil) = O(n)$.

Factor the small degrees $e_x \leq a$. For each rational prime $\pi \mid E$, put $d_\pi = \pi^{v_\pi(E)} \leq a$. Construct a degree-$d_\pi$ field K_π/K by enumerating at most $|K|^{d_\pi}$ monic polynomials and testing irreducibility. The pairwise coprime degrees make

$$F = \bigotimes_{\pi \mid E, K} K_\pi$$

a field. Its tensor basis gives arithmetic polynomial in E, hence exponential only in a. This avoids finding a degree-E irreducible by enumeration over $|K|^E$.

To embed a small field of degree $d \leq a$, compute $\ker(\mathrm{Frob}_K^d - I)$ in F. It is the unique d-dimensional subfield. Enumerating its $|K|^d$ elements finds a root of the small defining polynomial. Reuse a basis for each repeated degree. Phase tests involve at most a powers on a small generator. Source and target diagrams use the same constructed F after comparing E; no expensive isomorphism search between two exponential-degree presentations is hidden.

There are at most $|k|^{3 \dim C}$ contraction triples. Algebra/module work is on the original row, and the field arrays have total size $|k|^{O(\dim C + fa)}\,\mathrm{poly}(m)$. This proves the algorithmic estimate.

10.2 The contraction-triple source alternative

The field-quotient hypothesis used above is an outcome of an explicit test. Krull–Schmidt decomposition of each supplied contraction algebra gives either smaller centralizer factors or a unique factor large enough to act through a field. The calculation is over the prime field, so it remains valid when the distinguished older field is noncentral.

For a represented contraction algebra B, perform Krull–Schmidt over the prime working field k, including when its distinguished old field K is noncentral. Write $A = \bigoplus_i Q_i^{b_i}$, let $L_i = \operatorname{End}_B(Q_i)/\operatorname{rad}$ and set $g_i = [L_i : k]$, $\dim_k Q_i = g_i j_i$. Its centralizer has semisimple factors $\operatorname{Mat}(b_i, L_i)$ and ordinary prime-field natural degrees $d_i = g_i b_i$. If some d_i exceeds $f(a-r) > fa/2$, then $j_i = 1$, because $j_i \geq 2$ would exceed total original mass fa. The resulting one-dimensional residue-field module is a genuine field quotient of B. Only now does restriction of that quotient embed the distinguished K: its kernel on the field K is zero. The associated annihilator core is K-stable, and its k-dimension at least d_i is therefore K-dimension greater than $a-r$. Uniqueness follows from the same mass bound. If no such degree exists, every centralizer natural row has prime-field degree at most $f(a-r)$, and the contraction tuple is a geometric decomposition certificate. These alternatives are exhaustive and constructible by represented-algebra operations. No residue field is asserted to contain the old K before the field quotient has been obtained.

For moving labels the value-map kernel is an actual prime-field original subspace. Match a proper nonzero kernel by the whole explicitly realized frame call, retain its subalgebra preserving the subspace, and make a strict unbranched restriction. If the image is injective, for each global phase the value equation kills the large coefficient action; complete the whole remaining at-most-two-large-factor observation before reuniting phases. This is the uniform rule of Section 12; no uncharged phase branch enters outer recursion.

11 Smaller tensors from a common zero block

Low simultaneous rank produces subspaces on which every contraction has a small image. We turn this information into invariant subspaces of the two operator arguments. The resulting zero block makes two quotient tensors intrinsic, and makes the last quotient problem affine rather than quadratic in the choice of lifts. This is the geometric step that reduces argument dimensions.

Let C, B, V, L be finite-dimensional k-spaces and let the current tensor be $T : C \times B \times V \to L$. Put $c = \dim_k C$, $b = \dim_k B$, and $v = \dim_k V$. For $x \in C$, its contraction is $M_x : B \to V^* \otimes_k L$. Suppose a canonical boundary pair $X \leq B$, $E \leq V^* \otimes_k L$ has been matched completely and satisfies $M_x(X) \leq E$ for every $x \in C$. The current group still retains the full value-space action and all previously matched conditions.

Theorem 11.1 (The zero-block reduction)**.** *Assume*

$$h = \operatorname{codim}_B X + \dim E \leq \frac{1}{8}\min(b, v). \tag{11.1}$$

Assume that the explicitly realized old label action has the hereditary complementary-image bound and that current large natural types are taken on the recognized associated-graded argument action. Then the full tensor transporter reduces, with shared changes of basis and exact affine lifts, to completed degree-two calls and three-large-factor children for which at least one corresponding natural slot has degree at most h. The other slots do not increase. No condition involving c is needed.

Proof. Set

$$V_0 = \operatorname{im}(E \otimes L^* \longrightarrow V^*), \qquad U = \operatorname{Ann}_V(V_0).$$

Evaluation is equivariant for the whole boundary-pair stabilizing parent. Hence X and U are actual invariant original subspaces, and

$$T(C, X, U) = 0. \tag{11.2}$$

One must not infer $\dim V_0 \le \dim E$. Fix a newly selected quasisimple factor Q and take its full preimage H_Q under the action on all retained grades of the full original faithful representation, including inactive grades. Its kernel R is a normal p-group. Every simple H_Q-module kills R. The hereditary bound on the actual old label image, and perfectness of Q, therefore make every composition factor of L trivial as an H_Q-module. Tensor a composition series of L^* with E: every composition factor of $E \otimes L^*$ occurs in E, up to multiplicity. Evaluation makes V_0 a quotient of this module. Hence every natural constituent of V_0, and therefore of V/U, has prime-field degree at most $\dim E \le h$. This argument does not assume that R acts trivially on the old argument or on the old label module. Also $\dim(B/X) \le h$.

The two adjacent parts are intrinsic tensors

$$T_1 : C \times (B/X) \times U \to L, \qquad T_2 : C \times X \times (V/U) \to L.$$

Changing either quotient lift changes its value only by (11.2). Match T_1 completely and then T_2 inside its full returned parent coset. Afterward the remaining top on $C \times (B/X) \times (V/U)$ is an exact affine fiber. If quotient-lift changes are $s : B/X \to X$ and $t : V/U \to U$, its change is

$$T_2(-, s(-), -) + T_1(-, -, t(-));$$

the prospective quadratic term $T(-, s(-), t(-))$ is zero. Use the current subgroup image and kernel of these shears, rather than all formal Hom parameters. Lemma 7.2 supplies the exact current numerator and denominator at each step.

The first and top arms already have an original argument B/X of degree at most h. For the second arm, prepare the actual V/U row. Its bounded-factor pieces have at most two remaining large positions and are complete local calls. A large isotypic piece is $A_\tau \otimes_{K_\tau} M_\tau$ with $\dim_k A_\tau \le h$. Match its genuine one-axis component family by Theorem 8.1, with all C and X maps shared. Apply Lemma 8.2 immediately if needed. The resulting active natural row is A_τ, not its full isotypic original mass. The pullback along

$$A_\tau \otimes_k M_\tau \twoheadrightarrow A_\tau \otimes_{K_\tau} M_\tau$$

is injective on tensor coefficients, with unmatched-type and field-balancing relations imposed exactly. Thus each unresolved child has the required small row. Its degree is at most one eighth of the corresponding old operator row by (11.1).

Finally complete the affine top, correct by actual original translation lifts, and restore all invisible kernels. The two arms and top determine the entire original tensor, so the resulting coset is exact. □

11.1 The rank bounds needed by the reduction

The geometric proof compares the boundary with the two operator arguments. It makes no comparison with the coefficient argument C. This allows both rank alternatives from Section 12 to use the same theorem. For an off-diagonal low branch, the smallest coefficient dimension gives maximum stack rank $r < c/64$. The canonical boundary pair of Section 9 has $h \le 4r < c/16$, which implies (11.1). For a same-row field branch, let both operator rows be K^a, where $[K : k] = f$, and put $r_K = \lfloor a/32 \rfloor$. Zero field value means that every original labelled operator annihilates the same core

of codimension at most fr_K over k. Consequently the simultaneous stack has rank at most fr_K, and

$$h \le 4fr_K \le fa/8 = \tfrac{1}{8}\min(b, v),$$

even when $c \ll fa$. Requiring $h \le c/16$ here would not be supplied by the field argument and is unnecessary for either arm.

The zero identity (11.2) is retained while the smaller original tensor summands are formed. A correction $T - J(s)$ with s outside the actual allowed translation-parameter space can refill this zero block. Therefore canonical reduction to a tensor is performed inside a child only after that child's smaller original format has been certified.

11.2 The augmented contraction requires a separate argument

A small-boundary pair for an insertion-covariant augmented contraction family does not imply (11.2). Its output contains an anchor and may contain a nonzero graph representing a pure insertion. Section 13 proves separate alternatives: a proper original domain, a proper shear-fixed original output subspace, or an insertion-only degree-one terminal. Theorem 11.1 is applied only to an ordinary tensor or a tensor made ordinary by an exact prior correction.

12 Contractions and invariant subspaces

We have reached a tensor summand with one large natural occurrence in each argument. Permuted components and new multiplicities have already been processed. Our next task is to find a condition whose possible images are inexpensive to enumerate and whose stabilizer is substantially simpler. A contraction in a smallest argument provides such a condition unless all the resulting maps have low rank or nearly scalar action on a large subspace.

Write the simultaneous tensor as $T : C \times B \times V \to L$ over k. Put $m = \dim_k L$ and choose the coefficient argument so that

$$c = \dim_k C \le \min(\dim_k B, \dim_k V).$$

A choice of basis of L gives scalar components T_j for $j \in [m]$. For $x \in C$, the full contraction is

$$M_x : B \to V^* \otimes_k L, \qquad M_x(y)(z) = T(x, y, z), \quad y \in B, z \in V.$$

The component maps are denoted by $M_j(x)$. Their simultaneous rank is $\operatorname{rank}(M_x)$; it is not the sum of their separate ranks.

12.1 Different natural types

Let $E \le \operatorname{End}_k(W)$ be the current semisimple algebra with involution for the natural groups, represented on the direct sum W of their natural modules and the formal duals required for adjoints. In this representation each ordinary simple-algebra type is counted once. If $E \cong \prod_i \operatorname{Mat}(a_i, K_i)$ as an ordinary algebra, its natural degree is $\sum_i [K_i : k] a_i$. This degree measures the remaining independent natural actions. The actual subgroup, field records, and any restrictions on its matrices remain part of the state.

When the two operator arguments belong to distinct ordinary simple components, the contraction occurs in an off-diagonal matrix block. A large-rank map in that block forces a decrease in the natural degree of its centralizer.

Lemma 12.1 (A high-rank contraction)**.** *Suppose the two operator arguments have distinct ordinary simple types. If $x \in C$ has $\rho = \operatorname{rank}(M_x) \geq c/64$, there are at most p^c possible images to test for x. After its simultaneous bilinear contraction has been matched completely, the natural degree decreases by at least $\rho/2$.*

Proof. Let $B_x \leq \operatorname{End}_k(W)$ be generated by $\operatorname{End}_E(W)$, all the off-diagonal maps $M_j(x)$, and their adjoints. The projections onto the distinct simple components lie in B_x. Consequently its commutator ideal $I = B_x[B_x, B_x]B_x$ contains every $M_j(x)$ and its adjoint. The sum of the adjoint images has dimension ρ, so $\dim_k IW \geq \rho$.

Write the Krull–Schmidt decomposition of the B_x-module as $W = \bigoplus_i Q_i^{m_i}$, where each Q_i is indecomposable. Let $K_i = \operatorname{End}_{B_x}(Q_i)/\operatorname{rad}\operatorname{End}_{B_x}(Q_i)$, put $f_i = [K_i : k]$, and write $\dim_k Q_i = f_i j_i$. The semisimple quotient of the centralizer has factors $\operatorname{Mat}(m_i, K_i)$ and natural degree $\sum_i f_i m_i$. If $j_i = 1$, the action on Q_i factors through a field, so I annihilates it. Therefore

$$\rho \leq \dim_k IW \leq \sum_{j_i \geq 2} f_i j_i m_i \leq 2 \sum_i f_i (j_i - 1) m_i.$$

This is twice the decrease in natural degree. The complete bilinear transporter retains the actual current subgroup, its radical actions, and all kernels. For every subsequently selected large factor, Lemma 7.6 supplies an actual record-fixing subgroup mapping onto it. In this ordinary-contraction case that subgroup centralizes the displayed operators literally. The radical-filtration argument of Lemma 13.6 therefore bounds its new natural constituents by those of this centralizer, even if the new flag kernel acts nontrivially on the old row. Further invariant-subspace conditions can only decrease the simple-module degrees. Finally x has at most $|C| = p^c$ possible images. □

If all contractions have rank at most $r < c/64$, Proposition 9.2 supplies a canonical pair $X \leq B$, $E_0 \leq V^* \otimes L$ with $M_x(X) \leq E_0$ for every $x \in C$ and $\operatorname{codim}_B X + \dim E_0 \leq 4r$. Section 11 turns this pair into smaller tensor problems. Both alternatives use the simultaneous contraction, so neither introduces the number of labels into the exponent.

12.2 The same natural type in both operator arguments

When both operator arguments have the same natural type $A = K^a$, with $[K : k] = f$, the contractions can act like scalar operators on a large subspace. A single contraction need not distinguish this possibility. We compare the algebras generated by tuples of at most three contractions.

For a tuple S of at most three vectors in C, define

$$B_S = \operatorname{alg}_k(K, I, \{M_j(x) : x \in S,\ j \in [m]\}) \leq \operatorname{End}_k(A),$$

where alg_k means the unital k-algebra generated by the indicated operators. The copy of K is the distinguished scalar-operator subalgebra. It is not assumed to be central in B_S.

For $a < 64$, the bounded-matrix-size procedure in Theorem 5.17 completes the problem locally. Otherwise put $r_K = \lfloor a/32 \rfloor$. A simple centralizer factor of prime-field natural degree greater than $f(a - r_K) > fa/2$ must come from an indecomposable module of dimension one over its residue field. Indeed, two such residue-field dimensions would exceed the total dimension fa. This yields a unique field quotient of B_S acting on an annihilator subspace of K-dimension at least $a - r_K$.

It follows that one of two cases occurs. Either some tuple has every centralizer natural degree at most $f(a - r_K)$, or every tuple has this unique large field quotient. The first case gives a useful restriction: enumerate the images of the at most three vectors and match all their bilinear contractions simultaneously. There are at most $p^{3c} = p^{O(fa)}$ choices, and the resulting natural

components of both operator arguments have degree at most $f(a - r_K)$. The constant-fraction decrease pays for these choices. Radicals and multiplicities are retained and processed before recursion continues.

The second case gives a coherent field-valued map. The pair and triple algebras determine compatible embeddings of their field quotients. Section 10 proves that they all embed in one field F/K with

$$[F : K] \le \operatorname{lcm}(1, \dots, a) = \exp(O(a)).$$

The quotient values of the contractions define a k-linear map

$$\lambda : C \to \operatorname{Hom}_k(L^*, F) \cong F \otimes_k L. \tag{12.1}$$

It is determined up to one common Frobenius automorphism and is equivariant for the actual field and value-space actions. The dual L^* indexes scalar observations of the L-valued tensor. The construction and its arithmetic cost $p^{O(c+fa)}$ times a polynomial in the labels; it does not enumerate the elements of F.

12.3 Using the field-valued map

There are three possibilities for (12.1). Each has a different algorithmic role.

1. If $0 < \ker \lambda < C$, put $W = \ker \lambda$. This is a proper nonzero invariantly defined k-subspace of the original coefficient argument. Match it by the complete subspace subroutine. The complete subspace procedure is applied in the actual group, and its preimage contains every invisible original transformation. Irreducibility makes this subgroup restriction proper. This conclusion also applies to the quadratic orthogonal group in characteristic two; no replacement by its polar-form isometry group is made. The subspace W is only required to be k-linear.

2. If λ is injective, list the admissible common field automorphisms. For each one, match the whole map (12.1) by a degree-one or full Hom-module call, treating the coordinates of F as auxiliary multiplicity. A remaining coefficient transformation g_C fixing all other recorded actions satisfies $\lambda g_C = \lambda$, so $g_C = I$. Apply Lemma 7.6 to each newly selected large factor and the simultaneous old field and value records. Its record-fixing realization satisfies the same injective equation, so its coefficient-row projection is trivial. Thus the remaining tensor has at most two large natural occurrences. Complete it before considering the next field automorphism, and join all these complete answers locally.

3. If $\lambda = 0$, every labelled contraction annihilates its common large subspace. The simultaneous rank is at most fr_K. The canonical boundary pair then satisfies

$$\operatorname{codim}_B X + \dim E_0 \le 4fr_K \le fa/8.$$

   This is small relative to the two operator arguments, even when c is much smaller. The geometric reduction in Section 11 applies.

12.4 Contractions modulo pairing insertions

A cubic affine quotient can contain the translations classified by Proposition 6.4. Ordinary contractions then change when the tensor representative changes. Section 13 constructs projected contractions for which this change is a controlled scalar term. The same tuple argument either gives a constant-fraction decrease, a proper invariant subspace, or a nonzero quotient field value.

It uses at most six marked vectors, so its exponent is still linear in the corresponding argument degree.

If all quotient values are zero, the scalar extraction from Section 7.1 gives the exact tensor and parameter pair (7.1). Match its parameter quotient completely, then apply the ordinary contraction analysis to its tensor entry under the returned subgroup. This is a change of representation within the current call. Its insertion denominator is processed once.

For one paired occurrence and a larger independent argument, a different construction avoids enumerating vectors in the larger argument. It appends the pairing value to the contraction, making insertion translations into output shears. The resulting small-boundary pair supplies a proper argument subspace, a proper shear-fixed value subspace, or an insertion-only parameter problem. The full calculation is Proposition 13.3; its augmented tensor is used only for these alternatives.

Proposition 12.2 (Complete contraction alternatives)**.** *For a tensor summand with three large natural occurrences, the preceding procedures produce one of the following: a high-rank restriction whose branching is dimension-charged by natural-degree loss; a bounded-tuple restriction giving smaller natural arguments; a strict subgroup restriction without outer branching; a completed field-value or insertion-only base case; the exact tensor and parameter pair; or a common-zero-block reduction. A zero tensor condition is complete immediately.*

Proof. The two operator types are equal or distinct. The distinct-type case is exhausted by Lemma 12.1 and the small-boundary alternative. The equal-type tuple test is exhausted by smaller centralizer factors and the unique field quotient. A linear field-value map has proper nonzero kernel, is injective, or is zero. Proposition 6.4 and Section 13 supply all pairing, duality, twist, and unequal-dimension cases for translation quotients. Every imposed lower-degree or subspace condition is matched completely in the current subgroup. Images of marked vectors remain imposed until the original tensor condition is solved. Hence every genuine transporter occurs in a returned branch, and every branch still retains the exact original condition to be completed. □

13 Pairing insertions and their quotient equations

The translations that survive in a cubic tensor layer are pairing insertions. This section proves their classification and gives contractions that are well defined modulo these translations. It then treats the unequal-dimension case and proves the exact tensor-and-parameter reduction used by the main algorithm. The allowed translation space is retained throughout: a useful parameterization of all insertions does not make every parameter an available group transformation.

Throughout this section k is the prime field, K/k is a finite extension of degree f, and dimensions used for original costs are over k. When old defining fields do not contain the original input field, we take $k = \mathbb{F}_p$, as in Section 7. All original field-linearity equations remain imposed. A label family always means the whole supplied family with its actual common action. In particular, the parameters of an additive translation are not made independent by writing them in separate coordinates.

When the value coordinates can move, let L be their k-space and put $m = \dim_k L$. Scalar observations are indexed by the coordinate functionals in L^*; an index j ranges over $[m]$ after a basis is chosen. All formulas are applied simultaneously in these coordinates, with the induced L-action retained.

13.1 A full-module calculation of the insertion types

A translation comes from a Hom space between original argument quotients. To identify its possible cubic image, we calculate maps from one natural module into three natural modules. Currying converts this to a degree-two calculation, where the full matrix-module image algebra has already been determined.

Let L be the product of full natural classical groups corresponding to the recognized derived cores. In odd characteristic their required joint image comparison is Theorem 4.7. Natural dimensions are in the stable range; bounded natural dimensions are handled by the bounded-factor procedures. A permitted one-factor module is a natural module, its dual, or one of its specified Frobenius twists, with its actual defining-field realization. Fixed multiplicity spaces may accompany any factor.

The insertion classification is Lemma 6.4. In odd characteristic its proof uses the full degree-two intertwiner calculation; in characteristic two the proof by finite odd-order tori applies directly to submodules and does not assume semisimplicity. In both cases the one-factor intertwiners and invariant pairing tensors are computed by linear equations.

The lemma is constructive. Solve the actual one-factor and pairing intertwiner equations, form their three insertion maps, and compute their linear span and kernel. Solving the full Hom equations gives the same space by the proof. Every matrix in these calculations has polynomial dimension in the original degree and explicit label input.

Here is the resulting exhaustive list for a three-large-factor tensor summand. For three independent simple factors there is no insertion. For a paired simple factor and an independent free simple factor there is one pairing times a free-slot functional. For a full general-linear simple factor, only a natural twist and its matching dual twist can pair: the mixed word V, V, V^* has two positions, and the pure three-natural or three-dual word has none. For orthogonal and symplectic simple factors the form identifies a natural with its dual, giving three positions when all twists match and the eligible subset otherwise. For a unitary simple factor its dual is the involution-Frobenius twist: on an individual defining-field signature there are at most two eligible positions. These are exactly the one-pair and general-linear two-position patterns. There is no additional determinant identification in stable natural dimension. Together with Proposition 6.4, this classifies the actual translation image, rather than just a possible list of its composition factors.

13.2 Representing all twists over the original field

Individual Frobenius signatures need not become separate recursive observations. The following formulation keeps the original k-tensor on its original rows.

Let $A = K^a$. In the first-kind case let $Q : A \times A \to K$ be its nondegenerate symmetric or alternating form and put $q_0 = \mathrm{Tr}_{K/k} Q$. In the unitary case let K/K_0 be quadratic with involution σ, let h be Hermitian and linear in its first variable, and put $q_0 = \mathrm{Tr}_{K/k} h$. We assume $k \subseteq K_0$ in this latter notation; restriction to the prime field ensures this. The form q_0 is nondegenerate. Indeed, after choosing a nonzero value of Q or h, varying a K-scalar in one argument tests the nondegenerate field-trace pairing. In the Hermitian case it is symmetric, since trace is invariant under σ.

Lemma 13.1 (All pairing insertions over the base field)**.** *Write b for Q or h, as appropriate. All permitted first-kind or unitary insertions in the original k-trilinear ambient lie in the image of*

$$J : S \longrightarrow \mathrm{Hom}_k(A^{\otimes 3}, k), \qquad S = \mathrm{Hom}_k(A, K)^3, \tag{13.1}$$

$$J(\ell_1, \ell_2, \ell_3)(x, y, z) = \mathrm{Tr}_{K/k}(\ell_1(x) b(y, z) + \ell_2(y) b(x, z) + \ell_3(z) b(x, y)).$$

The statement holds jointly over all labels, with their actual actions. If N is the actual translation image, the allowed parameter space is $R = J^{-1}(N)$, including $\ker J$*, and remains exactly this space. The ambient S is a sum of actual natural-dual copies with passive field multiplicity; it adds no original argument.*

Proof. The commuting algebra of the full natural group is K. Using the nondegenerate form q_0 to identify a bilinear form with an endomorphism, the invariant k-bilinear forms are precisely

$$(y, z) \longmapsto \mathrm{Tr}_{K/k}(c\, b(y, z)), \qquad c \in K.$$

Every product of such a pairing with a k-linear functional u on the free slot occurs in (13.1) by setting $\ell = cu$. These products span all insertions by Lemma 6.4. This also includes all descended Frobenius occurrences. Each space $\mathrm{Hom}_k(A, K)$ is a direct sum of f copies of the underlying natural-dual k-module: choose a k-basis of the codomain K. Its dimension is polynomial in $\dim_k A$. The trace formula is equivariant under the actual form-preserving group by substitution. Semilinear maps, similitudes, value characters, and label actions give the corresponding actual action on S; they are retained, rather than replaced by independent scalar choices. The preimage R preserves exactly N, regardless of whether N is K-linear. □

For a naturally K-valued tensor summand, use the ordered k-coordinate values of its output, or equivalently all trace-dual output functionals. The lemma is then applied to that one entire ordered family. A correction from S may leave a narrower balanced or single-signature subspace. Its result is stored in the full original k-tensor ambient and is not projected back to the narrower subspace. In particular there is no same-size cubic call for each scalar-extension signature.

For a mixed general-linear word the corresponding simultaneous notation is an actual k-bilinear map $B : A \times A \to A$ with

$$J(\alpha, \beta)(x, y) = \alpha(x)y + \beta(y)x, \qquad \alpha, \beta \in \mathrm{Hom}_k(A, K). \tag{13.2}$$

This follows by using the trace evaluation pairing between A and its dual in the preceding argument. For a genuinely different free simple factor C, the one-pair parameters are maps $u : K \to C$ over k, not hypothetical vectors in a shared defining field.

13.3 Projected contractions and noncentral old fields

An insertion changes an ordinary contraction by several terms. Projecting away a small subspace containing the marked vector kills all but one of them, leaving a scalar multiple of the projection. Since that scalar operator already belongs to the generated algebra, the algebra and its field quotients are unchanged by allowed translations.

We first treat (13.1). Set $a \geq 64$ and $r = \lfloor a/32 \rfloor$. For each nonzero $x \in A$ use all marked nondegenerate K-subspaces H of dimension at most two containing x. A nonisotropic vector spans such an H. An isotropic vector has a partner y with $b(x, y) \neq 0$, and their two-dimensional Gram determinant is nonzero. This includes the alternating case, in which every nonzero vector uses a two-plane. Retain x and its actual auxiliary mark, rather than choosing a preferred partner. Let P_v be the orthogonal projection onto $H^{\perp}$ for $v = (x, H)$. There are at most $|K|^{2a}$ choices of marks for one vertex. The zero coefficient contributes zero operators and needs no mark.

For each original k-trilinear label define $L_{j,x}$ by

$$q_0(L_{j,x}y, z) = T_j(x, y, z), \qquad D_{j,v} = P_v L_{j,x} P_v.$$

Here $L_{j,x}$ is only k-linear. It need not commute with K. Define the contraction algebras on the original row by

$$B_v = \mathrm{alg}_k(K, I, P_v, \{D_{j,v}\}_j), \qquad B_Z = \mathrm{alg}_k\{B_v : v \in Z\}. \tag{13.3}$$

For moving labels use the image of the entire label map instead of individual j's. The distinguished copy of K is a constant subalgebra, not an assertion that it is central in B_Z.

Lemma 13.2 (Exact compression and scalar quotients)**.** *Adding* (13.1) *changes the compressed operator by*

$$D_{j,v} \longmapsto D_{j,v} + \ell_{1,j}(x)P_v. \tag{13.4}$$

Consequently all contraction algebras (13.3) *are exactly unchanged. Suppose that every contraction tuple with at most three vertices has a dominant field quotient $B_Z/P_Z = L_Z$ whose annihilator core $A_Z = \mathrm{Ann}_A(P_Z)$ has K-dimension at least $a - r$. The gluing of Section 10 then gives a common field F/K, unique up to one global K-Frobenius, and a k-linear map*

$$\lambda : A \longrightarrow F^{\{j\}},$$

whose entries, together with K, generate F. Its value at x is independent of the projection mark, and adding an insertion changes $\lambda_j(x)$ by $\ell_{1,j}(x)$. The degree $[F : K]$ is at most $\mathrm{lcm}(1, \dots, a)$.

Proof. Since P_v is self-adjoint for q_0,

$$q_0(P_v L_{j,x} P_v y, z) = T_j(x, P_v y, P_v z).$$

Both terms in the insertion involving $b(x, P_v y)$ or $b(x, P_v z)$ vanish. The remaining term is

$$\mathrm{Tr}_{K/k}(\ell_{1,j}(x) b(P_v y, P_v z)) = q_0(\ell_{1,j}(x) P_v y, z).$$

For a Hermitian form this uses its first-variable-linear convention. This proves (13.4). Its increment and negative already belong to the algebra containing K and P_v, so the contraction algebra is unchanged, including every additive correlation among labels.

The restriction of a quotient homomorphism $B_Z \to L_Z$ to K is injective: it is a unital homomorphism from a field. Thus L_Z contains the distinguished K, even if K is noncentral in B_Z. Its core is K-stable. For nested tuples, the two cores have total dimension greater than a and hence meet. The restriction of the larger field type is therefore the unique smaller dominant type. This is exactly the annihilator-core argument of Section 10. All quotient embeddings fix K, and the contraction union algebras give its required pair and triple compositum identities.

The image of P_v in a dominant field is an idempotent, hence 0 or 1. If it were 0, P_v would annihilate a core of dimension at least $a - r$ and would have rank at most r. This contradicts $\mathrm{rank}_K P_v \geq a - 2 > r$. Hence its image is 1, so the singleton quotient fields are generated by the compressed values and K.

For any k-linear L and any K-linear projection of codimension at most two,

$$\mathrm{rank}_k(L - PLP) \leq \mathrm{rank}_k((I - P)L) + \mathrm{rank}_k(PL(I - P)) \leq 4f.$$

Two vertices for the same x consequently differ by rank at most $8f$. For vertices at $x, y, x + y$, the additive defect $D_{j,x+y} - D_{j,x} - D_{j,y}$ has rank at most $12f$. These differences belong to their contraction pair or triple algebra. A nonzero image in its field quotient acts invertibly on the common core and has rank at least $f(a - r) > 12f$. Every displayed defect therefore has zero field value. The same argument with two vertices proves homogeneity under k-scalars. This establishes independence of the mark and genuine k-linearity; it does not assert an equality of compressed contraction algebras at $x, y, x + y$. The coherent embeddings give λ, and the preceding generation observation proves that its entries generate F over K. Under an insertion the same quotient diagram is used, and (13.4) has quotient value $\ell_{1,j}(x)$. □

If the original pencil is genuinely K-linear in its coefficient argument, the same H can be used at x and αx; its quotients fix K, giving K-linearity of λ. No such conclusion is needed for the entire original-field bundle. In particular an extracted scalar functional always belongs to $\mathrm{Hom}_k(A, K)$, whether or not it has the narrower form $Q(u, -)$.

The dominant-field hypothesis in the lemma is an exhaustive outcome, not an extra tensor assumption. Apply the contraction centralizer alternative of Section 10 in the original k-representation. Either some tuple has every ordinary natural degree at most $f(a-r)$, or its unique degree exceeding this bound is a multiplicity of a one-dimensional field module. The latter provides the stated quotient and K-stable core. Testing all tuples and constructing their fields costs $|k|^{O(fa)}$ times a polynomial in the labels. The former tuple is a dimension-charged decomposition: at most six marked original vectors cost $|K|^{6a}$, while each succeeding active row loses a fixed fraction. For projected operators the marked observation consists of the original marks, the projectors P_v, and all ordered classes $D_{j,v} + KP_v$. In the mark stabilizer this is an explicit invariant quotient of an ordered degree-two Hom module. The complete procedure of Theorem 5.17 therefore matches it under the current subgroup. It is a necessary observation: its scalar denominator may contain more than the actual correlated image of N. The full original tensor and N remain to be solved under its complete returned coset. Every genuine original transporter occurs in one of the marked target branches. Lemma 13.6 places every returned large perfect factor in the centralizer of B_Z, so the source centralizer bound applies to the actual successor. In a unitary realization this is the centralizer of B_Z intersected with that of B_Z^*; the added equations cannot create a larger ordinary natural constituent than those of the original centralizer. To see this last assertion, restrict each ordinary simple module of the old centralizer to the subalgebra imposed by the additional equations. The kernel of their direct sum is contained in the old nilpotent radical. Hence every new simple type occurs among these restrictions, and its k-dimension is at most the largest old simple module dimension. One must not claim that a class transporter fixes every $D_{j,v}$ literally.

13.4 The general-linear and balanced one-pair patterns

The preceding form calculation has two variants. A natural–dual pair uses an evaluation projection rather than a self-adjoint projection. When the third argument is an independent smaller space, contracting in it makes the insertion itself scalar, and no projection is needed. We give both formulas so that their marking costs are explicit.

For (13.2), put $L_{j,x}y = B_j(x, y)$. Mark a nonzero x and a K-linear functional ϕ with $\phi(x) = 1$, and put

$$P = I - x \otimes \phi, \qquad D_{j,(x,\phi)} = PL_{j,x}P.$$

The simultaneous natural and contragredient action transports this projection by conjugation. Since $Px = 0$, an insertion changes D by $\alpha_j(x)P$. Its rank is $a - 1$ and its image in a dominant field is 1. The compression error has k-rank at most $2f$, the same-point difference at most $4f$, and the triple additive defect at most $6f$. The proof of Lemma 13.2 applies without alteration with these smaller constants. Perform it on both $B(x, y)$ and $B(y, x)$ to recover both scalar functional arrays. Its actual faithful star realization retains A and A^* with the exchange involution, so every projection has its star-dual partner. A projection on A need not itself be self-adjoint. For a unitary two-position signature the same computation is valid in the actual unitary subgroup. Alternatively its entire original-field bundle is covered by (13.1).

For the one-pair pattern let X, Y be paired K-rows with nondegenerate K-bilinear $Q : X \times Y \to K$, and let the free row be C over k. The tensor summand and its insertions are

$$T_j : X \times Y \to C, \qquad T_j(x, y) \longmapsto T_j(x, y) + u_j(Q(x, y)), \quad u_j \in \mathrm{Hom}_k(K, C). \tag{13.5}$$

If $\dim_k C \le \dim_k X$, contract a coefficient $c^* \in C^*$ and identify X with Y^* by the trace pairing. The insertion becomes a scalar operator $s_j(c^*)I$, where trace duality uniquely gives

$$c^*(u_j(z)) = \mathrm{Tr}_{K/k}(s_j(c^*)z) \qquad (z \in K). \tag{13.6}$$

Thus the uncompressed contraction algebra containing K is already insertion-invariant. The direct gluing proof has zero additive rank defect. Its tuple marks cost $|k|^{O(\dim C)}$, dimension-charged by a constant-fraction decomposition of the paired row. If all field values are scalar, (13.6) recovers an actual map $u_j : K \to C$ with no assumed K-structure on C. All twists and parameter correlations are those of the original insertion map.

These constructions exhaust the fixed-signature cases listed after Lemma 6.4. Equal or unequal twists on one original first-kind or unitary simple factor can instead be processed at once by its trace-functional representation. Pure general-linear words with no pairing are already ordinary. The only remaining size case of (13.5) is treated next.

13.5 The augmented pencil for a larger free row

Contracting in the independent argument would be too expensive when it is larger than the paired arguments. Instead, append the pairing value as an extra output coordinate. Insertion parameters then act by a known output shear. A small-boundary pair for this augmented map gives an invariant subspace or identifies a tensor consisting entirely of insertions.

Proposition 13.3 (Unbalanced one-pair alternatives)**.** *In* (13.5)*, suppose* $\dim_K X = \dim_K Y = a \ge 64$ *and* $c = \dim_k C \ge fa$*. The exact insertion quotient has one of the following transitions: a high-rank mark with cost* $|k|^{O(fa)}$ *dimension-charged by ordinary natural-degree loss; a proper original-subspace restriction without branching on a paired row; a proper subspace restriction without branching on the free row with its entire label action; or an exact degree-one parameter quotient. No mark in the larger free row is required.*

Proof. For $x \neq 0$ mark $z \in Y$ with $Q(x, z) = 1$ and set

$$P_{x,z}(y) = y - zQ(x, y), \qquad M_x(y) = (T_j(x, y))_j.$$

Every insertion vanishes in $M_x P_{x,z}$. The marked compressed stack is therefore exactly quotient-invariant. Its marks cost at most $|K|^{2a}$, independent of c. If some stack has rank at least $fa/64$, Lemma 12.1 applies to these actual marked operators. It pays their $O(fa)$ marking branching exponent. The marks and actual returned subgroup remain imposed throughout the retained restriction sequence.

Otherwise put $r_0 = \lfloor fa/64 \rfloor$ and define the whole augmented linear pencil

$$\widetilde{M}_x : Y \longrightarrow C^{\{j\}} \oplus K, \qquad \widetilde{M}_x(y) = (M_x(y), Q(x, y)).$$

The last coordinate has rank f for $x \neq 0$, with kernel $\mathrm{im}\, P_{x,z}$. Rank–nullity on this coordinate gives $\mathrm{rank}_k \widetilde{M}_x \le r_0 + f$ for every x, also at $x = 0$. Equivalently, the discarded part factors through the single map

$$K \longrightarrow C^{\{j\}}, \qquad s \longmapsto (T_j(x, zs))_j.$$

This has rank at most f jointly over all labels. Pulling s outside T_j would be invalid for a merely k-bilinear tensor.

An insertion $u = (u_j)_j$ acts on the augmented output by the invertible shear

$$(d, s) \longmapsto (d + u(s), s).$$

The genuine row, field, and label transformations give their corresponding invertible block-diagonal maps, with the carried characters. Thus the augmented pencil is covariant under every actual quotient transporter. Proposition 9.2 gives a canonical pair $W \subseteq Y$, $E \subseteq C^{\{j\}} \oplus K$ such that

$$\widetilde{M}_x(W) \subseteq E \quad \text{for all } x, \qquad \operatorname{codim}_Y W + \dim_k E \leq 4(r_0 + f) \leq fa/8 < fa.$$

In particular $W \neq 0$. If $W \neq Y$, match this actual original k-subspace by the complete explicitly realized subspace procedure. It is a proper subspace restriction of the full paired natural group, which acts irreducibly on Y. There is no claim that W is K-linear and no unrestricted K-subspace transporter is substituted.

Suppose $W = Y$. Nondegeneracy of Q implies that the projection $E \to K$ is onto. Set

$$F_0 = E \cap C^{\{j\}}.$$

Every insertion shear acts identically on this kernel, so F_0 is a genuine covariant subspace for the actual free-row and label actions. If $F_0 \neq 0$ and labels are fixed, any first nonzero ordered coordinate projection is a nonzero original subspace of C of dimension at most $\dim E < fa \leq c$. Match it completely to obtain a proper subspace restriction. For moving labels retain the whole F_0 in the explicitly realized module $C \otimes L$. Each nonzero module for the full free natural group in this isotypic sum has dimension at least c, using its record-fixing realization from Lemma 7.6, which is trivial on L. Thus $0 < \dim F_0 < c$ ensures that its full-core stabilizer is proper. Match this entire subspace by the explicitly realized frame procedure under the current subgroup, including its label action, and prepare the complete returned original group by Theorem 3.11. Its stabilizer is a proper subgroup because some element of the old actual pure natural subgroup moves F_0. This gives strict unbranched progress even if the label p-action prevents F_0 from descending to the old diagonal quotient. This operation does not ask for a free transporter of an arbitrary auxiliary matrix space.

Finally, if $F_0 = 0$, projection identifies E with K. It is the graph of a unique k-linear map $u : K \to C^{\{j\}}$. Since it contains every $\widetilde{M}_x(y)$, the complete tensor summand is $T_j(x, y) = u_j(Q(x, y))$. Its information is exactly the parameter class of u modulo the actual correlated allowed translation parameters. The degree-one procedure matches this class and restores all actual original translation lifts and their kernel. It need not have u itself in the allowed translation space. These three low-rank cases are exhaustive. □

The augmented boundary pair is used only to extract W, the shear-fixed F_0, or the graph parameter. Matching the whole E by an ordinary unsheared frame procedure would lose the displayed insertion action. Nor does this augmented boundary pair assert an ordinary zero block of the original tensor.

13.6 Nonzero field values and exact scalar extraction

The quotient field values separate the remaining cases. A nonzero value yields a restriction, while scalar values in every permitted orientation determine the insertion parameters modulo the kernel of their parameterization. The pair consisting of the corrected tensor and the parameter class preserves the entire transporter.

Under Lemma 13.2 the intrinsic value of the insertion quotient is

$$\overline{\lambda} : A \longrightarrow (F/K)^{\{j\}}. \tag{13.7}$$

For moving labels the codomain is $\operatorname{Hom}_k(L^*, F/K)$, with the supplied whole L-action. The map is one linear map on the entire original coefficient row; the label length is not an exponent.

Lemma 13.4 (Fixed-label value bound)**.** *Suppose the old field and label actions are fixed. If* $t = \operatorname{rank}_k \overline{\lambda} > 0$*, at most* $\operatorname{char}(k)|k|^t \le |k|^{2t}$ *field phases can carry its source image to its target image. For each phase, imposing the actual ordered value equations removes at least* t *original* k*-dimensions from the pure coefficient action.*

Proof. The phase transporter, when nonempty, is a coset of the cyclic subgroup of $\mathrm{Gal}(F/K)$ preserving the source image U. Its image in $\mathrm{GL}_k(U)$ has order at most $|k|^t$. One elementary bound is as follows: the order of an element is the product of its semisimple order and a p-power from its largest Jordan block. For distinct irreducible factors of degrees d_i, the semisimple order divides $\prod_i (|k|^{d_i} - 1)$. If the largest block exponent is e, its p-power is at most $p^{\lceil \log_p e \rceil}$, which is at most $|k|^{d(e-1)}$ for that block when $e > 1$. The sum of these block degrees is at most t, giving the bound.

An element of the pointwise kernel on U sends each chosen field generator v among the coordinates of λ to $v + c_v$ with $c_v \in K$. Its pth power fixes each such generator and fixes K. It is therefore the identity on F. The cyclic pointwise kernel has order at most p, proving the phase bound. The pointwise kernel need not be trivial.

After a phase and one transporter have been fixed, the residual pure coefficient group satisfies $\mathrm{im}(g - I) \subseteq \ker_k \overline{\lambda}$. If its defining field is K_C, its image is a K_C-subspace, hence lies in the largest K_C-subspace W contained in this kernel. The group is the identity on the original quotient C/W, whose dimension is at least t. Its remaining natural action is explicitly realized on W, with its original shears and kernel retained. The number of phase choices is therefore dimension-charged by this actual original loss. □

For moving labels the displayed phase bound is not used: a phase can be compensated by a label transformation. The original-group procedure in the next subsection instead uses a proper kernel-subspace restriction or complete injective terminals.

If (13.7) is zero, every representative field value lies in K. Since these values generate the common field together with K, the field is K itself, with no additional gluing phase. Apply the construction to every eligible coefficient orientation. For (13.1) these are all three orientations, using the order of the paired variables appearing in that formula. For (13.2) there are two orientations. Each orientation is tested for a bad tuple or nonzero quotient value before claiming scalar extraction for the complete tensor summand. If all these tests have scalar values, they give a functional tuple modulo the insertion kernel. Write $\overline{S} = S/\ker J$, let $\overline{J} : \overline{S} \to M$ be the induced injective map, and denote the extracted value by $\overline{s}(T) \in \overline{S}$. Its exact laws are

$$\overline{s}(gT) = g\overline{s}(T), \qquad \overline{s}(T + Jr) = \overline{s}(T) + (r + \ker J). \tag{13.8}$$

The shift identity holds for every insertion parameter r for which $T + Jr$ remains in the scalar-extraction class. No equivariant parameter lift is required. This full shift identity follows from the same compression formula and is used for the insertion-valued affine cocycle, which need not lie in N. Indeed the compression identity extracts exactly the functional in its free position; its other insertion terms vanish. In the stable trace-functional and two-position cases J is injective: if $Jr = 0$, the same compression gives $\ell(x)P_v = 0$ for every free coefficient x, and $P_v \neq 0$ forces every extracted functional to vanish. The faithful quotient $\overline{S}$ also covers redundant parameter presentations without requiring an equivariant choice of a lift to S. In a one-pair tensor summand use trace duality (13.6) instead. The first law in (13.8) follows from the covariance of the whole contraction diagram and the uniqueness of its scalar quotient values. It includes the inherited semilinear and character action on S; no choice of a normalized similitude is necessary.

Proposition 13.5 (Complete insertion-pair encoding)**.** *Let $J : S \to M$ be an equivariant insertion map on the original ordinary ambient, let $N \subseteq \operatorname{im} J$ be the actual translation image, and put $R = J^{-1}(N)$ and $\overline{R} = R/\ker J \subseteq \overline{S}$. On the scalar-extraction class define*

$$C(T) = T - \overline{J}\,\overline{s}(T), \qquad v(T) = \overline{s}(T) + \overline{R}.$$

The pair $(C(T), v(T))$ is well defined on $T + N$ and is equivariant under the full current subgroup. For every source and target in this class,

$$\{g : gT - T' \in N\} = \{g : gC(T) = C(T'),\ gv(T) = v(T')\}. \tag{13.9}$$

It preserves the entire transporter and stabilizer, not just the existence of an equivalence. The assertion does not require $\overline{s}(T) \in \overline{R}$.

Proof. For $r \in R$, the second law of (13.8) shows $C(T + Jr) = C(T)$ and $v(T + Jr) = v(T)$. Its first law proves equivariance. These facts give one inclusion in (13.9). Conversely, matching its two entries gives

$$gT - T' = \overline{J}(g\overline{s}(T) - \overline{s}(T')) \in \overline{J}(\overline{R}) = N.$$

This also proves the equality of stabilizers. A residual discrepancy in N is corrected through the actual original translation map onto N; all of its kernel is restored as in Lemma 7.2. If the original current subgroup acts affinely with increments in $\operatorname{im} J$, write $c(g) = \overline{J}a(g)$ in the faithful parameter space. Then the two conditions are $gC(T) = C(T')$ and $g\overline{s}(T) + a(g) - \overline{s}(T') \in \overline{R}$. The full shift identity proves their equivalence exactly as above. Thus the insertion-valued cocycle remains in the one lower-degree residual even when its values lie outside N. No normalized complement to the original normal p-subgroup is required. □

In particular $C(T)$ is an ordinary original-field tensor even when the chosen correction is not induced by an allowed original transformation by itself. Its missing affine information is exactly $v(T)$. First match this degree-one value completely, and then process the whole $C(T)$ under that exact returned prefix. The projected small-rank estimates do not imply small rank of the uncompressed labelled stack: the omitted output borders can stack across labels. Ordinary contraction analysis is applied afresh to $C(T)$ inside the same node, rather than making that inference.

If a lower-degree part M_2 shares the translation image with the high part, use the single map and denominator

$$J_{\text{joint}} : M_2 \oplus S \longrightarrow M_2 \oplus M, \quad (x, s) \longmapsto (x, Js), \qquad R_{\text{joint}} = J_{\text{joint}}^{-1}(N).$$

Take its faithful parameter quotient by $\ker J_{\text{joint}}$. The extraction there is $(T_2, \overline{s}(T_3))$, its ordinary remainder is $(0, C(T_3))$, and its residual value is the one joint class modulo $R_{\text{joint}}/\ker J_{\text{joint}}$. The original complete preimage R_{joint} and every original lift kernel remain recorded. Its possible affine cocycle stays in the degree-two procedure. This is an immediate application of the preceding proof and preserves correlations that separate low-part and parameter solves would destroy.

13.7 Keeping field and value-space actions correlated

The earlier formulas were derived in coordinates, but the current group can still move the fields, forms, and value space. We now verify covariance under that full action. Perfectness and the hereditary order bound ensure that a newly selected large factor centralizes the relevant contraction algebra, without listing those complementary actions at every recursive step.

We now verify that these constructions apply to the actual carried state of Definition 7.5. Write H for its group, R_p for its actual normal original p-kernel, and $\overline{H} = H/R_p$. Its recognized product of derived natural classical groups is normal in this diagonal quotient. It need not have a normal complement in H. The structural preparation of Theorem 3.11 retains the old complementary actions and their permanent hereditary bound. In particular, Lemma 7.6 realizes each new large perfect quasisimple core inside the joint image on current row grades and all old complementary records, acting trivially on the latter. All calculations below take place in this joint image and are pulled back through its full kernel. The degree-two procedure of Theorem 5.17 is used throughout with its original-group return and explicitly realized whole label model.

Lemma 13.6 (Contraction centralizers under a complete stabilizer)**.** *Match a marked projected observation completely, retaining the entire field, form, character, exchange, and label action. The returned group normalizes its contraction algebra. For the perfect central product Q of the newly selected large quasisimple factors acting nontrivially on the old active rows represented in the contraction space W, an actual subgroup of the literal contraction centralizer maps onto Q. Consequently the natural-degree bounds for that centralizer bound the new natural constituents. No section of the original group extension is needed.*

Proof. An invertible change of the entire label basis leaves its generated operator algebra unchanged. Nonzero distinguished-field scalar changes do the same. Semilinear maps transport the field as a set, and a similitude transports the orthogonal projection because it transports both its image and kernel. Evaluation projections are covariant under the natural–dual action, and exchange records carry the adjoint partner. Thus the complete stabilizer normalizes the algebra.

Let α be the new action on all retained original argument grades, including inactive grades, and let $R = \ker \alpha$, and let β include every old complementary record, also those not descending through R. By Lemma 7.6, $K = \alpha^{-1}(Q) \cap \ker \beta$ maps onto Q. On the actual old operator row the marked equations for $g \in K$ are

$$gD_{j,v}g^{-1} = D_{j,v} + c_{j,v}(g)P_v.$$

The field and P_v are fixed, so $c : K \to \bigoplus_{j,v} K_+$ is an additive homomorphism. Its complete kernel K_0 centralizes every operator literally. The kernel of $K_0 \to \alpha(K_0)$ is $K_0 \cap R$, a normal p-group; hence every simple K_0-module kills it. The image $\alpha(K_0)$ is normal in Q, and $Q/\alpha(K_0)$ is a quotient of K/K_0, hence abelian. Perfectness implies $\alpha(K_0) = Q$. This remains true even when the new R acts nontrivially on the old row; no assertion that c factors through Q is required. The kernel and its original lifts are computable by Lemma 3.1. For ordinary marked observations $c = 0$ and already K is in the literal centralizer.

The old space W contains one representative of every relevant old active natural type, including each retained formal dual type once. It is sufficient for the simultaneous degree estimate: every active original interval has its actual natural–multiplicity factorization, and K_0 fixes the entire old multiplicity/complementary action. Its composition factors on such an interval are therefore copies of factors occurring on the corresponding summand of W. Original flag kernels add extensions, not additional simple types. Thus every relevant newly selected active natural type occurs among these factors of W, with the original field and Frobenius identifications retained. Inactive grades and any unrelated large factors remain in α and in all original preimages, but are not charged to this contraction.

Finally let B be the contraction algebra and $C = \mathrm{End}_B(W)$ its represented centralizer on the old row. Write $W = \bigoplus_i Q_i^{m_i}$ as a B-module and $K_i = \mathrm{End}_B(Q_i)/\operatorname{rad}\mathrm{End}_B(Q_i)$. Then $C/\operatorname{rad} C = \prod_i \mathrm{Mat}_{m_i}(K_i)$. Filter W by the radical of C. Each semisimple layer is a sum of copies

of the natural $K_i^{m_i}$, so every simple constituent for its subgroup K_0 occurs in a restriction of one of these modules and has prime-field dimension at most $\max_i [K_i : k]m_i$. Since K_0 maps onto the selected Q in the new graded action, the new natural constituents satisfy this same bound. More strongly, every distinct new natural simple type occurs among composition factors of the direct sum of one copy of each $K_i^{m_i}$. Its sum of distinct prime-field dimensions is therefore at most $\sum_i [K_i : k]m_i$. These are dimensions over $k = \mathbb{F}_p$, hence the required original prime-field degrees. Actual field, type, and Frobenius identifications merge repeated types before this sum is taken. This is the sum bound in exactly the same type convention as W, used in the off-diagonal natural-degree charge; there is no further duality factor in this bound. If adjoint equations are also imposed, they replace C by a subalgebra; restriction of these natural modules gives the same bound. All original kernels remain in the preimage when returning the transporter. □

For completeness, in a fixed-record branch the correlated scalar image can alternatively be enumerated at dimension-charged cost. On $U = \operatorname{im} P_v$ the equations are $gD_jg^{-1} = D_j + c_j(g)I$. A Sylow p-subgroup maps onto their additive image and has a nonzero fixed vector $u \in U$. Applying $g - I$ to every relation among $u, D_1u, \dots, D_\ell u$ shows that $(c_j(g))_j$ belongs to a K-linear space of dimension at most $\dim_K U - 1$. Thus its actual, possibly only prime-field-linear image has size at most $|K|^{\dim_K U - 1}$. At most three vertices still give $|K|^{O(a)}$. This enumeration is used only in the dimension-charged bad-tuple branch. For the full moving-record group the record may be crossed, so treating it directly as this additive homomorphism would be incorrect.

Lemma 13.7 (Proper invariant-subspace restriction)**.** *Let W be an intrinsic nonzero proper subspace of a current irreducible natural argument. Its complete transporter is obtained by Theorem 5.19, with the full original kernel. Its stabilizer is a proper subgroup. More generally, for a covariant subspace F_0 in a supplied supported module, the same conclusion holds whenever the current full natural subgroup does not preserve F_0.*

Proof. The subspace equations are exactly the supported frame equations. They are imposed in the actual supplied group, retaining its entire preimage, so that no statement about a section of the group extension is required. If the stabilizer were the whole group, its irreducible natural normal subgroup would preserve W, contrary to irreducibility. The general assertion has this noninvariance as an explicit hypothesis. The structure theorem is applied to the resulting actual subgroup. In particular, this argument applies to quadratic orthogonal groups in characteristic two and does not identify such a group with the full isometry group of its alternating polar form. □

An entire label subspace need not descend to H/R_p: R_p can act nontrivially on its labels. In that case the expression H_{F_0}/R_p is not used. The complete explicitly realized frame call returns the actual H_{F_0}, its intersection with R_p, and all lifts; structural preparation is performed on this actual group. The small nonzero F_0 of Proposition 13.3 still gives strict subgroup progress by its dimension argument. No complementary-quotient embedding is required for this alternative.

Apply these facts to a nonzero common-field value $\lambda : C \to F \otimes L$, or $\overline{\lambda} : C \to (F/K) \otimes L$. Every genuine transporter has a common field isomorphism on the first factor and the actual invertible label action on the second. Its original kernel is therefore intrinsic. If this kernel is nonzero and proper, match exactly this k-subspace and use Lemma 13.7; no field phase is selected and no claim of K_C-linearity is made.

If the value is injective, use the common field constructed for both source and target by Section 10. For each fixed field phase θ, solve the complete value equations by the explicitly realized full matrix-module procedure, with F as passive explicit coefficient multiplicity. If the old field action can move, use $k = \mathbb{F}_p$, enumerate all prime-field Frobenius powers of F, and impose the

actual restriction of θ to K in this same call. There are at most $f[F : K] = \exp(O(fa))$ such phases. All characters remain in the old correlated label model.

In the returned stabilizer for a fixed phase, a pure coefficient element fixing the labels satisfies $\lambda g_C = \lambda$, so injectivity gives $g_C = I$. A possible new large perfect factor is represented on the joint row-and-record image as $Q \times 1$ by Lemma 7.6; this is valid even if the original label representation does not descend through the row kernel. Its coefficient projection is therefore trivial by the same equation. Thus the entire current three-slot observation has at most two large slots under this returned subgroup. Complete that entire observation, including its affine ambiguity and every inherited previously matched condition, by Theorem 5.17 before proceeding to the next phase. Reunite only these completed terminal cosets in the original group. The phase costs add; they are not exported as recursive choices. This argument uses the supplied support realization of L. Pure-core triviality of an arbitrary unrepresented label action would not suffice.

13.8 Completion of the insertion reduction

We assemble the alternatives and explain how each advances the original recursive state. In particular, the scalar-extraction step is used once for its current denominator. Its output retains all the information needed to solve the original quotient condition, so the same denominator is not reintroduced by a change of representation.

Every operation above keeps the original filtration, current numerator and denominator, completed observation family, and actual translation lift map. The alternatives are now exhaustive. A tensor summand with no insertion is ordinary. Each permitted insertion pattern has a dimension-charged high-rank or bad-tuple alternative, a strict unbranched original subspace restriction, a nonzero field-value transition, or scalar extraction. The unbalanced one-pair case additionally has its exact insertion-only parameter terminal. Scalar extraction gives the complete pair of Proposition 13.5, including the joint lower-degree residual when necessary.

This last transition consumes the current insertion denominator once. It is a change of representation inside the node: the ordinary first entry is subsequently processed under the complete stabilizer obtained from the residual value. A new affine denominator can arise only after strict advancement of the retained original layer or a proper original subspace restriction or dimension-charged decomposition. It cannot be recreated by dropping the pair and restarting an abstract cubic quotient problem. This is the consumed phase in Definition 7.5, whose finite progress is counted in Section 14.

All coordinate spaces used here have polynomial size in the original degree and explicit labels, except the glued field coordinates. Those occur only inside complete local operations of cost $|k|^{O(d)} \operatorname{poly}(L)$. Marks in the projected same-row cases have branching exponent $O(fa)$ and their bad tuples have a constant-fraction row loss. Balanced one-pair marks are bounded by the free-row degree, which is no greater than the paired-row degree. Unbalanced marks use only the paired rows and their high-rank loss. Proper subspace restrictions and canonical pair encoding export no target lists. Together with the original-group lemmas, these are precisely the insertion alternatives and return properties used by Sections 12 and 14.

14 Correctness and running time

The preceding reductions each have a single-exponential local cost. To obtain the same bound for the complete algorithm, we must distinguish choices whose costs add from choices whose numbers multiply along a recursive path. Component-pair calls and completed field-automorphism cases

add. Images of marked vectors remain unresolved while smaller tensor problems are solved, so their branching must be charged to a decrease in argument dimension.

14.1 Retaining restrictions until a dimension decrease

Fix the dimensions (d_1, d_2, d_3) at the beginning of a call, and put $c_0 = \min_i d_i$. Set

$$\epsilon = 1/128.$$

Continue the current restriction sequence while every corresponding argument has dimension greater than $(1-\epsilon)d_i$. Components and newly created multiplicities are processed before this test. An argument that becomes smaller than this threshold gives a *geometric decrease*. The constant ϵ is below the $1/64$ loss supplied by the tuple analysis after bounded-size cases are removed, and below the losses from components, multiplicities, and the zero-block reduction.

There are at most six ordinary natural types among the three arguments and their paired duals. We retain their identifications throughout the restriction sequence. This matters because, for example, a natural and a dual general-linear module can become the same orthogonal or symplectic type after restriction.

Lemma 14.1 (Charging a sequence of restrictions)**.** *Suppose a vector or bounded-tuple image list has at most p^β choices, with β bounded by a constant times its proved loss of distinct ordinary natural degree. Suppose its cost is also at most $p^{C_0 c_0}$ for an absolute constant C_0. Include the direct argument-dimension charges for tuple decompositions and new multiplicity frames. If a retained sequence ends with dimensions (d_1', d_2', d_3') and a geometric decrease, then*

$$\sum \beta \le C \sum_{i=1}^{3} (d_i - d_i')$$

for an absolute constant C. If it ends by completing the tensor condition, then $\sum \beta = O(d_1+d_2+d_3)$.

Proof. Before the final geometric decrease, an old natural type can have at most one constituent of more than $(1-\epsilon)$ of its original dimension: two would exceed the original argument dimension. Its identifications with other retained types can therefore only merge along this continuation. Among at most six types there are at most five such mergers.

Between mergers, the loss of natural degree counted with paired duals is at most twice the sum of actual argument-dimension losses. The charges consequently telescope. Each merger contributes at most $C_0 c_0$, so their combined contribution is $O(c_0)$.

The final restriction can split an old type; its last vector-image charge is bounded directly by $O(c_0)$. At the geometric decrease, some argument loses at least ϵ of its entry dimension. The total dimension loss is thus at least ϵc_0, which pays all the bounded merger and final charges. A tuple decomposition already loses a fixed fraction of an operator argument. Lemma 8.2 pays each new frame by its actual argument loss. Proper-subspace restrictions have no outer branching, and injective field-value cases are completed locally. If the sequence instead ends in a complete solution, its whole entry dimension pays these same bounded charges. □

For example, restricting $\mathrm{SL}_a(k)$ to a diagonal copy of $\mathrm{SL}_{a/2}(k)$ on two equal summands leaves the isotypic dimension a unchanged. The multiplicity test immediately replaces this with a natural argument of dimension $a/2$ and two value coordinates. The algorithm therefore records a geometric decrease at this point.

14.2 Bounding work between geometric decreases

Every new invariant flag refines old intervals. For a new normal p-subgroup, intersect its augmentation flag with these intervals and retain all old inter-interval shears in the exact translation kernels. Along a path, at most $O(d)$ argument intervals can be created, and their induced tensor filtration has polynomially many layers. Previously matched conditions remain matched under the full triangular action on finer coordinates.

A step without geometric decrease or outer branching must do at least one of the following: strictly restrict the current original subgroup, properly refine a retained argument flag, match a previously unfinished condition, or complete the current reduction from an insertion quotient to its tensor and parameter pair. The last reduction cannot recur without one of the earlier changes. A strictly descending chain of subgroups of $\mathrm{GL}_d(p)$ has length at most

$$\log_2 |\mathrm{GL}_d(p)| = O(d^2 \log p).$$

Hence these sequences have polynomial length. Long auxiliary additive filtrations are handled within a completed degree-two call and do not create one outer call per coordinate.

In a component decomposition there is at most one dimension-compatible pair containing more than half the argument on each side. Its continuation is retained in the same call and its integer dimension strictly decreases. All other component-pair calls have a geometric decrease. The retained continuation therefore contributes a polynomial-length sequence with additive side calls.

14.3 Counting the complete recursion tree

The following statement isolates the combinatorial argument. A target-choice node is one that opens a list of possible images which remains imposed on unresolved descendants. An additive side call is a complete subproblem whose cost is added to the current computation.

Theorem 14.2 (Weighted recursion bound)**.** *Suppose a recursive procedure has the following properties.*

1. *A target-choice node has at most p^β children, and the sum of its branching exponents on any root-to-leaf path is $O(d)$.*

2. *A geometric decrease reduces one of three corresponding argument dimensions by a fixed factor and increases none. Between such decreases there is a polynomial-length sequence with at most one retained large-component continuation and polynomially many additive side calls.*

3. *Each component-pair call is computed once. Every other local list is completed before it returns, at cost*
$$p^{O(d)} \operatorname{poly}(d, \log p).$$

4. *The number of labels does not create outer branching, and all polynomial exponents are uniform.*

Then the total running time is $p^{O(d)} \operatorname{poly}(d, \log p)$.

Proof. At a target-choice node, assign weight $p^{-\beta}$ to every outgoing edge. The sum of these weights is at most one. By induction on a finite rooted tree, the sum of the products of edge weights over any prefix-free set of paths is at most one. Fixing the choices of additive side calls, each terminal target-choice path has weight at least p^{-Cd} for an absolute constant C. There are therefore at most p^{Cd} such paths.

Each positive integer argument dimension can decrease by a fixed factor only $O(\log(d+2))$ times. Contract the intervening polynomial-length sequences. Let $P(d, \log p)$ be a fixed polynomial bounding the number of additive side choices at one contracted node. There are at most

$$P(d, \log p)^{O(\log(d+2))} = p^{O(d)}$$

possible sequences of such choices. Here $(\log(d+2))^2 = O(d)$ and $\log(d+2)\log\log p = O(d \log p)$. One may fix this finite sequence before exposing the target choices; a choice not present on a particular branch produces no leaf. Thus adaptive target lists do not add a further factor.

The polynomial lengths of the contracted sequences bound the internal nodes as well. Multiplication by the complete local cost $p^{O(d)}$ changes only the constant in the exponent. The uniform polynomial dependence on the labels contributes one polynomial input factor. □

Lemma 14.1 telescopes across geometric decreases and supplies the first property, including the final completed restriction sequence. The component-pair precomputation and the preceding polynomial event count supply the second and third properties.

Lemma 14.3 (Presentations of the moving labels)**.** *Every moving label module used by the ordinary algorithm satisfies the second input version of Theorem 5.17, including its actual evaluation covers, summand maps, section maps, and local label-stabilizer image bounds. The total cover dimension is $O(d)$ and the number of tensor positions is at most eight. These properties persist under every restriction and return used above.*

Proof. Initially the tensor is scalar-valued and there is no moving label. All later labels come from an actual structural complementary block, a multiplicity factor of an original argument interval, or the component indices of such a block. For a complementary block retain the actual module and section maps supplied by Theorem 3.11; for a deleted-permutation block retain its actual permutation cover, whose coordinate summands have their genuine stabilizer actions. One does not replace that action by an independent permutation group. The local coordinate-stabilizer image is the bounded multiplicity image supplied by the structural theorem; the alternating permutation image is represented by its coordinate labels, rather than enumerated.

For a large isotypic interval $A \otimes_K M$, keep the actual group of factorizations and its scalar kernel from Lemma 3.12. Before a multiplicity frame is imposed, its total moving multiplicity image is bounded by Lemma 3.13, aggregating all equivalent occurrences. More explicitly, with $f_i = [K_i : k]$, natural dimension a_i, and total multiplicity m_i,

$$\sum_i f_i a_i m_i \le O(d), \qquad f_i a_i^2 > C_* d, \qquad \sum_i f_i m_i^2 = O(d).$$

Thus its joint image has order at most $p^{O(d)}$, including the retained field, graph, and scalar-form records. After a frame is fixed its multiplicity matrix is identity and only these actual correlated records remain. These are valid bounded-image evaluation summands. Passive fixed multiplicities require no group action or cover enumeration.

A component pair call selects actual source and target summands and uses their original index stabilizer. Its maps on the other arguments and all existing label summands are restrictions of the saved maps. The multiset combination returns an original subgroup; it does not create a new freely moving label space. Likewise a frame or a subspace call returns only its original transporter. Its auxiliary frame domain is eliminated, and is not appended as another moving label. Previous completed observations are conditions on that original subgroup, rather than new independent general-linear actions on their coefficient spaces.

The zero-block arms use restrictions and quotients of original intervals; the exact affine top uses their actual shear image. The insertion parameters are sums of natural duals with passive field multiplicity and the original pairing-line action. Their common scalar and Frobenius records were already part of the factorization data. Glued-field coordinates occur only in a completed local call after a field phase is specified, where they are fixed passive multiplicities; they are never exported as a moving evaluation module.

After an original-subgroup restriction, retain each existing evaluation cover with its restricted action and its same section maps. Its local stabilizer image is a subgroup of the saved bounded image, so its order bound cannot increase. All subgroup sections continue to obey the saved hereditary bound. New decompositions are made only inside remaining original argument intervals. Rebuild their factorizations directly from those intervals: successive multiplicity factors are combined into the actual total multiplicity module, rather than accumulated as fresh copies. This is the actual tensor-product evaluation module of the successive multiplicities, with its inherited diagonal action and the original field-balancing relations; no new section of a factorization extension is chosen. Its dimension bound follows by multiplying the successive factor dimensions within each original interval. For each original argument, the sum of natural and multiplicity cover dimensions is bounded by a constant times its dimension, since $a + m \leq 2am$ and the intervals are disjoint. The constant inflation of a permutation cover is used only on its original interval. Already saved covers are not covered a second time. Consequently the three arguments have total cover degree $O(d)$.

Each original argument needs at most one natural position and one multiplicity/evaluation position. Unmatched component types are eliminated by numerator equations and defining-field balancing is imposed by exact section maps. There are therefore at most six positions, with at most two additional temporary positions for the scalar coordinate or subspace frame in a completed call. Every map just described is an explicit linear map on the original intervals; pulling back the whole numerator and denominator retains all correlations. This verifies the precise second-version hypotheses, not merely triviality on one current core. □

14.4 Uniform representation size

For each original argument, rebuild the current factorization using one direct sum of natural spaces and one direct sum of multiplicity or evaluation spaces. More precisely, a sum of pieces $A_\tau \otimes_{K_\tau} M_\tau$ is a quotient of

$$\left(\bigoplus_\tau A_\tau\right) \otimes_k \left(\bigoplus_\tau M_\tau\right)$$

by the relations eliminating unmatched types and imposing K_τ-balancing on matching types. Pullback is injective on tensor coefficients. For three original arguments this requires at most six factor positions. A scalar or homogeneous coordinate and one temporary subspace frame bring the maximum to eight. The total degree of their original-space realizations remains $O(d)$.

The representation is rebuilt from the original rows and their current subquotient maps, so an exported child has coefficient and label description length $\mathrm{poly}(d, \log p)$. Previously expanded auxiliary spaces are not tensor-multiplied again. The lower-degree algorithm permits any fixed arity r, with cost $p^{O_r(d)}$, provided at most two positions have large natural action. Exponentially many coordinates of the glued field occur only inside completed local calls; their polynomial arithmetic is included in those calls and they are not exported as growing recursive value spaces.

For a complete successful union, choose a representative b_0 and generate its source stabilizer from the branch stabilizers and the ratios $b_0^{-1} b_j$ of successful representatives. Retain a generator only if it enlarges the generated subgroup. Each addition at least doubles its order, so $O(d^2 \log p)$

generators suffice. Constructive membership uses the original vector action, and all words remain shared straight-line programs. Reapplying the structural theorem to this actual returned group gives the data required by the next call.

14.5 Proof of the main theorem

Theorem 14.4 (Correctness of the recursive construction)**.** *Algorithm 7.7 is exhaustive, returns the complete original transporter, and satisfies the running time in Theorem 7.1.*

Proof. We induct on geometric argument decreases, using the finite subgroup, flag, and completed-condition progression for the intervening work. First we verify the recursive invariants.

At the initial state, the arguments are the original vector spaces, the translation denominator is zero, and the product group has its standard representation. Every completed return is an actual subgroup on these original spaces, so Theorem 3.11 applies within the retained flags. Lemma 7.4 constructs the exact coefficient quotient and normalizes its cubic action modulo the full insertion space. Lemma 7.3 computes the image of the actual flag kernel and all its lifts. Lemma 6.4 applies to precisely this image. Deleted-permutation denominators remain within completed lower-degree calls. The fixed hereditary threshold and Lemma 7.6 provide the actual joint-image core that fixes saved bounded value-space actions, as required by the field-value and zero-block reductions. This use does not split the original group extension.

A component-pair problem is solved under the actual index stabilizer with its full original preimage. It therefore retains the shared affine action and all kernels acting on other components. A multiplicity frame uses the genuine group of factorizations, realizes the natural argument as $A \otimes e_1$ inside the current argument quotient, and pulls coefficients back with all field-balancing equations. Both reductions retain the entire ordered value family.

After these reductions, each contraction is a complete supported lower-degree observation under the current subgroup. A proper-kernel restriction is its actual subspace stabilizer, and every chosen vector image remains a condition on unresolved tensor children. Scalar extraction is applied only after the insertion lemmas establish its permitted shift law. The resulting pair retains the joint lower-degree residual and original translation lifts. The augmented one-pair construction is used only for its proved invariant-subspace or insertion-only alternatives. In the zero-block reduction, the zero identity is retained while the two intrinsic quotient tensors and the affine final part are formed. Their smaller-argument certificates and actual shear fibers therefore satisfy Theorem 11.1. Every generated state again has the form in Definition 7.5.

Proposition 12.2 exhausts the contraction cases. Theorem 5.17 completes every value with at most two large occurrences. Every other exported child has a geometric decrease, except the unique large component continuation which retains its restrictions within the call. Complete pair transporters, the exact fiber sequence, actual translation correction, and complete-branch union show inductively that the output is the full original transporter with every kernel. Structural preparation of its actual source stabilizer completes the induction.

Finally Lemma 14.1, the polynomial event count, component-pair precomputation, and Lemma 14.3 and the uniform representation bound verify all hypotheses of Theorem 14.2. The total cost is $p^{O(d)}\,\mathrm{poly}(d, \log p)$. For an input over $q = p^e$ the faithful original representation has prime-field degree at most a constant times $e(n_1 + n_2 + n_3)$. Hence this bound is $q^{O(n_1+n_2+n_3)}\,\mathrm{poly}(L)$. For an arbitrary supplied subgroup H on the original arguments, first compute this complete transporter gS in the full product group. If it is empty, the H-transporter is empty. Otherwise compute $H \cap gS$ by Corollary 5.18 in the faithful block-diagonal action on the direct sum of the three original arguments. The two comparison copies have total degree $O(d)$, so this final intersection has the

same bound. Take its complete original preimage in the supplied comparison group, retaining every invisible kernel. The resulting coset is exactly the requested subgroup transporter. □

14.6 An explicit transporter for the canonical targets

Theorem 14.5 (Anchored insertion-quotient transport)**.** *Let H act on three original argument spaces and preserve specified pairings between eligible pairs of arguments, allowing their actual field and form-line actions. Let $J : S \to T$ be the corresponding insertion map and $N \leq \operatorname{im} J$ an explicit H-invariant prime-field subspace. For two tensors t, t', their complete H-transporter modulo N is computable in $p^{O(d)}\operatorname{poly}(L)$ time, where d is the original faithful row degree. In particular this solves each canonical target $C_j = T_j/J(T_j)$ under every subsequent actual subgroup, without any cohomology hypothesis on that subgroup.*

Proof. First suppose the pairings are prime-field valued and fixed. Adjoin an anchor coordinate to each argument and define

$$\tilde{t}((x,s),(y,u),(z,v)) = t(x,y,z) + sb_{23}(y,z) + ub_{13}(x,z) + vb_{12}(x,y).$$

Omit absent pairings; all coefficients with at least two anchors are zero. Retain the full preimage $R = J^{-1}(N)$, including $\ker J$. For $r = (\ell_1, \ell_2, \ell_3) \in R$, let the row shears send (x,s) to $(x, s+\ell_1(x))$, and similarly in the other positions. Substitution changes exactly the ordinary tensor coefficient by $J(r)$; there is no mixed shear term, because each anchored coefficient uses only the other two original coordinates. The other coefficients remain fixed. Thus the explicitly represented group generated by the lifted H and these additive basis shears is $R \rtimes H$, up to the retained invisible kernel, and its augmented tensor transporter projects onto precisely

$$\{h \in H : ht - t' \in N\}.$$

Every quotient transporter lifts because R maps onto N.

Run Theorem 7.1 in the full augmented argument group, and intersect its complete answer with this supplied semidirect group by Theorem 5.18. Project the witness and its entire stabilizer to H. The full preimage of N and all original kernel words have been retained, so this projection restores every invisible transformation, not merely one choice of a shear parameter.

For a defining-field-valued pairing b use an anchor $s \in K$ and the coefficient $\operatorname{Tr}_{K/k}(sb)$. Its shear parameter is an arbitrary k-linear map into K. A scalar, Frobenius, or form-line action on b is accompanied by the known dual anchor action. The baseline tensor is invariant and the same substitution proof applies. The constantly many anchors add only $O(d)$ dimensions. If the target forgets part of the original faithful H-action, retain that action in parallel as its graph and lift through the complete kernel. The group generators are an additive basis of R, never its element list; a parameter dimension $O(d^2)$ therefore causes no exponential degree increase.

For several independent canonical target coordinates with fixed labels, solve them consecutively in the full returned original coset, normalizing the next target by its witness. Their costs add and the final intersection is exact. If labels move, this statement is applied only after their actual common action has been fixed, or to the corresponding single explicitly realized simultaneous observation. It does not permit replacing correlated labels by independent coordinates. This is precisely the fixed-label target call used after the outside action has been specified in the supported cubic algorithm. □

15 Affine equivalence of cubic tensor subquotients

We now use Theorem 7.1 and the canonical-target results of Section 6, which depend only on the already proved quadratic algorithm and its intersection and frame consequences. In particular, no invocation below is an application of ordinary tensor isomorphism to an arbitrary cubic denominator.

Lemma 15.1 (Detection on a semisimple module)**.** *Let C be a finite group and let M be a semisimple $\mathbb{F}_pC$-module. Suppose a list of modules T_j contains in its socles every simple type of M outside a specified class, and contains no simple type from that class in any of its socles. Stack bases of all spaces* $\mathrm{Hom}_C(M, T_j)$ *into a map*

$$F : M \longrightarrow \bigoplus_j T_j^{\oplus n_j}.$$

Then $\ker F$ *is exactly the sum of the simple summands of M in the specified class. The map is obtained by linear systems on generator matrices.*

Proof. A nonzero map from a simple module into any module is injective and its image is contained in that module's socle. Thus every simple summand of the excluded class is killed by every map. For each other simple type choose an embedding in some $\mathrm{Soc}(T_j)$. Given a nonzero vector in an isotypic semisimple summand, a C-linear projection onto a suitable simple copy followed by this embedding does not kill that vector. Such a map belongs to $\mathrm{Hom}_C(M, T_j)$ and is a linear combination of the computed basis maps. Hence the stacked map is injective on the complementary semisimple summands. Solving $f\rho_M(g) = \rho_{T_j}(g)f$ for group generators computes the entire Hom space, not just maps extended from an ambient tensor space. □

Lemma 15.2 (A quadratic presentation of the remaining simple types)**.** *Let W_2 be a specified direct sum of all degree-at-most-two tensor words on the current natural spaces, and let M be a semisimple C-module whose simple types all occur among the composition factors of W_2. Then M has an explicit invariant section presentation in passive copies of W_2, of polynomial coefficient size. The construction introduces no new moving representation space and remains equivariant after restriction to any subgroup of C.*

Proof. Let A_2 be the image of $\mathbb{F}_pC$ in $\mathrm{End}(W_2)$. An element of the group algebra annihilating W_2 annihilates every composition factor of W_2, and therefore annihilates M, since M is semisimple with only those types. Hence M is an actual A_2-module.

This action is computable without enumerating the group algebra. Generate the simultaneous image algebra on $W_2 \oplus M$ by repeatedly taking spans and products of generator matrices. Projection to its W_2 block is injective by the annihilator assertion and surjective onto A_2, so linear algebra transfers the M-block action to a basis of A_2. The number of span enlargements and all matrix dimensions are polynomial in $\dim W_2 + \dim M$.

Choose a basis $w_1, \dots, w_n$ of W_2. The column map

$$A_2 \longrightarrow W_2^{\oplus n}, \qquad a \longmapsto (aw_1, \dots, aw_n)$$

is an injective map of left A_2-modules. Choose a vector-space basis $m_1, \dots, m_\ell$ of M. The map

$$A_2^{\oplus \ell} \longrightarrow M, \qquad (a_1, \dots, a_\ell) \longmapsto \sum_i a_i m_i$$

is surjective, since $1 \in A_2$. Combining the two maps realizes M as an invariant section of $W_2^{\oplus n\ell}$, with numerator the embedded free module and denominator the computed kernel. Copies are

passive, and W_2 has degree at most two on the same natural spaces. Every map is C-equivariant, hence equivariant for its subgroups. □

Proposition 15.3 (A complete cubic quotient)**.** *At a cubic problem with a bounded number of selected evaluation summands, one can either finish the complete original affine equation or compute a complete original transporter on a quotient of positive dimension h, using $p^{O_r(d)}\operatorname{poly}(L, \log p)$ time and only earlier complete algorithms. In the latter case, its entire original preimage, including the full joint-action kernel, is retained. The remaining equation is one affine problem in the caller's presentation, of dimension $\dim Q - h$.*

Proof. Use $\pi : H \to J$, $C \trianglelefteq J$, $K = \ker \pi$, and $E = c(K)$ from Lemmas 5.11 and 5.12, after centering the source at zero. If $\overline{Q} = Q/E$ is zero, translation correction completes the equation. Otherwise let A be the image algebra of C on $\overline{Q}$, let $R = \operatorname{rad} A$, and put

$$M = \overline{Q}/R\overline{Q}, \qquad U = \{q \in Q : q + E \in R\overline{Q}\}. \tag{15.1}$$

The image algebra and its radical are computed by Lemma 3.15 on this explicit coefficient module. They are not treated as new physical representations. The quotient M is semisimple as an A-module and hence as a C-module. Since R is nilpotent, $M \neq 0$: if $R\overline{Q} = \overline{Q}$, iteration would give $R^k\overline{Q} = \overline{Q}$ for every k, contradicting nilpotence. Set $h = \dim M > 0$.

Normality of C in J implies that conjugation by J preserves its image algebra and its Jacobson radical. Hence $R\overline{Q}$ and U are invariant under the entire current linear action. We have $E \leq U$ and $\dim U = \dim Q - h$. The affine action on M factors through J.

Enumerate every actual coset of C in J. For each representative u, normalize the target by the descended affine action of u^{-1} and work solely under C. Every evaluation space and every isotypic multiplicity space is then a fixed label space. The head M is a semisimple section of cubic words on the specified natural representations. Apply Theorem 6.12 to these words. Its list of canonical modules T_j has the following precise properties: their socles contain all simple types not occurring in the universal quadratic presentation W_2, contain none of the latter types, and $H^1(C, T_j) = 0$. Their presentations are the specified literal cubic words modulo pairing-insertion spaces; the theorem is not a statement about arbitrary denominators.

Compute the map F of Lemma 15.1. For each component $f : M \to T_j$ of F, the projected actual cocycle $f(c)$ has an origin $b_f \in T_j$. It is found by the linear system

$$f(c(g)) = gb_f - b_f$$

on generators of the full current C. Consistency is guaranteed by the stated cohomology theorem. Therefore its affine equation is the linear equation

$$gb_f = f(t) + b_f$$

when the source is centered at zero. These equations are imposed successively using Theorem 14.5, always under the entire subgroup returned by the preceding equation, with exact target normalization by its witness. All origins have been computed for the full C before these restrictions; no cohomology vanishing is asserted for an arbitrary later stabilizer.

There is no independent motion of the Hom-space labels in this step. All maps are C-equivariant, and the computation takes place inside a single normalized coset case under C. Its successive stabilizers are subgroups of C, so those labels remain fixed. Outside elements are introduced only when completed coset cases are reunited. The canonical-target theorem uses its completed ordinary

three-tensor and earlier intersection algorithms, retaining their insertion, scalar, and original kernels. Its fixed enlargement of natural degree is covered by the uniform $O_r(d)$ bound.

If the high equations are consistent, their complete answer is the transporter on $M/\ker F$. Lemma 5.1 gives the exact remaining affine equation on $\ker F$. By Lemma 15.1, this is the semisimple sum of the types occurring in W_2. Lemma 15.2 provides its actual quadratic section, including its entire denominator. Apply the already established Theorem 5.17 to the residual cocycle under the whole subgroup returned by the high equations. This is an application of the supplied-group quadratic theorem, not an application of a full-classical-group result to that arbitrary subgroup. It completes the head equation for this coset case. If C is trivial, use equality tests for the bounded list of coset representatives.

Complete every coset case and combine their head answers by Lemma 5.3. This gives the complete head transporter in J. Its original preimage in H is the complete head transporter there. In particular it includes every element of K: those elements translate by vectors in $E \le U$, so act identically on M. At this stage one must not replace K by $\ker(c|_K)$ or demand that the chosen witness solve the full equation on Q.

Choose any original lift of the completed head witness. Apply Lemma 5.1 to the original quotient $Q \to Q/U = M$, using the original affine action and the caller's section presentation. This produces exactly one remaining problem on U, of dimension $\dim Q - h$. It enforces all the translations in E as well as the other remaining equations. No such problem was opened separately inside a coset case for the head.

The number of cosets of C in J is $p^{O_r(d)}$. The image algebra, Hom spaces, origins, and free-module presentation have polynomial coefficient size. There are polynomially many completed canonical-target and quadratic calls, all at natural degree $O_r(d)$. Thus the work before the single remaining fiber has the required common bound. The argument also proves its complete-preimage output contract. □

Theorem 15.4 (Complete cubic affine transport)**.** *For every fixed arity r, a supplied-group affine transporter on an arbitrary invariant subquotient of a tensor presentation of degree at most three can be computed deterministically in*

$$p^{O_r(d)}\,\mathrm{poly}(L, \log p)$$

time. The output is empty, or an original witness and generators of the entire original source stabilizer. Arbitrary correlated denominators, quotient-valued cocycles, repeated representations, the specified field and duality identifications, and retained original invisible kernels are allowed. The evaluated-factor version of Theorem 5.17 is allowed with at most three value occurrences and the same fixed total arity.

Proof. The exact evaluation, filtration, cover, and balanced-family constructions apply unchanged at degree three. Their number of tensor positions is fixed. At bounded label count, Proposition 15.3 either completes the problem or supplies a complete positive-dimensional quotient and exactly one original fiber. The fiber uses its own actual subgroup and current-value preparation, while retaining only the caller's selected evaluation summands. It therefore satisfies the same input conditions and has strictly smaller direction dimension.

Apply Proposition 5.16. Its terminal assumption is precisely the full-preimage and single-fiber statement just proved. The acyclic proof order is:

(1) affine transport under the full classical product;

(2) supplied-group quadratic transport;

(3) intersection and frame transport;

(4) complete ordinary three-tensor isomorphism;

(5) the canonical cubic target transporters;

(6) supplied-group cubic transport.

Only the one strictly smaller original fiber invokes the last algorithm again. Every earlier algorithm is a completed call, so the hierarchy has fixed depth for the common local cost. The full-output assertions follow from the exact quotient/fiber identities, complete unions of coset answers, and translation-kernel reconstruction already proved. □

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